

# Captured weight and boundary leakage bound the error of sample-based spectral functions

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Sample-based quantum diagonalization is moving to dynamics. Given the subspace the shots populated, how wrong is the spectral function built on it? We prove a two-sided bound on the relative  $L_1$  error of Rayleigh–Ritz reconstruction: with  $w$  the captured Born weight of the probe, the error is at least  $1 - w$  and at most  $(1 + \sqrt{w})(\sqrt{1 - w} + \hat{\Lambda}(\eta)/\eta)$ ,  $\hat{\Lambda}$  the normalized coupling of  $H$  across the subspace boundary at resolution  $\eta$ , from the Ritz vectors. Shots buy the first term down, not the second. On the subspace that *is* the probe’s Born support, where that term vanishes identically, the relative error is still 0.33–0.43 at  $\eta = 0.18t$  and 0.90–1.15 at  $0.05t$ , against a ceiling of 2. No bound indexed on the captured weight alone can do this: an analytic counterexample forces any such constant to its trivial value; none of 606 subspaces violates the bound proved here. On Hubbard sectors to dimension 10 306 296 ours is worse than the trivial bound at every published fraction, at all five sizes, by 1.66–8.61, converged in depth at  $L = 12$ , so those reconstructions are uncertified, though independent code recovers the fractions to 7%. The obstruction is structural: at  $L = 8$  the probe’s Krylov space touches every determinant of the sector. What survives is a calibration: over sixteen  $(L, \eta)$  cells spanning five decades of true error, the bound crosses the trivial one at a true relative error  $1.2\text{--}7.3 \times 10^{-3}$ , median  $2.8 \times 10^{-3}$ , computed without the answer. No orbital-rotation invariant substitutes: one-body fermionic magic collapses to  $4 \text{tr}[\gamma(1 - \gamma)]$ , constant along an orbit where the determinant support runs from  $O(K)$  to  $2^K$  at bond dimension 2, and on fibers where our bound separates. Within each of six  $(L, N)$  strata its rank correlation with that support is exactly  $-1$ , exact  $p = 3.3 \times 10^{-13}$ ; pooled, the same thirty points give  $\rho = -0.11$ , a Simpson reversal, not an absence of signal. We claim no quantum advantage.

## I. THE OPERATIONAL QUESTION

A sample-based quantum calculation returns a list of bitstrings. Diagonalizing the Hamiltonian inside the span of the configurations they name yields an energy, and—as this work and, now, several others show—a frequency-resolved response as well. What it does not yield is any statement about how wrong that response is. This paper is built around the question a user is left with once the shots are spent: *given the subspace those shots populated, how wrong is the spectral function built on it?*

The answer splits in two, and the split is a theorem rather than a metaphor. Let  $\phi$  be the probe vector,  $A(\omega)$  the spectral function broadened to resolution  $\eta$ ,  $\mathcal{S}$  a set of determinants and  $A_{\mathcal{S}}$  the Rayleigh–Ritz reconstruction on its span. The  $L_1$  error is bounded below by the Born weight the subspace failed to capture, and above by that same weight together with the Hamiltonian coupling that leaks across the boundary of the subspace, averaged over the resolution window (Theorem 1, Sec. III). The two terms of the upper bound are two budgets, and only the first is bought by drawing more bitstrings. That is measured rather than argued: on the subspace that *is* the Born support of the probe,  $\mathcal{S}^* = \text{supp}(\phi)$ —of size exactly  $(\frac{1}{2} + \frac{1}{L})$  of the sector, fixed by the probe and not by any ranking, so that the captured weight is unity by construction with no tie-breaking and no zero-amplitude padding—the first term vanishes identically while the

relative  $L_1$  error is still 0.43 at  $L = 6$  and 0.33 at  $L = 8$  at  $\eta = 0.18t$ , rising to 1.15 and 0.90 at  $\eta = 0.05t$ , against a hard ceiling of 2. (These are the periodic rings; the open chain on which Sec. IV measures the same witness gives 0.28 at  $L = 6$ , and the two lattices are never pooled into one range in this paper.) The second budget is a property of the boundary of the subspace in the basis in which the device measures, and the rest of this paper is an account of what can and cannot be said about it.

Four bodies of work bound what is new here, and we state them before stating our own claims. (i) *Concurrent and independent*. Patel, Rangi and Tam [107] construct Green’s functions from sampled Krylov subspaces for dynamical mean-field theory [80], at the same system size this work reaches, and cite it as “a closely related sample-based approach to dynamical spectral functions, constructed directly from bitstring-sampled subspaces.” Their construction is independent of ours; we make no priority claim against it, and none is needed. (ii) *Classical dynamical DMRG arrived first*. Benthien and Jeckelmann [117] computed the dynamical spin and charge structure factors and the one-particle spectral function of the half-filled one-dimensional *extended* Hubbard model in the Mott regime—three channels, on the  $V = 0$  line the plain Hubbard model, and *all three* are channels this paper reports: their spin factor, our  $S^{zz}(q, \omega)$ , at  $L = 100$ ; their charge factor, our  $S(q, \omega)$ , at  $L = 120$ ; and their  $A(q, \omega)$ , our  $A(k, \omega)$ , at  $L = 90$ —nineteen years ago, and the correction-vector literature treats the Hubbard chain at  $L = 48$  with  $\eta = 0.1t$ , and reaches  $L = 64$  in the Heisenberg chain and  $48 \times 2$  sites in the  $t$ – $J$  ladder [73]. The sentence in the abstract of versions 1–2,

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that the ground-state energy is where classical methods have caught up and dynamics is therefore the defensible target, is *inverted* in one dimension and is withdrawn: here, dynamics is where the classical methods arrived first. (iii) *Hardware*. Frequency-resolved responses of correlated lattice models have already been measured on devices far larger than the one used here—a retarded spin Green’s function of the one-dimensional Fermi–Hubbard model on an IBM Heron processor at up to 30 qubits [108]; an angle-resolved-photoemission-like  $A(k, \omega)$  on a 27-site chain at 54 qubits [109];  $S(q, \omega)$  of a 20-site Heisenberg chain benchmarked against a real copper-sulfate crystal [26];  $S(q, \omega)$  on 50 qubits against neutron scattering in  $\text{KCuF}_3$  [110]; and quench spectroscopy of  $L = 101$  spins with no ground-state preparation at all [111]. Our own device run is at  $L = 6$ , and the subspace it recovers spans its entire sector, so its agreement with exact diagonalization is exact by coverage rather than a test of the method (Sec. IX). (iv) *The invariant versus basis-dependent dichotomy*. Tarabunga *et al.* [59], six weeks before version 1 of this work, set out the same division on general grounds: occupation entropies are Gaussian invariants and lower-bound the non-Gaussian gate count, while participation entropies in the natural-orbital basis are basis dependent and upper-bound the classical simulation cost. Leone and Bittel [112] gave a Gaussian monotone of the same class seventeen days before version 1 of this work. The priority for the dichotomy is theirs; what this work adds is the constructive instance those papers do not contain, and the extension of that division from the determinant support to the leakage term above (Sec. VI).

What versions 1 and 2 of this preprint published in place of Theorem 1 was different. They carried a weight-only error bound, Theorem 1(iii), and invoked it as the certificate governing the reconstruction. It is false—by an analytic two-level counterexample, which forces its constant to the trivial value  $2\|\phi\|^2$ , and by counting, since from  $L = 8$  onward every operating point those versions published lies below the combinatorial floor at which its second term has any content at all. It fails wherever it was invoked: in 126 of 126 subspaces of the original sweep with captured weight above 0.99, and in 363 of 363 such subspaces of the pooled sweep behind Fig. 17. We retract it explicitly, together with the sentence that repeated it in prose in the discussion, and replace it by Theorem 1: proved from the resolvent identity alone, violated by none of the 606 subspaces of that sweep, and stated with the three hypotheses—common energy origin, normalization and support of the  $L_1$  norm, and exactness of the reference—that a user must check against the pipeline that produced  $A_S$  (Sec. IIID).

A certificate that is never evaluated is a claim and not a result, and the retracted one was never evaluated: neither of the two quantities it is written in is computed anywhere in the reproduction repository. We evaluate the replacement on the subspaces this work actually used, at  $L = 6, 8, 10, 12, 14$  (sector dimensions 300 to 10 306 296), and report the outcome negative end first. *At every pub-*

*lished fraction, at each of the five sizes evaluated, the leakage branch of the bound is worse than the trivial bound  $\|\phi\|^2(1 + w_S)$ —vacuous by factors of 1.66 to 8.61—and at  $L = 12$  that verdict is converged in Krylov depth, so it cannot be moved by computing harder. The reconstructions reported in this paper are therefore, at present, uncertified. Two measurements replace that sentence rather than soften it. The obstruction is structural: the coordinate support of the Krylov space  $\mathcal{K}(H, \phi)$  is 296 of 300 determinants at  $L = 6$  and all 3920 of 3920 at  $L = 8$ , so at  $L = 8$  the leakage is strictly positive for *every* proper coordinate subspace and no leakage-type certificate in this basis can be non-vacuous there at low fraction. That support is now measured at  $L = 8, 10, 12$  and 14—every size this work reports—and at each it is the whole sector at the declared threshold, by two estimators that agree wherever both ran. And the certificate is calibrated: across all sixteen  $(L, \eta)$  cells of the sweep its crossing of the trivial bound occurs at a true relative  $L_1$  error between  $1.2 \times 10^{-3}$  and  $7.3 \times 10^{-3}$ , median  $2.8 \times 10^{-3}$ , on a table whose true errors span five decades—computed from  $H, S, \phi$  and  $\eta$  alone, without the answer (Sec. V). The reconstruction itself is untouched: an independent implementation, with different code, normalization and reference, recovers the published acceptance fractions at the four sizes it covers, the largest discrepancy being 7%—at  $L = 12$ , against the coarse replication-grid value 0.18, and under 2% against the 0.17 the scan itself reports. What the measurement retires is the word *certified* applied to those fractions, not the fractions themselves.*

Section III leaves one question open, and the second half of this paper answers it: can the a posteriori certificate be replaced by an a priori predictor, computed from the state before any circuit runs? The natural candidates are the efficiently measurable monotones of fermionic non-Gaussianity, and on number-conserving states the one-body member of that family collapses to a single one-particle-density-matrix invariant,  $\mathcal{F}_1 = 4 \text{tr}[\gamma(1 - \gamma)]$  (Eq. (17)), equal to twice the effectively-unpaired-electron index for spin-restricted natural orbitals and not for the spin-polarized members of the suite. An invariant of the occupation spectrum cannot see a boundary. Along a geminal orbit every order of it is fixed while the determinant support runs from  $O(K)$  to  $2^K$  at bond dimension 2 (Theorem 3); and on fibers where every invariant *and* the captured weight are frozen to  $10^{-14}$ , the subspace-aware certificate still separates a non-vacuous value from the trivial bound (Proposition 1). The statement is one of non-determination and non-vacuity, not of non-existence: an invariant upper bound on the leakage does exist, and it is the trivial one. The same discipline applies to our own statistics, which were reported the wrong way up. Analysed within the six  $(L, N)$  strata it contains, the Hubbard interaction sweep gives a rank correlation of  $\mathcal{F}_1$  with the determinant support of exactly  $-1$  in all six: stratified Kendall  $\tau = -1.000$ , zero concordant pairs out of 60, exact one-sided permutation  $p = 3.3 \times 10^{-13}$ . That is not an absence of relation but a deterministic

monotone one with the opposite sign, whose sign is not stable from family to family—exactly what an orbital invariant compared against a basis-dependent count must do. Both pooled readings of versions 1–2 are withdrawn in Sec. VI E; one of the two was in our own favour.

*We claim no quantum advantage in this paper.* No result reported here would support one; the classical-simulability boundary is charted rather than contested (Sec. IX); and the one quantitative statement this work makes about its own cost—that the basis-dependent bond dimension empirically tracks the determinant support—argues, wherever that bond dimension is small, for a tensor network rather than for a sampler.

The paper is organized around that division. Section II fixes the primitive and states, channel by channel, which reconstructions carry a declared shot budget and which are exact Haydock references. Section III proves the bound and retracts its predecessor; Sec. IV isolates what the subspace structure does certify—moment exactness through order  $2K + 1$  and no further, with the proof the code actually realizes (Theorem 2) and a non-circular shot count replacing the published one (Proposition 2). Section V measures where the certificate is informative; Sec. VI closes the search for a cheaper a priori substitute; Sec. VII prices support, resolution and shots; Sec. VIII validates the primitive against the opposite finite-size behaviour of the charge and spin scales of the Hubbard chain (Table XI, Fig. 10); and Sec. IX reports the device runs, the scope of the molecular suite, and the region in which this work’s own thesis argues against a quantum advantage. The systems throughout are the half-filled one-dimensional Hubbard model at  $U/t = 8$ —rings and open chains at  $L = 4$  to 14, by exact diagonalization and matrix-free Lanczos—and a suite of nineteen molecular active spaces. Three different counts share the symbol  $\mathcal{S}$  in this literature and are kept apart here: the static ground-state support of Sec. VI, the dynamical support of Sec. VII, and the set of configurations the sampler actually returned, which is the  $\mathcal{S}$  of Theorem 1.

## II. ONE SAMPLING PRIMITIVE, FOUR CHANNELS: WHAT IS RECONSTRUCTED FROM SHOTS, AND WHAT IS NOT

This section states the measurement primitive, fixes the one definition the rest of the paper turns on, and then says — channel by channel, figure by figure — which objects here were reconstructed from a finite number of measurements and which are full-sector references computed classically. That line is routinely drawn nowhere in the text, and a single clause of an abstract is enough to draw it in the wrong place. It is drawn here, explicitly, before any result is quoted.

### A. Construction

For a Hamiltonian  $\hat{H}$  with  $N$ -electron ground state  $|0\rangle$  of energy  $E_0$ , the single-particle spectral function has an electron-addition and an electron-removal branch,

$$A_p(\omega) = \sum_n |\langle n^{N+1} | \hat{c}_p^\dagger | 0 \rangle|^2 \delta(\omega - (E_n^{N+1} - E_0)) + \sum_m |\langle m^{N-1} | \hat{c}_p | 0 \rangle|^2 \delta(\omega - (E_0 - E_m^{N-1})), \quad (1)$$

energies measured from  $E_0$ ; the chemical potential  $\mu = \frac{1}{2}(E_0^{N+1} - E_0^{N-1})$  places the Fermi level mid-gap for the particle-hole-symmetric Hubbard model at half filling. The subspace is built in three steps. (1) Prepare the response seeds  $|\phi^\pm\rangle = \hat{c}_p^\dagger |0\rangle, \hat{c}_p |0\rangle$  in the  $(N \pm 1)$  sectors. (2) Time-evolve each,  $|\phi^\pm(t_k)\rangle = e^{-i\hat{H}t_k} |\phi^\pm\rangle$ , and measure computational-basis bitstrings at each of the  $K+1$  snapshots; the union of the *distinct* bitstrings returned is the determinant subspace  $\mathcal{S}^\pm$ . (3) Form  $\hat{H}_{\mathcal{S}^\pm} = P_{\mathcal{S}^\pm} \hat{H} P_{\mathcal{S}^\pm}$ , diagonalize it as  $\sum_m \tilde{E}_m^\pm |\tilde{m}^\pm\rangle \langle \tilde{m}^\pm|$ , and evaluate the Lehmann sum inside  $\mathcal{S}^\pm$ . The retarded Green’s function is then assembled entirely on the classical processor,

$$G_{pq}(\omega) = \sum_m \frac{\langle 0 | \hat{c}_p | \tilde{m}^+ \rangle \langle \tilde{m}^+ | \hat{c}_q^\dagger | 0 \rangle}{\omega - (\tilde{E}_m^+ - E_0) + i\eta} + \sum_m \frac{\langle 0 | \hat{c}_q^\dagger | \tilde{m}^- \rangle \langle \tilde{m}^- | \hat{c}_p | 0 \rangle}{\omega - (E_0 - \tilde{E}_m^-) + i\eta}, \quad (2)$$

whose imaginary part is the Lorentzian-broadened  $A_{pq}(\omega)$  at resolution  $\eta$ . The seed selects the channel: charged seeds  $\hat{c}_q^\dagger |0\rangle$  in the  $(N \pm 1)$  sectors give  $A(\omega)$  and, by Fourier transform of  $G_{pq}$ ,  $A(k, \omega)$ ; number-conserving density and spin seeds  $\hat{n}_q |0\rangle, \hat{S}_q^z |0\rangle$  in the neutral  $N$  sector give  $S(q, \omega)$  and  $S^{zz}(q, \omega)$ . The processor measures *only* bitstrings. Figure 1 schematizes the primitive — what one circuit and one readout can address; Table I states separately what was actually run for each channel here.

### B. Relation to quantum subspace and Krylov methods

In exact arithmetic the span of the time-evolved seeds is the order- $K$  polynomial Krylov space [7, 27], and diagonalizing  $\hat{H}$  inside a subspace containing it is the route of quantum subspace expansion [28], filter diagonalization [30] and real-time quantum Krylov algorithms [29], whose error theory is now characterized [31]. What distinguishes the present method is that these states are used not to *estimate matrix elements* [32, 34–37, 101] — the

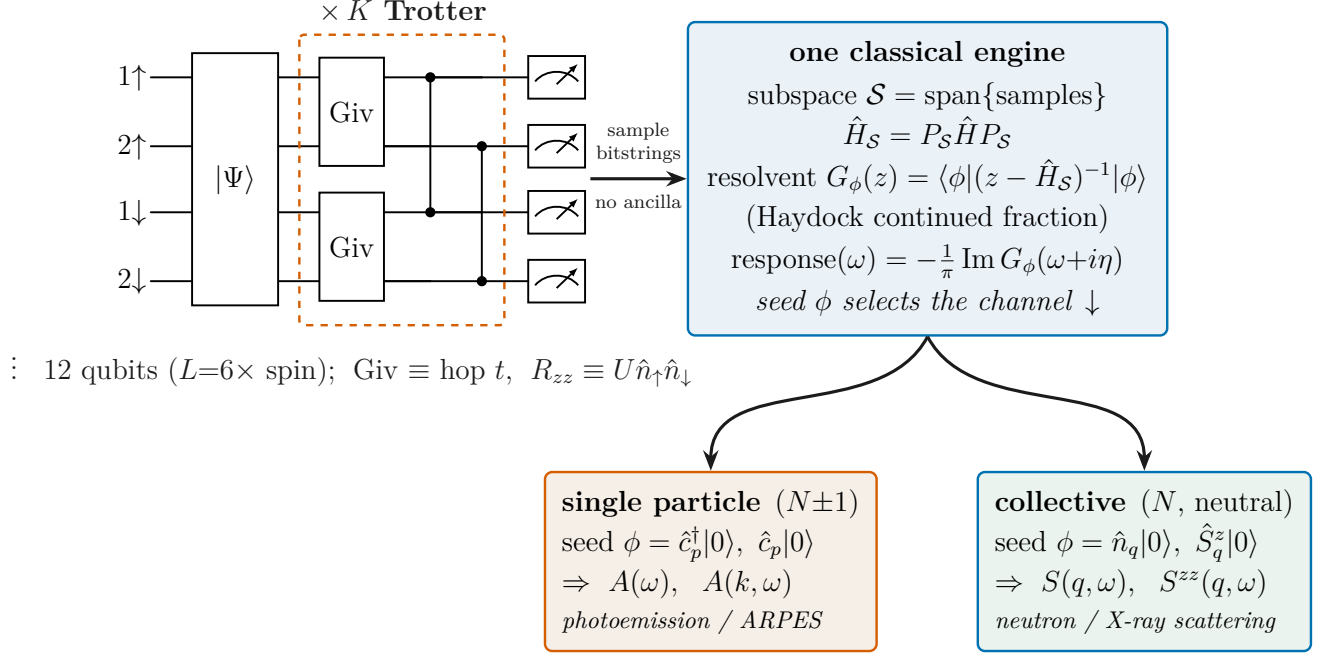
TABLE I. **Provenance of every dynamical object in this paper: the subspace it was built on, and the measurements that built it.** “Full sector” means no subspace selection of any kind — the continued fraction runs on the whole sector at the stated recursion depth  $n_\ell$ , so the row is a reference against which reconstructions are scored, not a reconstruction. “Observed bitstrings” means the subspace is the union of the distinct bitstrings actually returned by  $T$  draws split evenly over the  $K+1$  snapshots, which is the protocol of Sec. II; one *stream* is one independent pseudorandom realization of that draw. “Born ranking” means the subspace is the top fraction of the sector under the accumulated Born weight of the evolved state, which is the  $T \rightarrow \infty$  limit of the same protocol and carries no shot budget. Sector dimensions are exact; budgets are per channel and per stream.

| Channel and system                                                                        | Shown in    | Determinant subspace                                                                                                                                                            | Shots $T$                                                                                                                                    |
|-------------------------------------------------------------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| $A(\omega)$ ; Hubbard $L=6$ , $U/t=8$ , $\eta=0.15t$ ; dim = 300                          | Fig. 2      | observed bitstrings; $ \mathcal{S}  = 245$ of 300 in panel (a), 242.5 on average in panel (b)                                                                                   | $5.2 \times 10^4$ per stream ( $4 \times 10^3$ at each of 13 snapshots, $K=12$ ), 8 streams                                                  |
| $A(\omega)$ , orbital projected; 19 molecules at dissociation; dim = 2–300                | Table XIV   | observed bitstrings; the fraction $ \mathcal{S} /D$ is the last column of that table                                                                                            | $2.16 \times 10^5$ per molecule ( $8 \times 10^3$ at each of 27 snapshots, $K=26$ ), one stream                                              |
| $A(\omega)$ ; $N_2$ along dissociation, CAS(6,6)                                          | Fig. 13(a)  | full CAS sector                                                                                                                                                                 | none                                                                                                                                         |
| $A(\omega)$ ; hardware, <code>ibm_fez</code> , $L=6$ , $U/t=4$ , $\eta=0.15t$ ; dim = 300 | Fig. 14(a)  | recovered from device bitstrings; $ \mathcal{S}  = 300$ of 300, the entire sector, so the agreement with the exact curve is exact by coverage and is not a fidelity measurement | $3.5 \times 10^5$ : $5 \times 10^4$ on each of 7 circuits, $k=0$ (the $t=0$ reference) to $k=6$                                              |
| $A(k, \omega)$ ; $L=12$ , $\eta=0.18t$ ; dim = 731 808                                    | Fig. 4      | full sector, $n_\ell = 150$ per momentum                                                                                                                                        | none                                                                                                                                         |
| $A(k, \omega)$ ; $L=8$ , $\eta=0.18t$ ; dim = 3920                                        | Fig. 3(a,b) | Born ranking, 3332 of 3920 per channel, over $K=16$ snapshots                                                                                                                   | none: (a)–(b) are the $T \rightarrow \infty$ limit; at finite shots the same configuration takes $5.2 \times 10^6$ per channel (see text)    |
| $A(\omega)$ ; $L=8$ , $\eta=0.15t$ ; dim = 3920                                           | Fig. 3(c)   | observed bitstrings; sector fraction $0.835 \pm 0.0025$ at the largest budget                                                                                                   | up to $2.60 \times 10^6$ over 19 snapshots ( $K=18$ ), 8 streams                                                                             |
| $S(q, \omega)$ ; $L=12$ , $\eta=0.18t$ ; dim = 853 776                                    | Fig. 11     | full $N$ -particle sector, $n_\ell = 200$                                                                                                                                       | none                                                                                                                                         |
| $S^{zz}(q, \omega)$ ; $L=12$ , $\eta=0.10t$ ; dim = 853 776                               | Fig. 12     | full $N$ -particle sector, $n_\ell = 200$                                                                                                                                       | none                                                                                                                                         |
| Required sector fraction versus $L$ ; $L=4$ –14, $\eta=0.15t$                             | Fig. 9(b,c) | both selections, plotted together: Born ranking (open squares) and observed bitstrings (filled circles)                                                                         | measured: $2.4 \times 10^4$ , $2.6 \times 10^5$ , $3.1 \times 10^6$ , $1.9 \times 10^7$ at $L = 6, 8, 10, 12$ (Table IX, 4–6 seeds per size) |

step requiring Hadamard tests or quantum equation-of-motion measurements on the device — but as a *sampling distribution* selecting a determinant subspace [1–6, 9], extended since to embedded fragments and to auxiliary-field post-processing [10, 11], and here lifted to the  $(N \pm 1)$  manifold and to a frequency-resolved observable. That lineage stops at energies and, at most, discrete levels: the closest published results are a QSCI quasiparticle band structure returning discrete poles [12], a molecular spectrum built from a neutral-sector autocorrelation with classically projected long-time dynamics [38], and, from the measured-overlap side rather than the sampled one, a dynamical structure factor of a Kitaev spin liquid obtained by quantum subspace expansion [33], or a hybrid

quantum–classical framework for many-fermion response and structure [39]. Closest on lattice models, Nogaki *et al.* [13] apply symmetry-adapted SQD to ground-state *energies*, and Wray *et al.* [96] measure how SQD converges on a variable-length cuprate chain. The readout is that of SQD, the circuits are those of sample-based Krylov diagonalization [6, 8, 42], and all dynamical information is assembled classically through Eq. (2) by one continued-fraction resolvent [40, 41].





**FIG. 1. One sampling primitive, four dynamical responses.** A shallow, ancilla-free circuit—state preparation and a Trotter layer repeated  $K$  times (Givens rotations for the hopping  $t$ , native  $R_{zz}$  for the on-site  $U \hat{n}_\uparrow \hat{n}_\downarrow$ )—is measured in the computational basis; no Hadamard test or controlled unitary is run on the device. The sampled bitstrings define a sector-specific subspace  $\mathcal{S}$  and feed a single classical engine—a Haydock continued-fraction resolvent [41]  $G_{\phi}(z) = \langle \phi | (z - \hat{H}_{\mathcal{S}})^{-1} | \phi \rangle$ , whose single-particle case ( $\phi = \hat{c}_p^\dagger |0\rangle$ ) is the Green’s function of Eq. (2). The seed  $\phi$  then selects the channel: particle addition/removal seeds  $\hat{c}_p^\dagger |0\rangle, \hat{c}_p |0\rangle$  in the  $(N \pm 1)$  sectors give the single-particle spectral functions  $A(\omega)$  and  $A(k, \omega)$  (photoemission/ARPES), while number-conserving density and spin seeds  $\hat{n}_q |0\rangle, \hat{S}_q^z |0\rangle$  in the  $N$  sector give the dynamical structure factors  $S(q, \omega)$  and  $S^{zz}(q, \omega)$  (neutron [44] and resonant inelastic X-ray [45] scattering)—all from *one* sampling primitive, the seed selecting the sector and hence the subspace. The four channels are realized in Figs. 2 ( $A(\omega)$ ), 4 ( $A(k, \omega)$ ), 11 ( $S(q, \omega)$ ), and 12 ( $S^{zz}(q, \omega)$ ).

### C. What the subspace reconstruction is: Rayleigh–Ritz, not restriction

Step (3) is the definition the rest of this paper turns on, and it is stated before it is used. With  $P_{\mathcal{S}}$  the orthogonal projector onto the span of the determinants in  $\mathcal{S}$  and  $\phi$  the seed,

$$A_{\mathcal{S}}(\omega) = \frac{\eta}{\pi} \left\| (\omega + E_0 + i\eta - H_{\mathcal{S}})^{-1} P_{\mathcal{S}} \phi \right\|^2, \quad H_{\mathcal{S}} = P_{\mathcal{S}} \hat{H} P_{\mathcal{S}}, \quad (3)$$

the spectral function of the *Rayleigh–Ritz operator*  $P_{\mathcal{S}} \hat{H} P_{\mathcal{S}}$ , whose poles are the Ritz values  $\tilde{E}_m$  of Eq. (2) — not eigenvalues of  $\hat{H}$  — and not the restriction of  $A$  to a subset of its own eigenpairs. The reproduction repository computes exactly this, at the small sizes by forming and diagonalizing the submatrix  $\hat{H}[\mathcal{S}, \mathcal{S}]$  and at the large ones by a matrix-free restricted matrix–vector product inside the same continued fraction; the consequence is the mechanism of this paper: *spectral weight the subspace*

*retains is not kept in place, it is moved in  $\omega$ .* Truncation does not delete Lehmann poles, it displaces every surviving one, so the weight a subspace captures cannot by itself bound how far the reconstruction has moved — which is why no weight-only bound survives contact with Eq. (3) (Sec. IIID), and why Theorem 1 carries a second term indexed on the coupling  $P_{\mathcal{S}} \hat{H} Q_{\mathcal{S}}$  across the boundary of  $\mathcal{S}$ , with  $Q_{\mathcal{S}}$  the complementary projector — exactly what Eq. (3) discards. Two corollaries are used

later. A determinant acts on  $A_S$  through  $H_S$  even carrying no seed amplitude: at sector fraction 0.85, 30.2% of the selected determinants at  $L = 6$  and 36.9% at  $L = 8$  lie outside  $\text{supp}(\phi)$ , add exactly nothing to the captured weight, and still move the Ritz spectrum. And on the smallest subspace of *unit* captured weight,  $\mathcal{S}^* = \text{supp}(\phi)$  of size exactly  $(\frac{1}{2} + \frac{1}{L}) \dim$ , the relative  $L_1$  error of Eq. (3) is still 0.43 at  $L = 6$  and 0.33 at  $L = 8$  at  $\eta = 0.18t$  (Sec. IIID).

#### D. Provenance: what was reconstructed from shots, channel by channel

Table I gives, for every dynamical object in this paper, the subspace it was built on and the measurements that built it. Three readings the word *sampled* invites are closed off by it.

*The neutral channels are references, not reconstructions.*  $S(q, \omega)$  and  $S^{zz}(q, \omega)$  are computed here by a continued fraction over the *entire*  $N$ -particle sector of 853 776 determinants at  $L = 12$ , at recursion depth  $n_\ell = 200$ : no subspace is selected and no shot is drawn, and the generating scripts say so in their own headers. The primitive *targets* these two channels — that is what Fig. 1 asserts, and what the seed algebra supports — and targeting a channel is not reconstructing it from measurements; they were not, nor were the  $A(k, \omega)$  of Fig. 4 or the  $N_2$  series of Fig. 13. A reference at finite recursion depth is not the exact spectral function either, so its depth is quoted with it, and we quote no error that lies within a factor 20 of the drift of its own reference.

*What the Born ranking is, and what it is not.* The two objects it would be easiest to call sampled without a budget — Fig. 3(a,b) and the dashed curve of Fig. 9(b) — select  $\mathcal{S}$  by sorting the accumulated Born weight  $\sum_k |\langle d | e^{-i\hat{H}t_k} \phi \rangle|^2$  of the evolved state over the  $K+1$  snapshots and keeping the top fraction. That weight is the distribution the device draws from, not the ground-state amplitudes and not the exact Lehmann residues, so the ranking is *not* a clairvoyant oracle: the repository contains one of those, in `amortized_recovery.py`, and labels it as such there. It is the infinite-shot limit  $T \rightarrow \infty$  of the declared protocol, approached from above in error by any finite budget — at matched subspace size finite

sampling is worse by  $1.50 \pm 0.25$  at  $L = 6$  and  $1.96 \pm 0.12$  at  $L = 8$  (Table X) — and in that limit it selects nothing the device could never return: no determinant it ranks carries exactly zero accumulated weight at  $L = 6, 8, 10$  or  $12$ , and at  $L = 12$  no determinant of the whole sector does. What it is not is a finite-shot experiment, and it must not be called one. The replacement is measured and quoted with its budget: at  $L = 8$ ,  $T = 2.60 \times 10^6$  shots over  $K+1 = 19$  snapshots reach sector fraction  $0.835 \pm 0.0025$  at relative  $L_1$  error  $(4.81 \pm 0.27) \times 10^{-3}$  over eight independent shot streams (Fig. 3(c)), and the measured ladder from  $2.4 \times 10^4$  to  $1.9 \times 10^7$  shots for  $L = 6$  to  $L = 12$  is Fig. 9(c) and Table IX. The literal configuration of Fig. 3(a,b)— $\eta = 0.18t$ ,  $K = 16$ , all sixteen channels—has since been run at finite shots:  $5.2 \times 10^6$  per channel buy panel (b)’s accuracy, but on a *larger* subspace, so at the announced fraction the penalty is a measured 2.80 and not the 2.05 Table X would transfer.

*What the a posteriori checks establish.* The branch sum rules  $\int A_p^\pm d\omega$  are identities of the ground state, not measurements of a reconstruction. Integrated on the grid the figures use, at  $L = 6$  and  $\eta = 0.15t$ , the exact and reconstructed addition branches carry 0.48659 and 0.48647 — agreement to  $2.4 \times 10^{-4}$  relative, both falling 2.7% short of the analytic  $1 - \langle \hat{n}_p \rangle = 0.5$  because Lorentzian tails of width  $\eta$  leave the finite frequency window. That is a genuine consistency check on a zeroth moment, and not an error bound. The certificate is the two-sided bound of Theorem 1, and where it is non-vacuous is measured, not asserted, in Sec. V — an essential property given that the classical hardness of a specific instance is empirical rather than a theorem (Sec. IX). What is guaranteed in advance is narrower and survives intact: a subspace containing the order- $K$  Krylov space reproduces that branch’s exact spectral moments through order  $2K+1$ , and is realized with high probability by a shot budget polynomial in the target support and independent of the Hilbert-space dimension (Theorem 2 and Prop. 2, Sec. IV).

#### E. The two budgets

The two halves of this paper are the two terms of one inequality. With  $Q_S$  the projector complementary to  $P_S$  and  $w_S = \|P_S \phi\|^2 / \|\phi\|^2$ , the upper branch of Eq. (7) reads

$$\|A - A_S\|_1 \leq (\|\phi\| + \|P_S \phi\|) \left( \underbrace{\|Q_S \phi\|}_{\text{what shots buy}} + \underbrace{\Lambda_S(\eta)/\eta}_{\text{what they do not}} \right).$$

The captured-weight term  $\|Q_S \phi\| = \|\phi\| \sqrt{1 - w_S}$  is what shots buy: it vanishes as the sampler exhausts the Born support of the probe, and Table I prices that exhaustion

in measurements. The leakage term  $\Lambda_S(\eta)/\eta$  is what they do not buy. On the subspace that *is* that support —  $\mathcal{S}^* = \text{supp}(\phi)$ , of size exactly  $(\frac{1}{2} + \frac{1}{L}) \dim$ , fixed by the probe

and not by any ranking, so that  $w_S = 1$  exactly by construction, with no tie-breaking and no zero-amplitude padding — the weight term is identically zero and the relative error is still 0.43 at  $L = 6$  and 0.33 at  $L = 8$  at  $\eta = 0.18t$ , rising to 1.15 and 0.90 at  $\eta = 0.05t$ .  $\Lambda_S(\eta)$  is defined by the resolvent at resolution  $\eta$  and exists only inside the spectral half of this work: the obstruction theorem of Sec. VI takes it as its subject and would have no subject without Sec. III. Conversely, Sec. III leaves open precisely the question Sec. VI answers — whether the a posteriori certificate can be replaced by an a priori predictor computed from the state. Neither half can be removed without leaving the other incomplete in a way we can name.

### III. TWO BUDGETS: WHAT THE SHOTS BUY AND WHAT THEY DO NOT

The object a sample-based calculation delivers is a subspace, and the object a user wants is an error bar

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$$A(\omega) = \frac{\eta}{\pi} \|R(z)\phi\|^2, \quad \text{so that} \quad \int_{\mathbb{R}} A(\omega) d\omega = \|\phi\|^2 \quad (4)$$


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for every  $\eta > 0$ , the sum rule holding because the Poisson

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$$\phi_P = P\phi, \quad \phi_Q = Q\phi, \quad w_P = \frac{\|\phi_P\|^2}{\|\phi\|^2}, \quad H_P = PHP, \quad R_P(z) = (z - H_P)^{-1}|_{\text{ran } P}, \quad (5)$$


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with Ritz pairs  $H_P|\tilde{m}\rangle = \tilde{E}_m|\tilde{m}\rangle$  and  $c_m = \langle \tilde{m}|\phi\rangle$ . The reconstruction of Eq. (3) is  $A_P(\omega) = (\eta/\pi)\|R_P(z)\phi_P\|^2$ , and it obeys the same sum rule with  $\|\phi_P\|^2$  on the right. When  $P = P_S$  is diagonal in the determinant basis — the only case a device produces — we write  $S$  for the

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$$\Lambda_P(\eta)^2 := \frac{\eta}{\pi} \int_{\mathbb{R}} \|QH R_P(z)\phi_P\|^2 d\omega = \sum_{m,m'} c_m \bar{c}_{m'} \langle g_{m'}, g_m \rangle \frac{2i\eta}{(\tilde{E}_{m'} - \tilde{E}_m) + 2i\eta}, \quad (6)$$


---

the second equality being exact and proved below; the sum is real, the  $(m, m')$  and  $(m', m)$  terms being complex conjugates. Two features of Eq. (6) carry the practical content of this paper. It is assembled from objects the reconstruction has already produced — the Ritz vectors and their coefficients — plus one matrix-vector product  $H|\tilde{m}\rangle$  per Ritz vector, so evaluating it costs a small multi-

on the response built inside it. This section proves the bound that connects them. It is two-sided, it holds for *any* orthogonal projector and therefore for any set of configurations a sampler might return, and its upper branch separates into two terms with different economics: one falls as more bitstrings are drawn, the other does not move at all. Everything after it is an evaluation of that separation — Sec. V measures where the resulting certificate says something, Sec. VI shows that no invariant of the state can replace it, and Sec. VII prices both terms in shots.

#### A. The objects

Fix the  $(N \pm 1)$  sector, let  $H$  be the Hamiltonian restricted to it,  $E_0$  the ground-state energy of the  $N$ -electron sector, and let  $\phi$  be the probe —  $\hat{c}_p^\dagger|\Psi_0\rangle$  or  $\hat{c}_p|\Psi_0\rangle$  for the charged channels,  $\hat{n}_q|\Psi_0\rangle$  or  $\hat{S}_q^z|\Psi_0\rangle$  for the neutral ones. Write  $z = \omega + E_0 + i\eta$  and  $R(z) = (z - H)^{-1}$ . The  $\eta$ -broadened spectral function of Eq. (2) is

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kernel integrates to one on each Lehmann pole. Let  $P$  be an orthogonal projector on the sector,  $Q = I - P$ , and

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configuration set and  $w_S, \Lambda_S$  for the quantities below.

The magnitude the theorem is indexed on is the Hamiltonian coupling that crosses the boundary of the subspace, averaged over the resolution window rather than over the spectrum. With  $g_m = QH|\tilde{m}\rangle$ ,

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ple of the diagonalization that produced  $A_P$  and *does not require*  $A$ . And it vanishes exactly when  $\text{span}(S)$  contains the Krylov space  $\mathcal{K}(H, \phi)$ , which is a far more demanding condition than it looks: Sec. V E measures it.

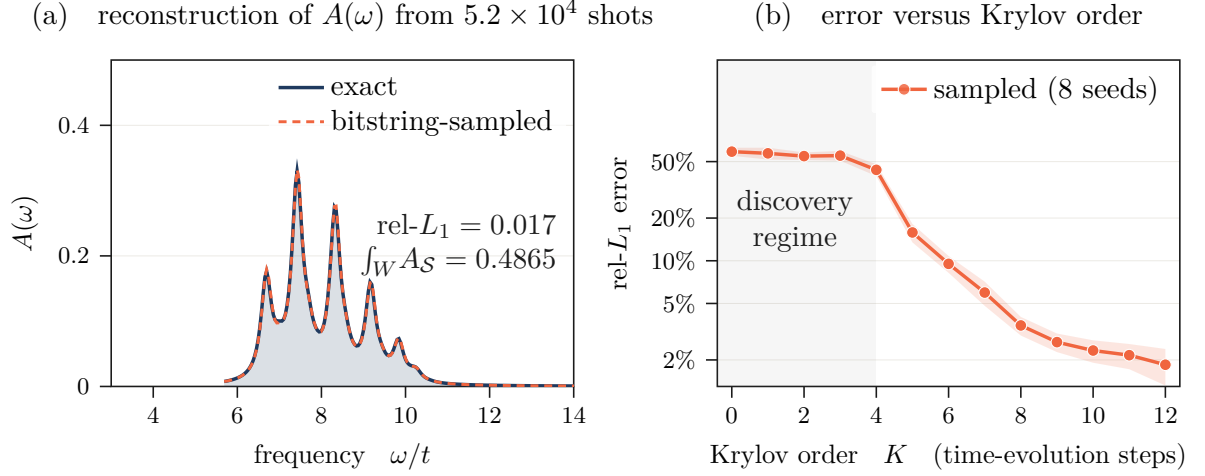


FIG. 2. **The method validated at  $L = 6$ , and its error against Krylov order.** (a) The bitstring-sampled single-particle spectral function reproduces the exact  $A(\omega)$  (Hubbard  $L = 6$ ,  $U/t = 8$ ; broadening  $\eta = 0.15t$ ) to relative- $L_1$  0.017—the exact (filled) and reconstructed (dashed) curves are indistinguishable. The analytic addition-branch sum rule  $\int A^+ d\omega = 1 - \langle n_p \rangle = 0.5$  is an identity of the ground state and not a check on the reconstruction; integrated over the plotted window the exact and reconstructed curves give 0.486588 and 0.486471, agreeing with each other to  $2.4 \times 10^{-4}$  relative and both falling 2.7% short of 0.5 because Lorentzian tails leave the window (Sec. VIII). (b) Convergence of the relative- $L_1$  error with the Krylov order  $K$  (number of time-evolution steps sampled), seed-averaged over eight independent runs (band = standard deviation),  $\eta = 0.15t$ , so the budget is  $T = 4000(K+1)$  and reaches  $5.2 \times 10^4$  at  $K = 12$ . The error falls by a factor 32 across the thirteen points, from 0.59 at  $K = 0$  to 0.0185 at  $K = 12$ , with one non-monotone step at  $K = 3$  inside the discovery phase, while the mean subspace returned grows from 75 to 242 determinants of the 300-dimensional sector. No geometric rate is fitted to the window  $4 \leq K \leq 9$  here (Sec. IV B): one is not implied by moment exactness, and inside that window the decay in  $K$  is confounded with the growth of  $|S|$ .

### B. The bound

**Theorem 1** (a posteriori leakage certificate). *Let  $H = H^\dagger$  act on the  $(N \pm 1)$  sector, let  $\eta > 0$ , let  $P$  be any*

*orthogonal projector on that sector, and let  $A$ ,  $A_P$ ,  $w_P$  and  $\Lambda_P(\eta)$  be as in Eqs. (4)–(6). Then*

$$\|\phi\|^2(1 - w_P) \leq \|A - A_P\|_1 \leq \min\left\{\|\phi\|^2(1 + w_P), (\|\phi\| + \|\phi_P\|)\left(\underbrace{\|\phi_Q\|}_{\text{what shots buy}} + \underbrace{\Lambda_P(\eta)/\eta}_{\text{what they do not}}\right)\right\}, \quad (7)$$

equivalently, dividing by  $\|\phi\|^2$  and writing  $\hat{\Lambda}_P = \Lambda_P/\|\phi\|$ ,

$$1 - w_P \leq \frac{\|A - A_P\|_1}{\|\phi\|^2} \leq \min\left\{1 + w_P, (1 + \sqrt{w_P})(\sqrt{1 - w_P} + \hat{\Lambda}_P(\eta)/\eta)\right\}.$$

Three hypotheses carry it. They are properties of the *pipeline* that produced  $A_P$  rather than of the inequality, each is load-bearing, and each is checked against this paper’s own pipeline in App. C.

**(H1) Common origin.:**  $A$  and  $A_P$  are referred to the same  $E_0$ . A shift common to both cancels exactly (verified to 1.000000 for shifts of  $10^{-3}$ ,  $10^{-2}$ ,  $10^{-1}t$ ). If instead  $A_P$  is referred to  $E_0 + \delta$ , then  $\|A - A_P^{(E_0 + \delta)}\|_1 \leq \|A - A_P\|_1 + \|\phi_P\|^2(4/\pi) \arctan(|\delta|/2\eta)$ , an additive term veri-

fied to  $2 \times 10^{-5}$  in all 48 cases tested and *which the certificate does not control*. A ground state obtained by Rayleigh–Ritz on 60% of the  $N$ -electron sector at  $L = 8$  gives  $\delta = 6.7 \times 10^{-4}t$ , which adds 0.0024 to a bound of 1.2036 (+0.20%) while multiplying the *measured* error by 16. No cheap a posteriori control of  $\delta$  is available: residual (Weinstein) and Temple bounds [127, 128] computed on ground-state subspaces of 20–80% of the  $N$  sector at  $L = 6$  give arctan terms of 1.76, 1.64, 1.40 and 0.92, all vacuous against the ceiling of 2.



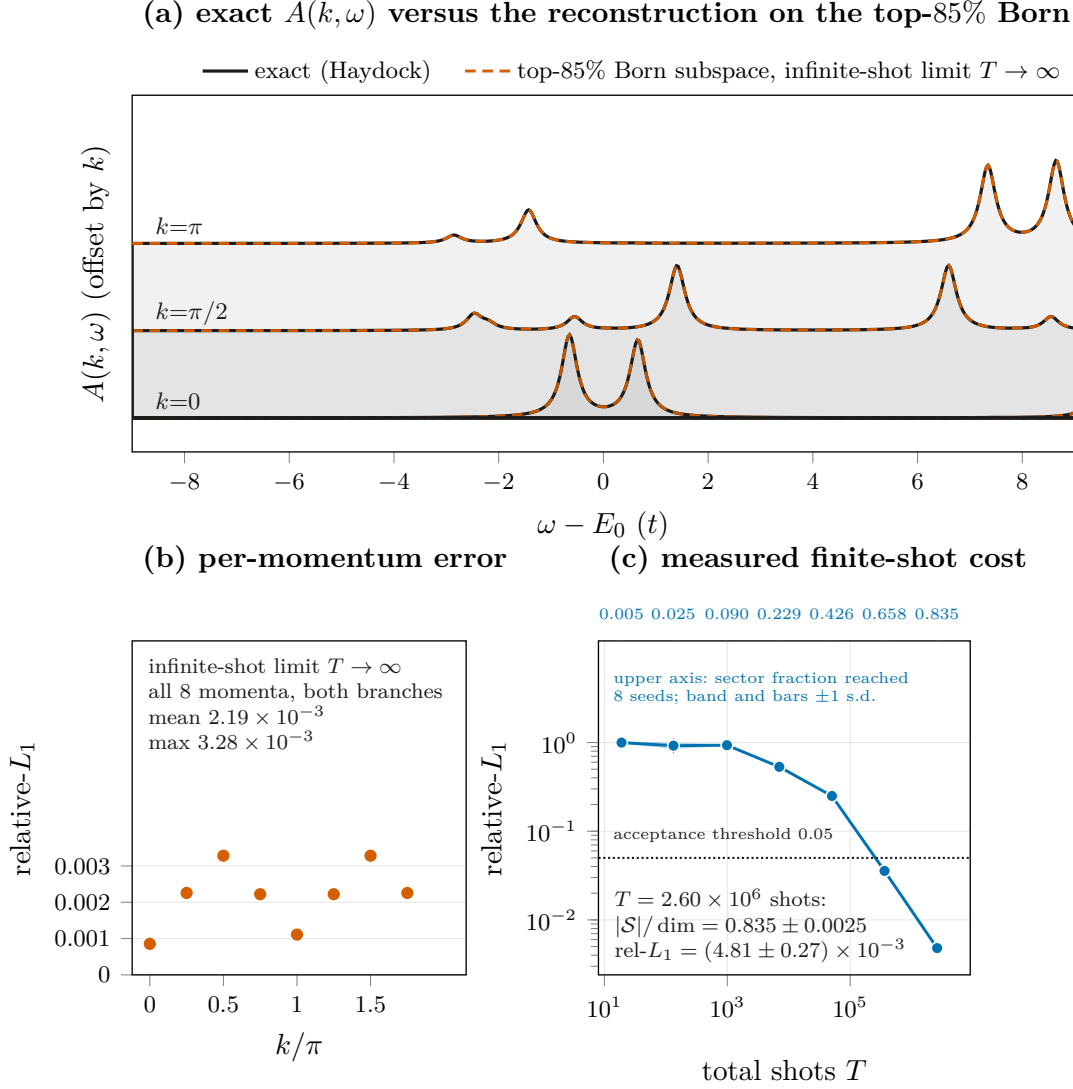


FIG. 3. **The momentum-resolved spectral function on a Born-ranked subspace, and what that subspace costs in shots.** (a) Exact Haydock  $A(k, \omega)$  (solid, shaded) and the reconstruction obtained by diagonalising the top-85% of the  $(N \pm 1)$  sector under the time-averaged Born weight of the evolved state (dashed), at three momenta ( $k = 0, \pi/2, \pi$ ; Hubbard chain,  $L = 8$ ,  $U/t = 8$ ,  $\eta = 0.18t$ ,  $K = 16$  time slices,  $\delta t = 0.5$ ; curves offset vertically by  $k$ ). The subspace is 3332 of the 3920 determinants of each channel. The ranking in `src/sampled_akw.py` is taken on the *exact* Born distribution, so panels (a) and (b) are the **infinite-shot limit**  $T \rightarrow \infty$  of the protocol, not a finite-shot experiment; no shot budget is attached to them. (b) Relative- $L_1$  error of that reconstruction at all 8 physical momenta, addition and removal branch summed: mean  $2.19 \times 10^{-3}$ , maximum  $3.28 \times 10^{-3}$ . (c) What a finite shot budget buys, measured. Same chain and the same sector dimension ( $L = 8$ ,  $\dim(N+1) = 3920$ ) but a separate run: local seed  $c_{0\uparrow}^\dagger |\Psi_0\rangle$ ,  $\eta = 0.15t$ ,  $K = 18$  time slices,  $T$  shots split evenly over the 19 snapshots, subspace = the support observed in those  $T$  shots, 8 independent seeds. Markers are the seed mean; the shaded band and the error bars are  $\pm 1$  s.d. over the 8 seeds (at most 8.1% of the mean at any budget, hence thinner than the line). Upper axis: the sector fraction  $|S|/\dim$  reached at that budget; the dotted line is the acceptance threshold  $\text{rel-}L_1 = 0.05$  used by the resource scan. At the largest budget measured,  $T = 2.60 \times 10^6$  shots, the observed subspace covers  $0.835 \pm 0.0025$  of the sector and the reconstruction reaches  $\text{rel-}L_1 = (4.81 \pm 0.27) \times 10^{-3}$ . **The shot budget of (a)–(b), measured.** Panel (c) is at another resolution and seed, so the configuration of (a)–(b) was run at finite shots separately: same  $\eta = 0.18t$  and  $K = 16$ , all 8 momenta, both branches, the subspace being the determinants observed in  $T$  multinomial draws per channel, 4 seeds, 32 (channel, seed) pairs. At  $T = 5.2 \times 10^6$  shots per channel— $8.3 \times 10^7$  over the 16 channels—the reconstruction reaches  $\text{rel-}L_1 = (2.11 \pm 0.26) \times 10^{-3}$ , maximum  $2.57 \times 10^{-3}$ : panel (b)’s accuracy. It reaches it on a *larger* subspace, covering 0.882 of the addition sector against the 0.850 of (a)–(b), so the budget buys that accuracy together with three extra points of fraction, and not the accuracy at the fraction this figure announces. At half the budget,  $T = 2.6 \times 10^6$ , the observed subspace covers 0.839—just under the 0.850 of (a)–(b)—and the error is  $(6.12 \pm 0.50) \times 10^{-3}$ , a factor 2.80 above (b). That factor, and not the 2.05 transferred from Table X ( $L = 8$ ,  $\eta = 0.15t$ ,  $|S| = 2180$ ), is the finite-shot penalty at this configuration; the transferred value understated it by 37%.

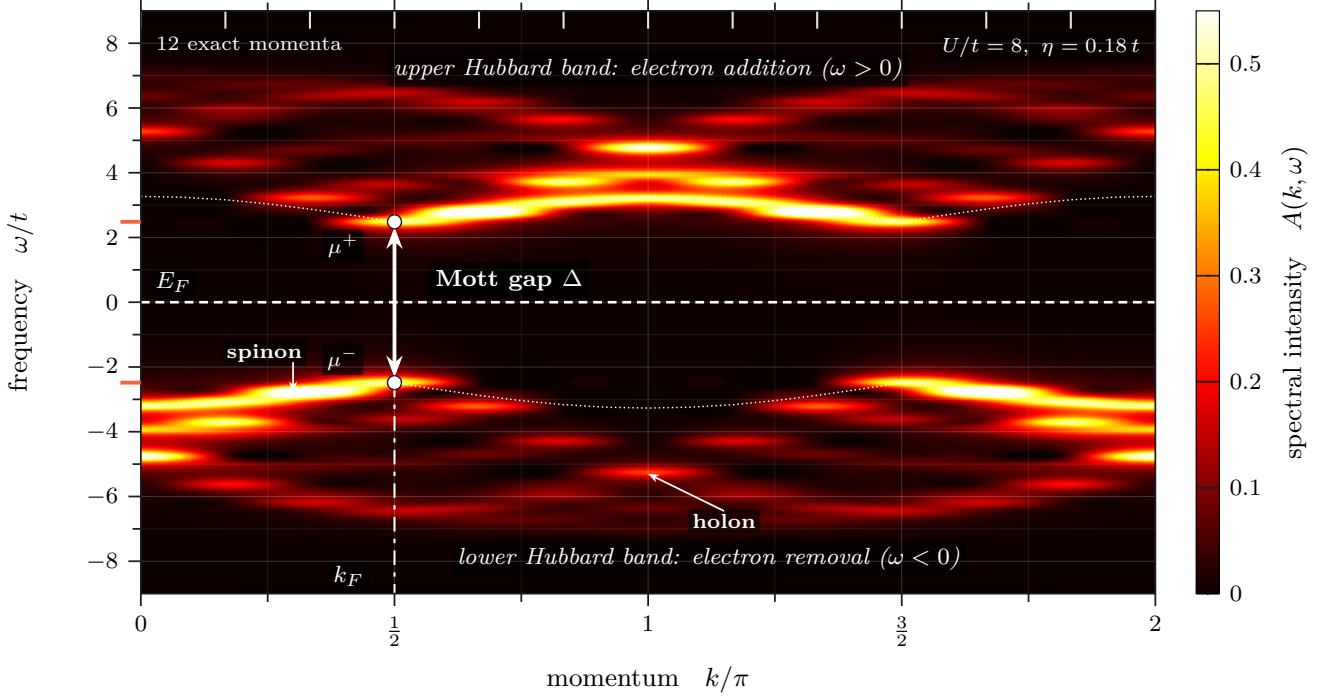


FIG. 4. **Momentum-resolved spectral function: the dynamical target.** The single-particle spectral function  $A(k, \omega)$  of the half-filled Hubbard chain ( $U/t = 8$ ,  $L = 12$ , Lorentzian broadening  $\eta = 0.18t$ ), computed from the resolvent  $\langle \phi | (z - \hat{H})^{-1} | \phi \rangle$  by a Haydock continued fraction [41] of depth  $n_\ell = 150$  over the full  $(N \pm 1)$  sectors—a finite-depth continued fraction, that is a 150-node Gauss quadrature of the spectral measure [40] and not the Lehmann sum, so the word *exact* is not used of it here (Sec. VIII)—the frequency- and momentum-resolved observable that sample-based diagonalization targets but its energy-only output cannot itself deliver. The two dispersing Hubbard bands—electron removal (lower,  $\omega < 0$ ; photoemission) and electron addition (upper,  $\omega > 0$ ; inverse photoemission)—are separated by the correlation-driven *Mott gap*  $\Delta = \mu^+ - \mu^- = 4.97t$  between the highest occupied state ( $\mu^-$ , lower edge) and the lowest unoccupied state ( $\mu^+$ , upper edge), obtained from the exact  $(N \pm 1)$  ground-state energies (ticks at  $\pm 2.484t$ ); the broadened spectral peaks of the plotted  $A(k_F, \omega)$  sit slightly wider, at  $\pm 2.49t$  (a  $4.99t$  splitting), the  $\sim 0.02t$  excess being the  $\eta = 0.18t$  Lorentzian broadening; the thermodynamic Lieb–Wu value [104] is  $4.68t$ , the  $L = 12$  excess being finite-size. The gap is *direct* at the Fermi momentum  $k_F = \pi/2$ . Particle-hole symmetry places the Fermi level at mid-gap ( $\omega = 0$ ), verified to double-precision round-off, and each momentum obeys the analytic spectral sum rule  $\int A(k, \omega) d\omega = 1$  (the plotted broadened grid recovers it to  $\sim 2\%$  within the  $\pm 9t$  window). The lower Hubbard band is not a single quasiparticle branch but *spin-charge separated*: a narrow spinon branch (bandwidth  $\pi J/2 = 0.79t$  [105],  $J = 4t^2/U$ ; dashed, validated against the exact peak dispersion to  $< 0.1t$ ) rides a broad holon (charge,  $\sim 4t$ ) band—the hallmark fractionalization of the 1D Mott insulator [104]. At  $L = 12$  that structure is three to seven poles per momentum (Table XII); the words appropriate to the thermodynamic limit are not used for it. The momentum axis is a periodic cubic-spline interpolation of the 12 physical momenta (ticks, top). Faithfully *sampling* this full momentum-resolved observable—as opposed to the ground-state energy—requires a large fraction of the  $(N+1)$  sector (Fig. 9), the cost quantified in Sec. VII.

**(H2) Normalization and support of the  $L_1$  norm.:**

The bound is on  $\int_{\mathbb{R}} |A - A_P| d\omega$  normalized by  $\|\phi\|^2 = \int_{\mathbb{R}} A d\omega$ . A relative error normalized by  $\int_W A$  over a finite window and evaluated by quadrature — the convention of this work’s own resource scan — is a different quantity. Measured here,  $\|\phi\|^2 / \int_W A = 1.0100, 1.0309, 1.0373$  at  $\eta = 0.05, 0.15, 0.18t$  ( $L = 6$ ) and  $1.0110, 1.0278, 1.0335$  ( $L = 8$ ), with a numerator ratio of  $1.0040$ – $1.0115$  across the 204 configurations of Sec. V. The two conventions are never interchanged silently in this paper.

**(H3) Exact reference.:**  $A$  is the exact spectral function of the full sector. If the reference is itself a Lanczos reconstruction of finite depth  $n_\ell$ , the theorem does *not* bound  $\|A^{(n_\ell)} - A_P^{(n_\ell)}\|_1$ . Every measured error in this work is therefore labelled with the depth of its reference and with that reference’s own drift, and we quote no error lying within a

factor 20 of that drift.

*Proof.* Write  $u = R(z)\phi$  and  $u_P = R_P(z)\phi_P$ , extended by zero outside  $\text{ran } P$ .

(1) *The split.* Since  $u_P \in \text{ran } P$  and  $(z - H_P)u_P = \phi_P$  there,  $(z - H)u_P = zu_P - PHu_P - QHu_P = \phi_P - QHu_P$ ; subtracting this from  $(z - H)u = \phi$  gives

$$u - u_P = R(z)[\phi_Q + QH R_P(z)\phi_P]. \quad (8)$$

This is the Feshbach–Löwdin partition of the resolvent [122, 123], written as an exact identity rather than as an expansion: the first source term is the part of the probe the subspace never held, the second is the part of the Hamiltonian that carries amplitude out of it. Only  $P^2 = P$ ,  $P^\dagger = P$ ,  $H = H^\dagger$  and  $\eta > 0$  are used; no property of the determinant basis, of the ranking, or of the sampler enters anywhere below.

(2) *Pointwise.*  $A - A_P = (\eta/\pi)(\|u\| + \|u_P\|)(\|u\| - \|u_P\|)$  and  $\|\|u\| - \|u_P\|\| \leq \|u - u_P\|$ , so  $|A - A_P| \leq (\eta/\pi)(\|u\| + \|u_P\|)\|u - u_P\|$  at every  $\omega$ .

(3) *Integration.* By Cauchy–Schwarz in  $\omega$ ,

$$\|A - A_P\|_1 \leq \frac{\eta}{\pi} \left[ \int (\|u\| + \|u_P\|)^2 d\omega \right]^{1/2} \left[ \int \|u - u_P\|^2 d\omega \right]^{1/2}.$$

The first bracket is at most  $(\pi/\eta)^{1/2}(\|\phi\| + \|\phi_P\|)$ , by Minkowski in  $L^2(d\omega)$  and the two sum rules. For the second, apply Minkowski to Eq. (8): its first term obeys the sum rule with seed  $\phi_Q$ , contributing exactly  $(\pi/\eta)^{1/2}\|\phi_Q\|$ ; its second obeys  $\|R(z)v\| \leq \eta^{-1}\|v\|$ , since  $H$  is self-adjoint and  $\text{Im } z = \eta$ , and the definition (6) then contributes at most  $(\pi/\eta)^{1/2}\Lambda_P(\eta)/\eta$ . Multiplying the three factors gives the leakage branch of (7).

(4) *The other two branches.*  $A$  and  $A_P$  are nonnegative with known integrals, so  $\|A - A_P\|_1 \leq \int A + \int A_P = \|\phi\|^2(1 + w_P)$  under no hypothesis at all; and  $|\int A - \int A_P| \leq \|A - A_P\|_1$  gives the lower bound  $\|\phi\|^2(1 - w_P)$ .

(5) *The closed form.*  $R_P(z)\phi_P = \sum_m c_m(z - \tilde{E}_m)^{-1}|\tilde{m}\rangle$ , so

$$\|QHR_P(z)\phi_P\|^2 = \sum_{m,m'} \frac{c_m \bar{c}_{m'} \langle g_{m'}, g_m \rangle}{(z - \tilde{E}_m)(\bar{z} - \tilde{E}_{m'})}.$$

With  $x = \omega + E_0$  the remaining integral is  $\int dx [(x - \tilde{E}_m + i\eta)(x - \tilde{E}_{m'} - i\eta)]^{-1}$ , which has one pole in each half plane; closing above picks up  $2\pi i/[(\tilde{E}_{m'} - \tilde{E}_m) + 2i\eta]$ , and multiplying by  $\eta/\pi$  gives Eq. (6).  $\square$

**C. What the two branches are, and what they are not**

*The separation is the economics.* The first summand of the leakage branch,  $\|\phi_Q\| = \|\phi\|\sqrt{1 - w_P}$ , is the Born

weight of the probe the subspace failed to hold: it is what more shots buy, and Table I prices that purchase channel by channel. The second,  $\Lambda_P(\eta)/\eta$ , reads the off-diagonal block  $PHQ$  and is untouched by the sampler — drawing further bitstrings enlarges  $S$ , and enlarging  $S$  moves the boundary rather than removing it. Section III D turns this from a reading into a measurement: on  $S^* = \text{supp}(\phi)$ , where the first summand vanishes identically and by construction, the relative  $L_1$  error is still 0.43 at  $L = 6$  and 0.33 at  $L = 8$  at  $\eta = 0.18t$  on the periodic rings, against a ceiling of 2.

*A certificate, not an error estimate.* The gap between the bound and the measured error is large, and it is not the accident of a constant. At *fixed*  $P$  the looseness is stable — within  $\pm 17\%$  ( $L = 6$ ) and  $\pm 25\%$  ( $L = 8$ ) over a decade in  $\eta$  — but across subspaces it varies by a factor 51, and at the threshold where the leakage branch first improves on the trivial one it is  $3 \times 10^2$  to  $2.4 \times 10^3$  (Sec. V D). Calling Eq. (7) an error estimate would be refutable in one line. What it is, is an exclusion; Sec. V reports what it excludes, and at what fraction of the sector.

*Only one of the two terms is a posteriori.*  $\Lambda_P(\eta)$  is computable from  $H$ ,  $\text{ran } P$ ,  $\phi$  and the Ritz vectors alone. The weight term is not:  $w_P$  needs the exact decomposition of the probe norm, and estimating it against an already-truncated  $\Psi_0$  is biased in the optimistic direction. The complete bound is therefore *not* evaluable by a user who

does not already have the answer, and we never claim that it is; Sec. VD gives the conservative substitution that makes the decision itself a posteriori in the regime where it is used.

*What object is certified.* If the resolvent inside  $\text{ran } P$  is itself evaluated by a Krylov recursion of depth  $n_\ell$ , then  $\Lambda$  computed on that Krylov space bounds the error of the depth- $n_\ell$  reconstruction *and of nothing else*. The convenient inequality  $\Lambda_K \geq \Lambda_S$  that would transfer the statement to the dense  $A_S$  is false: 1169 of 6000 random triples violate it, at a minimum ratio of 0.5258. Every table of Sec. V carries the depth of the object it certifies for that reason. Full reorthogonalization is mandatory throughout: without it  $P_K$  is not a projector — the orthogonality defect reaches 0.238 at  $L = 8$ , where the captured weight then reads 1.4868, an impossible value — and  $\Lambda$  is inflated thirtyfold.

*Relation to the classical error theory, and the scope of the novelty claim.* Nothing in the proof is new as analysis, and we say so before a referee does. The split (8) is Löwdin partitioning [122], the same algebra as the Feshbach projection of scattering theory [123]; and sixty years of Rayleigh–Ritz and Krylov error analysis for resolvents and matrix functions [31, 40, 124] contains every rung of steps (2) and (3), along with far sharper residual, Temple and Kato–Temple estimates for individual eigenpairs [127–129], and the loss-of-orthogonality analysis of Paige [126] that makes step (5) conditional on reorthogonalization, and the polynomial-restart convergence theory of Beattie, Embree and Sorensen [125].

Nor is “a posteriori” ours to claim, and we name the prior art rather than wait to be shown it. Stopping criteria for rational matrix functions of Hermitian matrices, computed from the residual alone, are Frommer and Simoncini’s [130]; Jia and Lv’s title is this section’s claim word for word—*a posteriori error estimates of Krylov subspace approximations to matrix functions* [131]—and Frommer and Schweitzer carry that line to Stieltjes functions [132]; and five months before this version, on *quantum* Krylov, Oliveira, Ziemis and Glaser introduce “two new metrics, the imaginary and unitary filters, that successfully assess the reliability of the obtained solutions without any knowledge of the true eigenspectrum” [133]. That is the framing of this section, established earlier and by others, and the priority is theirs.

What we did not find in that literature is the conjunction of three things, and it is the whole of the novelty claimed here. First, a closed form for  $\Lambda_P(\eta)$  on a subspace of *coordinates* carrying no Krylov structure, no invariance and no spectral-gap assumption—so that the estimate attaches to the set of determinants a sampler returned, which is not the span of the iterates of any recursion, and Eq. (8) is evaluated on that set directly. Second, that the bound is two-sided on the same object: the lower branch fixes what more shots can buy and the upper branch what they cannot, and Sec. V uses the distance between them rather than either alone. Third, a *measured* calibration of where that bound crosses the trivial one, over sixteen

$(L, \eta)$  cells whose true errors span five decades (Sec. VD), in place of an asymptotic statement about the crossing. Whether such a bound is *useful* at the fractions sampling reaches is a separate question; it is not settled by the proof, and Sec. V answers it — in the negative — by measurement.

#### D. The captured weight is not the controlling magnitude

The certificate one would want for a sampled reconstruction indexes the error on the captured weight alone,

$$\|A - A_S\|_1 \leq 2\|\phi\|^2(1 - w_S) + C\rho^{-K_S}, \quad (9)$$

with  $w_S = \|P_S\phi\|^2/\|\phi\|^2$  and  $K_S$  the largest Krylov order contained in  $\text{span}(S)$  (we set  $K_S = -1$  when  $\phi \notin \text{span}(S)$ ). It is false, for every constant  $C$  and every  $\rho$ , and it fails hardest in exactly the regime a sampling experiment aims for. (An inequality of this form was stated as Theorem 1(iii) of arXiv:2608.16436v1–v2, of which the present text is version 3; Theorem 1 replaces it and keeps the one inequality of that argument that was correct, with the direction it always had.)

*A two-level counterexample forces the trivial constant.* Take  $H = \begin{pmatrix} 0 & g \\ g & 0 \end{pmatrix}$ ,  $\phi = e_0$ ,  $S = \{0\}$ ,  $E_0 = 0$ . Then  $P_S\phi = \phi$ , so  $w_S = 1$  exactly and the weight term vanishes identically; and  $\mathcal{K}_0 = \text{span}\{\phi\} \subseteq \text{span}(S)$  while  $\mathcal{K}_1 \not\subseteq \text{span}(S)$ , so  $K_S = 0$  exactly and the second term is the bare constant  $C$ . The exact  $A$  is a pair of Lorentzians of weight  $\frac{1}{2}$  at  $\pm g$ ; the reconstruction, built on  $H_S = P_S H P_S = 0$ , is one Lorentzian of weight 1 at the origin, so  $\|A - A_S\|_1 \rightarrow 2\|\phi\|^2$  as  $g/\eta \rightarrow \infty$ . Since  $g$  is free, (9) requires  $C \geq 2\|\phi\|^2$ : false if  $C$  is a constant, vacuous if it may depend on the instance. With  $w_S < 1$  strictly the failure is unbounded: in a three-level family the violation diverges as  $\delta^{-2}$  in the displacement  $\delta$  of the discarded component, measured to  $7.6 \times 10^7$  (App. A). By contrast  $\|A - A_S\|_1 \leq \|\phi\|^2(1 + w_S)$  holds for every  $S$  under no hypothesis at all,  $A$  and  $A_S$  being nonnegative with known integrals: that trivial bound is the only surviving member of the family (9), and Theorem 1 keeps it explicitly, as the first branch of its minimum.

*The violation is systematic, and worst where this work operated.* The pooled sweep behind Fig. 17 contributes 621 subspaces from four Hamiltonian families and eight construction rules, written by two independent implementations; 15 reproduce their target to double-precision round-off and define no ratio. On the 606 that remain the weight-only bound fails on 468, and restricted to the regime this work operates in,  $w_S \geq 0.99$ , it fails on 363 of 363, the mildest by a factor 2.10 (App. A). The compliant cases are subspaces so poor that the relative error is already of order unity, where the bound asserts nothing. In the configuration of Fig. 3 itself —  $L = 8$ ,  $\eta = 0.18t$ , the 3920-dimensional  $(N+1)$  sector, momentum seeds  $\hat{c}_{k\uparrow}^\dagger|\Psi_0\rangle$ , both branches, eight momenta, subspaces grown



in order of accumulated Born weight — the median violation factor at sector fractions 0.50, 0.70, 0.85, 0.95 is 18.5, 23.6, 26.5, 29.9, the 64 rows spanning 15.1 to 57.3 overall and 15.1 to 33.7 at the published fraction 0.85. Those medians *rise* monotonically, by a factor 1.6, while the measured error over the same range falls by three orders of magnitude, from the  $10^{-1}$  level at the lowest fraction to  $1.0 \times 10^{-4}$  at 0.95: the weight-only bound fails worst exactly where the reconstruction is best, which is the opposite of what an error bound is for. That is a defect of the functional form and not a pathology at an edge of parameter space, and it is therefore measurable: Fig. 17 plots it against sector fraction and against  $\eta$ , beside the replacement bound on the same subspaces.

Three properties of that sweep are declared with it, because each of them limits how it may be read. Its ranking uses  $K = 16$  slices, one protocol step short of the  $K = 18$  of the resource scan of Sec. VII. Its three largest fractions lie above the support floor  $\frac{1}{2} + \frac{1}{L}$  derived below, so the ranking there is selecting among determinants of *zero* seed amplitude, where the order is arbitrary: ten different tie-breakings of one such subspace move the measured error by  $\pm 20\%$ , which is why the violation factors are quoted to three figures at most and why the rise across fractions is read as a direction and not as a rate. (That  $\pm 20\%$  is the spread of the *measured error* over tie-breakings of a zero-amplitude cut. It is a different quantity from the  $6.6 \times 10^{-4}$  by which ties move the *bound* in the sweep of Sec. V, where the cut falls below the support floor and only two determinants are tied.) And the error at fraction 0.85 has two independent determinations that differ by a factor 2.3, so only its order of magnitude is quoted. None of this touches the conclusion: the two-level counterexample is analytic, and the counting argument of the next paragraph but one uses no ranking, no protocol and no numerics at all. The sweep is corroboration of a conclusion that does not depend on it.

*Why it fails: the retained mass is displaced, not discarded.* The reconstruction is Rayleigh–Ritz on  $H_S = P_S H P_S$  (Sec. II), not the restriction of  $A$  to a subset of its own eigenpairs: the Ritz energies  $\tilde{E}_m$  are not eigenvalues of  $H$ , so weight the subspace *keeps* is nonetheless moved in  $\omega$ . The obvious sketch elides exactly this. Having correctly noted  $\int A = \|\phi\|^2$  and  $\int A_S = \|P_S \phi\|^2$ , it bounds the  $L_1$  discrepancy by twice the missing weight — a step legitimate when  $[P_S, H] = 0$ , where  $A_S$  is  $A$  with the outside contributions deleted and  $\|A - A_S\|_1 = \|\phi\|^2(1 - w_S)$  identically, and controlling nothing otherwise. The two-level example is the extreme: nothing is discarded there, and every unit of retained weight is displaced by  $g$ . What the integrals do give, with no commutator hypothesis, is  $|\int A - \int A_S| \leq \|A - A_S\|_1$ , that is  $\|\phi\|^2(1 - w_S) \leq \|A - A_S\|_1$  — a *lower* bound, with 0 violations across the same 239 subspaces on which the upper form fails. It survives intact as the left-hand half of Theorem 1: the captured weight does say how wrong the reconstruction must be, it never said how wrong it can be.

*The two terms are mutually exclusive, and one is off at every operating point here.*  $S$  is a set of determinants, so  $P_S$  is diagonal in the working basis and  $P_S \phi = \phi \iff \text{supp}(\phi) \subseteq S \iff w_S = 1$ , while  $K_S \geq 0$  means precisely  $\phi = H^0 \phi \in \text{span}(S)$ . Hence the exclusion lemma

$$K_S \geq 0 \iff w_S = 1, \quad (10)$$

verified with 0 discrepancies across 239 subspaces, of which 195 have  $K_S = -1$ . The two terms of (9) are therefore never both informative: either  $w_S = 1$  and the whole bound is the second term — the two-level counterexample — or  $K_S = -1$  and  $C\rho^{-K_S} = C\rho$ , an undefined constant with no decay in it. Counting settles which branch applies here. The seed is  $\phi = \hat{c}_{0\uparrow}^\dagger |\Psi_0\rangle$  at half filling, so  $\text{supp}(\phi)$  is the set of  $(N+1)$  determinants with orbital  $0\uparrow$  occupied and  $|\text{supp}(\phi)|/D = n_\uparrow/L = \frac{1}{2} + \frac{1}{L}$  exactly; by (10),  $K_S \geq 0$  demands  $|S|/D \geq \frac{1}{2} + \frac{1}{L}$ , a floor of 0.625, 0.600, 0.583, 0.571 at  $L = 8, 10, 12, 14$  against the sector fractions 0.5561, 0.3625, 0.1700, 0.0800 at which the resource scan of Sec. VII operates (Fig. 9; the coarser replication grid of Table VIII reads 0.180 at  $L = 12$  for the same operating point). From  $L = 8$  on, every operating point of that scan lies below its own floor, so  $K_S = -1$  by counting alone, with no numerics, no ranking and no protocol, and any point below the same line inherits the verdict (App. A). Neither  $w_S$  nor  $K_S$ , we note, is computed in any file of the reproduction repository: a certificate written in quantities the pipeline never forms can be quoted but never applied, which is how a statement false by counting survives unexamined. Theorem 1 carries no such bookkeeping and is held to the opposite standard. Its two terms are defined and simultaneously active for every  $S$ ; the weight term is the half this work cannot compute without  $\Psi_0$  —  $\|\phi_Q\| = \|\phi\|\sqrt{1 - w_S}$ , and estimating  $w_S$  against an already-truncated  $\Psi_0$  is biased optimistically — while  $\Lambda_S(\eta)$  is an *a posteriori* quantity assembled from the Ritz vectors the reconstruction already forms plus one matrix–vector product, and is evaluated, subspace by subspace, in Sec. V.

*The magnitude, not the constant.* Two measurements at the same, exactly unit, captured weight show that (9) was indexed not on a bad constant but on the wrong quantity. First, let  $S^* = \text{supp}(\phi)$ , the smallest determinant subspace of unit captured weight — fixed by the seed, with no ranking, no tie-breaking and no zero-amplitude padding — of size exactly  $(\frac{1}{2} + \frac{1}{L})D$ . Its weight term is identically zero, and its relative  $L_1$  error is 0.43 at  $L = 6$  and 0.33 at  $L = 8$  on the periodic rings at  $\eta = 0.18t$ , rising to 1.15 and 0.90 at  $\eta = 0.05t$ , against a hard ceiling of  $1 + w_S = 2$ . (Relative here and throughout this subsection means divided by  $\|\phi\|^2$ , the normalization of Theorem 1; dividing instead by the in-window integral of the exact  $A$ , as the repository’s own relative  $L_1$  does, moves them by about 1% at  $\eta = 0.05t$  and by 2.6 to 3.3% at  $\eta = 0.18t$ , measured in App. C, and the two conventions are not interchanged anywhere in this paper without saying so.) Second,  $S^*$  is the  $K = 0$  rung of the Krylov ladder  $S_K = \bigcup_{a \leq K} \text{supp}(H^a \phi)$ , every rung of



which contains  $\phi$  and so carries  $w_S = 1$  identically, while its moment reach grows:  $S_K$  reproduces the exact spectral moments through order  $2K+1$  — to the  $10^{-15}$  level, failing as it must at  $2K+2$  — whereas coordinate subspaces of the same size drawn at random miss already  $\mu_0$ , in all of 300 draws (Sec. IV). Along a family of constant, exactly unit captured weight the error therefore runs from order unity to moment-exact through order  $2K+1$ . The captured weight, identical on every rung, does not order them.

What does is elementary. Writing  $\mu_j, \mu_j^S$  for the moments of the unbroadened measures of Sec. IV, every determinant subspace with  $w_S = 1$  reproduces  $\mu_0$  and  $\mu_1$  exactly, and the first moment that can fail is  $\mu_2$ , with deficit

$$\mu_2 - \mu_2^S = \|Q_S H P_S \phi\|^2, \quad Q_S = I - P_S, \quad (11)$$

the squared coupling of the Hamiltonian across the boundary of  $S$  ( $E_0$  drops out because  $Q_S \phi = 0$ ), zero exactly when  $S$  also holds  $\text{supp}(H\phi)$  — the  $K_S \geq 1$  condition. That boundary coupling is the magnitude the replacement is indexed on: it controls the leakage term through  $\Lambda_S(\eta) \leq \|P_S H Q_S\| \|\phi_S\|$ , and Theorem 1 averages it over the resolution window rather than over the spectrum. The captured weight says what the subspace *contains*; the error is set by what the Hamiltonian does to the seed at the subspace’s *edge*. Both halves of this paper follow from that displacement: Sec. V measures where the boundary term is small enough for the certificate to say something, and Sec. VI shows why no orbital-rotation invariant can

predict it in advance.

#### IV. WHAT THE STRUCTURE DOES CERTIFY: MOMENT EXACTNESS

Section IIID rules out an error bound the subspace construction was supposed to satisfy. It rules out nothing about the construction itself. One structural guarantee does hold, and it is the one that says what a Krylov-adequate subspace actually buys: a finite list of spectral moments, exactly. We state it with the proof the code realizes, measure its exact reach, and separate it from two claims that attach themselves to it and do not follow from it.

##### A. The statement that survives, and the proof it actually has

Write  $\mathcal{K}_K = \text{span}\{\phi, H\phi, \dots, H^K \phi\}$  for the order- $K$  Krylov space of the seed  $|\phi\rangle = \hat{c}_p^\dagger |0\rangle$ , and let  $d\mu$  and  $d\mu_S$  be the unbroadened Stieltjes measures whose Lorentzian convolutions are the exact  $A$  and the reconstruction  $A_S$ , the Lehmann sum evaluated inside  $\text{span}(\mathcal{S})$  as in Sec. II (App. B).

**Theorem 2** (Moment exactness). *Let  $P$  be the orthogonal projector onto  $\text{span}(\mathcal{S})$  and  $H_S = PHP$ . If  $\text{span}(\mathcal{S}) \supseteq \mathcal{K}_K$ , then*

$$\int \omega^j d\mu_S(\omega) = \int \omega^j d\mu(\omega) = \langle \phi | (H - E_0)^j | \phi \rangle, \quad 0 \leq j \leq 2K + 1. \quad (12)$$

*Proof.* Because  $\mathcal{K}_K \subseteq \text{span}(\mathcal{S})$  we have  $PH^a \phi = H^a \phi$  for  $0 \leq a \leq K$ , whence by induction  $H_S^b \phi = PH^b \phi$  for

$0 \leq b \leq K + 1$ . Then for any  $a \leq K$  and  $b \leq K + 1$ , using only  $P = P^\dagger = P^2$  and  $H = H^\dagger$ ,

$$\langle \phi | H_S^{a+b} | \phi \rangle = \langle H_S^a \phi | H_S^b \phi \rangle = \langle H^a \phi | PH^b \phi \rangle = \langle PH^a \phi | H^b \phi \rangle = \langle \phi | H^{a+b} | \phi \rangle, \quad (13)$$

and every  $j \leq 2K + 1$  is such an  $a + b$ . Both measures are shifted by the same  $E_0$ , so the identity survives term by term in the binomial expansion of  $(H - E_0)^j$ .  $\square$

That proof is three lines of Rayleigh–Ritz, and it is not the proof this statement is usually given. The usual route argues that an order- $(K+1)$  quadrature “matches the first  $2K + 1$  moments exactly”, and  $2K + 1$  is exactly what Table II measures: the *reach* is right, the route is not. What does not hold here is the identification of  $A_S$  with a Gaussian-quadrature rule built by a nonorthogonal Lanczos/Krylov construction. No such object is built anywhere

in this work: every reconstruction diagonalizes  $P_S H P_S$  on a coordinate subspace, so  $d\mu_S$  carries  $|\mathcal{S}|$  nodes, not  $K + 1$ —in the  $K = 1$  row of Table II, 280 of them, where a genuine 280-node Gauss rule for  $d\mu$  would be exact through order 559 [40] while the reconstruction fails at order 4. Those 555 orders measure the distance between the two objects, not an error in the published reach. The conclusion is right, the route is not, and the correct route is also the more general one: it uses only that  $P$  is an orthogonal projector, so it covers the Krylov projector of Sec. V as well, and it needs no Lanczos basis and hence

no conditioning assumption on one [31] (App. B 1).

Table II measures Eq. (12) on the construction this work uses—the union of the determinant supports of  $\{\phi, H\phi, \dots, H^K\phi\}$ —for the open Hubbard chain at  $L = 6$ ,  $U/t = 8$ , in the  $(N+1, S_z=+1)$  sector of dimension  $D = 300$ . Through order  $2K+1$  the moments agree to a relative  $10^{-14}$ ; at order  $2K+2$  they part company by eleven orders of magnitude at  $K = 0$  and seven at  $K = 1$ . The order  $2K+1$  is the exact reach of the construction, not merely a sufficient claim, and the two rows that exhibit that reach are the two on which the ladder has not yet filled the sector.

A negative control fixes what the table is about. Over 300 coordinate subspaces of the *same* size  $|\mathcal{S}| = 200$  drawn uniformly from the sector, the zeroth moment alone is already wrong by 3.7% to 84% of  $\|\phi\|^2$  (median 33%), every single draw failing, against  $4 \times 10^{-16}$  for the Krylov-support subspace of identical size. Two thirds of the sector buys nothing by itself; two thirds of the sector chosen as  $\bigcup_{a \leq K} \text{supp}(H^a\phi)$  buys  $2K+2$  moments exactly. What Theorem 2 certifies is *which* determinants the subspace holds, not *how many*—a structural property, decidable from the subspace alone, before any exact reference exists.

One caveat of scale belongs with the table. The ladder terminates once the subspace contains the cyclic subspace generated by  $\phi$ , and there the reconstruction is exact: at  $L = 6$  that is  $K = 2$  on the open chain, and already  $K = 1$  on the periodic rings of Sec. VII, where it stalls at 296 of 300 determinants with  $A_{\mathcal{S}} = A$  to  $10^{-14}$  (App. B 2). Three rows is all this validation system has, and we claim nothing about the large- $K$  behaviour of  $\|A - A_{\mathcal{K}_K}\|_1$ . Within the reach it covers, the identity is exact and sharp at the next order.

### B. What moment exactness does not buy

Equation (12) invites a corollary: “consequently, at fixed broadening  $\eta > 0$ , the smoothed reconstruction converges geometrically,  $\|A - A_{\mathcal{K}_K}\|_1 \leq C(\eta) \rho(\eta)^{-K}$ .” It does not follow, and it is not a corollary here. Moment exactness is a list of  $2K+2$  linear identities, and the  $L_1$  distance between two Lorentzian-broadened measures is not a finite linear functional of them; converting one into the other needs a polynomial approximation theorem for the kernel  $x \mapsto \mathcal{L}_\eta(\omega - x)$  together with a tail estimate, and moment exactness supplies neither. The natural tool, a Bernstein-ellipse bound, is also where the resolution enters, through the poles of the kernel at  $x = \omega \pm i\eta$  (App. B 3); and neither  $C(\eta)$  nor  $\rho(\eta)$  is derived or exhibited anywhere in this work or in the deposit. We therefore state the claim as what it is.

**Conjecture 1** (Geometric convergence in the Krylov order). *Fix  $\eta > 0$  and let  $A_{\mathcal{K}_K}$  be the Rayleigh–Ritz reconstruction on  $\mathcal{K}_K$ . Are there  $C(\eta) < \infty$  and  $\rho(\eta) > 1$ , uniform over a class of Hamiltonians containing the models studied here, such that  $\|A - A_{\mathcal{K}_K}\|_1 \leq C(\eta) \rho(\eta)^{-K}$  for every  $K$ ?*

One constraint on any answer is already available. The two-level family of Sec. III D has  $\mathcal{K}_0 = \text{span}\{\phi\}$ , and by direct integration at  $\eta = 0.15$  its  $\|A - A_{\mathcal{K}_0}\|_1$  is 1.983 at  $g/\eta = 2 \times 10^2$  and 1.998 at  $g/\eta = 6.7 \times 10^3$ , rising to  $2\|\phi\|^2$ . Evaluating Conjecture 1 at  $K = 0$  therefore forces  $C(\eta) \geq 2\|\phi\|^2$ : the same family that forces the trivial constant on the weight-only bound forces it here on the prefactor. Whatever content the conjecture has must be carried by  $\rho(\eta)^{-K}$  alone, and in particular it cannot be non-vacuous at the small  $K$  at which a shot-limited subspace operates.

One figure of this work is read against that conjecture and not against a theorem. Panel (b) of Fig. 2 fits a geometric collapse  $\text{rel-}L_1 \sim \rho^{-K}$  with  $\rho \approx 1.7$ , and a fit of that shape is easily reported as “precisely the geometric convergence guaranteed by the moment-exactness theorem”. It is guaranteed by no such thing—that is the corollary ruled out above—and the fit does not carry the claim on its own either. The panel is one of the two places in this work where the sampling is genuine and its budget is declared—multinomial draws at 4000 shots per time step, eight seeds,  $\eta = 0.15 t$ , so  $T = 4000(K+1)$  and  $T = 5.2 \times 10^4$  at the last point—and the fit uses six of its thirteen measured points, inside a window selected after the fact ( $4 \leq K \leq 9$ , set as a literal in `spectral_validation_max.py`). Across that window the mean subspace the sampler returns grows from 179 to 236 determinants of the 300-dimensional sector, so decay in the Krylov order is confounded with growth in  $|\mathcal{S}|$ ; and the  $K$  of the abscissa is the number of measured time slices, not the  $K_{\mathcal{S}}$  of the weight-only bound, which is  $-1$  at all thirteen points by Eq. (10). No rate is fitted to that window here, and the caption claims none. What the panel does show, over all thirteen points, with its eight-seed band and with  $T$  attached, is that the error falls by a factor 32 as the measured Krylov order is climbed at fixed  $\eta$ , from 0.59 at  $K = 0$  to 0.0185 at  $K = 12$ , with a single non-monotone step at  $K = 3$  inside the discovery phase; that is the statement its caption now makes.

There is a sharper reason not to read Eq. (12) as an error bound, and Table II contains it. For a coordinate subspace  $P_{\mathcal{S}}$  is diagonal, so  $\text{span}(\mathcal{S}) \supseteq \mathcal{K}_K$  at any  $K \geq 0$  already forces  $\text{supp}(\phi) \subseteq \mathcal{S}$ , that is  $w_{\mathcal{S}} = 1$  exactly—as the fourth column records for all three rows. Moment exactness here is strictly stronger than unit captured weight, and unit captured weight is precisely the condition under which Sec. III measures the error to be  $O(1)$ . The two meet in the first row: the  $K = 0$  member of the ladder *is* the witness  $\mathcal{S}^* = \text{supp}(\phi)$  of Sec. III, of size exactly  $(\frac{1}{2} + \frac{1}{L})D$  and fixed by the probe rather than by any ranking. It is moment-exact through order 1, it holds the entire Born weight of the probe, and at  $\eta = 0.18 t$  its relative  $L_1$  error is 0.28 on the open chain of Table II at  $L = 6$ , and 0.43 at  $L = 6$  and 0.33 at  $L = 8$  on the periodic rings on which Sec. III reports it: a measured range of 0.28–0.43 at unit captured weight, whose lower end is the very system on which the method validation of Sec. II is performed. (All three are  $\|A - A_{\mathcal{S}}\|_1$  divided by  $\|\phi\|^2$ ;

TABLE II. **Moment exactness and its exact reach.** Open Hubbard chain,  $L = 6$ ,  $U/t = 8$ ; seed  $|\phi\rangle = \hat{c}_{0\uparrow}^\dagger|0\rangle$ ;  $(N+1, S_z=+1)$  sector of dimension  $D = 300$ ;  $\mathcal{S} = \bigcup_{a \leq K} \text{supp}(H^a \phi)$ . The largest Krylov order inside  $\text{span}(\mathcal{S})$ ,  $K_{\mathcal{S}}$ , and the captured Born weight  $w_{\mathcal{S}}$  are measured, not assumed.  $\epsilon_j$  is the *relative* discrepancy of the  $j$ -th moment, quoted as the largest value over six equivalent implementations of the same identity; the absolute discrepancy at order  $2K+2$  follows in parentheses. Only relative residues are quoted below that order, because  $\mu_j$  spans five decades between  $j = 0$  and  $j = 6$  and the absolute roundoff residue is route-dependent (App. B2). At  $K = 2$  the ladder has filled the sector, so  $\text{span}(\mathcal{S})$  carries every eigenvector of nonzero Lehmann weight,  $A_{\mathcal{S}} = A$  identically, and there is no order- $2K+2$  failure to exhibit; that row is kept to mark where the ladder terminates, and its single entry is the roundoff level of the moments.

| $K$ | $ \mathcal{S} $ | $ \mathcal{S} /D$ | $w_{\mathcal{S}}$ | $K_{\mathcal{S}}$ | $\max_{j \leq 2K+1} \epsilon_j \longrightarrow \epsilon_{2K+2}$ (absolute)         |
|-----|-----------------|-------------------|-------------------|-------------------|------------------------------------------------------------------------------------|
| 0   | 200             | 0.667             | 1                 | 0                 | $1.8 \times 10^{-15} \longrightarrow 9.5 \times 10^{-4}$ ( $3.10 \times 10^{-2}$ ) |
| 1   | 280             | 0.933             | 1                 | 1                 | $3.2 \times 10^{-15} \longrightarrow 6.0 \times 10^{-8}$ ( $1.34 \times 10^{-4}$ ) |
| 2   | 300             | 1.000             | 1                 | full sector       | $7.5 \times 10^{-15} \longrightarrow$ no failure: $A_{\mathcal{S}} = A$            |

dividing instead by the in-window integral of the exact  $A$ , the convention of the repository’s own relative  $L_1$ , moves them by less than 1% and changes none of them at the two figures quoted; the two conventions are never interchanged silently.) Exactness of  $2K + 2$  numbers is what the structure certifies; the curve is certified, at fixed  $\eta$  and on the subspace in hand, by Theorem 1, whose own non-vacuity is measured rather than assumed (Sec. V).

### C. The shot count, corrected and strengthened

The second surviving part of Theorem 1 counts the shots that populate such a subspace. Bitstrings are drawn from the time-averaged Born mixture  $p(x) = \frac{1}{K+1} \sum_{k=0}^K |\langle x | e^{-iHt_k} \phi \rangle|^2 / \|\phi\|^2$ , and the target is to capture every configuration with  $p(x) \geq \varepsilon$  at confidence  $1 - \delta$ . As published the answer is  $T = O(\varepsilon^{-1} \log(|\mathcal{S}|/\delta))$ , which is circular:  $|\mathcal{S}|$  is the random output of the very  $T$  draws the bound is meant to fix.

The repair is free. With  $\mathcal{N}_\varepsilon = \{x : p(x) \geq \varepsilon\}$ , each member contributing at least  $\varepsilon$  to a probability distribution,  $|\mathcal{N}_\varepsilon| \leq 1/\varepsilon$  *deterministically*; a union bound over  $\mathcal{N}_\varepsilon$  of the event “ $x$  never drawn in  $T$  trials” then gives  $\Pr[\mathcal{N}_\varepsilon \not\subseteq \mathcal{S}] \leq \varepsilon^{-1} e^{-\varepsilon T}$ , so

$$T \geq \varepsilon^{-1} \log(1/(\varepsilon\delta)) \quad (14)$$

suffices (Prop. 2). This is non-circular, and wherever the sampler returns more than  $\varepsilon^{-1}$  distinct configurations its logarithm is the smaller of the two as well.

What Eq. (14) does *not* say belongs in the same breath, and in the main text rather than in an appendix. The parameter  $\varepsilon$  thresholds a Born probability; it is not a spectral accuracy, and no step of the argument connects it to  $\|A - A_{\mathcal{S}}\|_1$ . Arbitrarily many configurations may each carry  $p(x) < \varepsilon$  while jointly carrying most of the mixture; and, decisively, even at  $w_{\mathcal{S}} = 1$  exactly—nothing left to capture at any  $\varepsilon$ —the relative error is  $O(1)$ , as Sec. III measures and the first row of Table II exhibits. Capturing the high-probability configurations is a statement about the sampler, not about the spectrum.

Two compositions of the bound follow easily from it, and neither is made here. “A sampling-capture bound

polynomial in  $|\mathcal{S}|$  and independent of the Hilbert-space dimension” is a pair of clauses that are literally true and jointly empty, because this same work measures  $|\mathcal{S}| \sim e^{1.05L}$  for these lattices, Eq. (20), so a budget polynomial in  $|\mathcal{S}|$  is exponential in  $L$ , and independence of the Hilbert-space dimension is no saving when the quantity the budget is polynomial in grows exponentially itself. And the fall of the required sector fraction with  $L$  is not “the favourable face of the polynomial-shot bound”: it is not a face of that bound at all. Equation (14) relates a shot budget to a Born-probability threshold and contains no statement about  $|\mathcal{S}|/D$ , about  $L$ , or about any trend in either.

What stands in their place is the narrower statement Eq. (14) does support, and it is the stronger one for being narrow: that many shots buy, at confidence  $1 - \delta$ , every configuration above Born weight  $\varepsilon$ , and nothing further. What the resulting subspace costs, and what it is worth, are separate measurements, reported as three independent prices—support, resolution and shots—in Sec. VII.

### V. WHAT THE CERTIFICATE CAN AND CANNOT ASSERT: THE NON-VACUITY THRESHOLD OF THE LEAKAGE BOUND

Throughout this section we call  $\text{FR}^*$  the *non-vacuity threshold of the certificate of Theorem 1*: the sector fraction above which the leakage branch of the bound first improves on the trivial bound  $\|\phi\|^2(1 + w_{\mathcal{S}})$ . We do not call it a frontier of the method, and it is not one. It is a property of the inequality chain that proves Theorem 1, measured here for the first time, together with one structural obstruction that no leakage-type certificate avoids. Two measurements separate the two contributions. First, at fixed system and fixed accuracy target the threshold sits far above the fraction the reconstruction actually needs: at  $L = 8$ ,  $\eta = 0.18t$ , a relative  $L_1$  error of 0.05—the acceptance criterion used throughout this paper—is reached at  $\text{FR} \approx 0.52$ , while the certificate does not reach 0.05 until  $\text{FR} \approx 0.98$ , and the intervening shift of  $\approx 0.45$  in fraction is the looseness of the proof. Second, the threshold cannot be lowered by choosing a better subspace: from  $L = 8$  on, the coordinate support of the Krylov space  $\mathcal{K}(H, \phi)$  is the

whole sector—measured at  $L = 8, 10, 12$  and  $14$ , which is every size this work reports—so  $\Lambda_S > 0$  strictly for every proper coordinate subspace, and a Krylov-diagonal oracle ranking gives a *worse* certificate than the Born ranking the device realizes. Section [VE](#) measures both. What is a property of the method, and is new, is the calibration of

$$\frac{\hat{\Lambda}_S(\eta)}{\eta} < \tau(w_S) := \frac{1 + w_S}{1 + \sqrt{w_S}} - \sqrt{1 - w_S}, \quad \tau(1) = 1, \quad (15)$$

so a subspace holding the entire Born weight of the probe certifies something as soon as its leakage, in units of the resolution it was asked for, drops below unity. This section is that one comparison, made 204 times.

### A. What is evaluated

We evaluate Theorem [1](#) on the subspaces of Fig. [9](#) at  $L = 6, 8, 10$  and  $12$ —sector dimensions 300, 3920, 52920, 731808—with the ranking protocol of Sec. [VII](#) ( $K = 18$ ,  $\delta t = 0.5$ , 16 substeps, 19 snapshots), over fractions 0.20–0.95 and  $\eta = 0.10$ – $0.25 t$ . Those subspaces are the  $T \rightarrow \infty$  limit of that protocol and not finite-shot draws, and every statement below inherits that limit: the marginal determinant at the  $L = 12$  threshold carries Born probability  $1\text{--}2 \times 10^{-9}$ , so at the budgets of Table [IX](#) that subspace barely exists as a sampled object. Of the 204 evaluations, 187 are converged in Krylov depth, 6 marginal and 11 not converged, at a total cost of 2874s with a peak of 1.64 GiB. *No row flagged not converged is quoted anywhere in this section*; the tables mark the ones that are marginal.

To that grid we add twenty evaluations that are not on it. A grid point near a published fraction is not the published subspace, and the difference is not cosmetic: this sweep reads 0.80, 0.56, 0.40 and 0.18 against published fractions of 0.8200, 0.5561, 0.3625 and 0.1700. The rows of Table [III](#) are placed instead at the integer determinant count  $|S| = \text{round}(\text{frac} \times \text{dim})$  that Fig. [9](#) reports—246, 2180, 19183, 124407 and 824504 determinants at  $L = 6, 8, 10, 12$  and  $14$ —at the same four resolutions and under the same protocol, at depths  $n_\ell = 245$  (exhausting the subspace), 900, 700, 300 and 250, with orthogonality defect at most  $4.9 \times 10^{-15}$  and  $|w_K - w_S|$  at most  $6.4 \times 10^{-15}$ . All twenty pass the depth test, and all twenty clear the reference-drift rule of App. [C](#), the weakest— $L = 14$  at  $\eta = 0.10 t$ , where the reference itself drifts by  $1.9 \times 10^{-3}$  on halving its depth—by a factor 51. They cost 129s of core time up to  $L = 12$  and 2238s of wall time at  $L = 14$ , and they add the size the sweep never reached.

Three conventions are fixed before any number is read. Full reorthogonalization is used throughout, without which the projector is not a projector: the orthogo-

nal defect is at most  $4.7 \times 10^{-15}$ , and  $P_K \phi = \phi_S$  to  $1.1 \times 10^{-14}$ . What is certified is the object the code forms— $\Lambda$  on  $\mathcal{K} = \text{Krylov}(H_S, \phi_S, n_\ell)$  bounds the error of  $A_K$  at that depth and of nothing else, the convenient inequality  $\Lambda_K \geq \Lambda_S$  that would extend it to the dense  $A_S$  being false (1169 of 6000 random triples violate it, minimum ratio 0.5258); at  $L = 12$  the certificate is evaluated at  $n_\ell = 300$ –500 on up to 658627 determinants, and  $A_S$  is never formed. And depth is certified per point by the internal indicator  $\Lambda_{\text{in}}/\Lambda_{\text{out}}$  rather than assumed [[31](#), [41](#)]: recomputing  $\text{FR}^*$  with  $L = 12$  capped at  $n_\ell = 300$  moves it by at most 0.0024, an eighth of the narrowest grid interval.

Three hypotheses carry the bound, each a statement about the pipeline rather than about the theorem, and all three are checked against this pipeline in App. [C](#) along with the ranking, tie and support declarations of the cut: (H1)  $A$  and  $A_P$  share  $E_0$ —satisfied here by construction, and worth +0.20% on the bound but a factor 16 on the measured error when it is not; (H2) the  $L_1$  norm is taken on the whole line and normalized by  $\|\phi\|^2$ , differing from the repository’s windowed convention by factors 1.010–1.037; (H3) the reference is exact, which enforces the rule that no measured error is quoted within a factor 20 of its own reference drift. One asymmetry runs through every table: of the two terms of Eq. [\(7\)](#) only the leakage is a posteriori, assembled from the Ritz vectors the reconstruction already produces plus one matrix–vector product each, while the weight term needs  $\Psi_0$ , and estimating  $w_S$  against an already-truncated  $\Psi_0$  is biased optimistically (Sec. [III](#)). What the tables report is accordingly one object and not a family of them: a bound on  $\|A - A_K\|_1/\|\phi\|^2$  for the reorthogonalized Krylov reconstruction at a declared depth, against a reference whose drift is declared and enforced.

### B. The threshold, measured: at every published fraction the certificate is vacuous

The result is negative and we state it first. Evaluated on the published subspaces themselves, the leakage branch of the bound exceeds the trivial bound  $\|\phi\|^2(1 + w_S)$  at all five sizes and all four resolutions, by factors running



from 1.66 ( $L = 6$ ,  $\eta = 0.25t$ ) to 8.61 ( $L = 14$ ,  $\eta = 0.10t$ ), and by 2.40 to 4.61 at the  $\eta = 0.18t$  of the momentum-resolved channel; Table III carries the twenty values. They are monotone in both arguments—rising with  $L$  at every resolution, falling with  $\eta$  at every size—so the trend runs against us: the certificate is furthest from saying anything at the largest sector and the finest resolution, which is where the method is aimed.

The largest classically converged case carries the evidence. At  $L = 12$  and the published subspace itself— $|S| = 124407$  of 731808 determinants—the captured weight is  $w_S = 0.998622$ , so Eq. (15) asks for  $\hat{\Lambda}_S/\eta < \tau(w_S) = 0.9625$  and the measured value is 4.236 at  $\eta = 0.18t$ , putting the leakage branch at 8.544 against a trivial branch of 1.9986. That is converged in depth: the depth indicator is  $\Lambda_{\text{in}}/\Lambda_{\text{out}} = 2.8 \times 10^{-3}$  and the bound moves by  $3.7 \times 10^{-7}$  relative across the depth plateau, while at the neighbouring grid point  $\text{FR} = 0.18$  the same quantity was checked twice more— $\Lambda_{\text{out}}$  changing by  $7.5 \times 10^{-7}$  between depths 250 and 300, and an independently written re-implementation moving it by  $8.6 \times 10^{-5}$  between depths 100 and 300. Since  $\Lambda$  decreases with depth, the bound at any shallower depth—including the depth 150 at which Fig. 9 is drawn—is larger still, so the vacuity is not an artifact of Lanczos truncation in either direction. Nor is it peculiar to these subspaces: over the 606 live subspaces of the pooled sweep of App. A, Fig. 17, the leakage branch is the smaller of the two on only 61. **The reconstructions this paper reports—Fig. 3, Fig. 9 and the operating points of Secs. VII and VIII—are therefore, at present, uncertified.**

One row is quoted with two caveats attached.  $L = 14$  is the size at which Fig. 9 reports its headline fraction of 0.08, and the certificate had never been evaluated there; it now is, at  $|S| = 824504$  of 10306296 determinants—one determinant above the round( $\text{frac} \times \text{dim}$ ) = 824503 of the deposited grid—with  $w_S = 0.998388$  and a leakage branch of 17.21, 11.18, 9.21 and 6.50 against a trivial branch of 1.9984. Ground state, seed and order there are the repository’s stored checkpoint—residual  $2.7 \times 10^{-9}$ , the order verified to be a permutation of the sector—and the  $K = 18$  ranking protocol was *not* re-run at that size; it was re-run at  $L = 10$ , where the two selections agree as sets to 1.000000. And the depth indicator  $\Lambda_{\text{in}}/\Lambda_{\text{out}} = 0.0453$  at  $\eta = 0.10t$  is the loosest value quoted anywhere in this section, comparable to the  $4.7 \times 10^{-2}$  of the last row of Table VI. Neither caveat is repaired, and the row is reported because it makes the verdict of this section worse rather than better.

One column of Table IV is a positive control and has to be read before the rest. Applying this paper’s own acceptance criterion to the same sweep—independent code, the theorem’s normalization, the whole line rather than a window, a deep reference rather than one recomputed at the subspaces’ own depth—returns 0.816, 0.553, 0.356 and 0.168 at  $\eta = 0.15t$  for  $L = 6$ –12, against the published 0.82, 0.56, 0.36 and 0.18: agreement to 7% at worst, and to under 2% at every size once each value is compared

with the fraction the scan itself reports rather than with the two-figure entry of the replication grid (0.5561, 0.3625, 0.1700 at  $L = 8, 10, 12$ ). None of the three ingredients is shared. Figure 9 is thereby reproduced independently.

The last column of Table III is that control taken one step further, and it is the sharper of the two: instead of re-deriving the fraction it evaluates the relative  $L_1$  error at the published subspace, at the resolution and against the threshold that define it ( $\eta = 0.15t$ ,  $\text{rel-}L_1 < 0.05$ ). The five values—0.0477, 0.0478, 0.0484, 0.0485 and 0.0463—all sit just below the threshold and are flat to within 5% of one another across a sector dimension running from 300 to 10306296,  $L = 14$  included. The criterion that defines the published fractions is therefore reproduced at the subspaces themselves, at every size the figure plots above  $L = 4$ , again with nothing shared. What fails in this section is not the reconstruction and not that curve; it is the word *certificate* attached to the fractions at which the curve is read.

In shots the distance between those two columns is a budget and not a rhetorical gap (Table V): reaching the non-vacuous band multiplies the shots per snapshot needed for the marginal determinant by 7.5–10 at  $L = 8$  and by 20–35 at  $L = 12$ , and only by 1.4–1.6 at  $L = 6$ , where the sector is small. Driving the certified value down to 0.10 costs more again—a further factor 241 at  $L = 8$  (fraction 0.95) and 634 at  $L = 10$  (fraction 0.90), the corresponding  $L = 12$  fraction resting only on evaluations that failed the depth test, so that no budget is quoted for it. Where the previous version of this work asserted a regime of validity it had never evaluated, this section reports the measurement: four sizes, four resolutions, 204 evaluations, every bracket delimited by converged rows, and a shot budget attached to each.

### C. The strongest certified statements, and what this paper’s own criterion would cost

Table VI collects what the sweep does certify, taking at each size the strongest statement whose bracket is converged—at  $L = 12$ , that 60% of a 731808-dimensional sector gives relative  $L_1 \leq 0.717$ , tightening to  $\leq 0.218$  at 70% and  $\eta = 0.25t$ . Each row is a statement about  $A_K$  at the declared depth, and none of them uses the exact spectrum.

The concession that belongs with that table is one a referee would make, so we make it first and in full. At the acceptance criterion this paper uses everywhere else—relative  $L_1 < 0.05$ —the certificate requires at least 0.98 of the sector at  $L = 8$ , measured at integer subspace size rather than interpolated: the certified value is 0.084 at 3802 of 3920 determinants, 0.060 at 3830 and 0.039 at 3855, so the crossing lies in  $[0.977, 0.983]$ ; on our grid it is not reached at all at  $L = 6$ ,  $L = 10$  or  $L = 12$ . At such fractions diagonalizing the whole sector is already what one would do instead, and a certificate informative only there does not certify the regime in which sampling



TABLE III. **The certificate on the subspaces this paper publishes, including the size the sweep never reached.** Each row is Theorem 1 evaluated at the integer determinant count  $|\mathcal{S}| = \text{round}(\text{frac} \times \text{dim})$  that Fig. 9 reports, not at the nearest point of the grid of Table IV. Columns 5–8 are the vacuity factor—the leakage branch of Eq. (7) over the trivial branch  $\|\phi\|^2(1 + w_S)$ —so a value above 1 means the certificate says nothing the trivial bound does not. All twenty evaluations are the  $T \rightarrow \infty$  limit of the ranking protocol and pass the depth test; controls and costs are in App. C. The last column is the measured relative  $L_1$  error at the resolution and against the threshold that *define* the published fraction ( $\eta = 0.15t$ ,  $\text{rel-}L_1 < 0.05$ ), from code sharing neither reference, normalization nor recursion depth with the scan that produced the fractions. At  $L = 14$  the count is one determinant above the deposited 824 503 and the ranking was not re-run; see the text.

| $L$ | $\text{dim}(N+1)$ | published subspace |                 | vacuity factor at $\eta/t =$ |      |      |      | rel- $L_1$<br>at $\eta = 0.15t$ |
|-----|-------------------|--------------------|-----------------|------------------------------|------|------|------|---------------------------------|
|     |                   | fraction           | $ \mathcal{S} $ | 0.10                         | 0.15 | 0.18 | 0.25 |                                 |
| 6   | 300               | 0.8200             | 246             | 4.54                         | 2.94 | 2.40 | 1.66 | 0.0477                          |
| 8   | 3 920             | 0.5561             | 2 180           | 5.65                         | 3.69 | 3.05 | 2.17 | 0.0478                          |
| 10  | 52 920            | 0.3625             | 19 183          | 6.50                         | 4.28 | 3.54 | 2.52 | 0.0484                          |
| 12  | 731 808           | 0.1700             | 124 407         | 8.04                         | 5.20 | 4.27 | 3.00 | 0.0485                          |
| 14  | 10 306 296        | 0.0800             | 824 504         | 8.61                         | 5.59 | 4.61 | 3.26 | 0.0463                          |

TABLE IV. **Where the certificate stops being vacuous, and what the reconstruction actually needs, measured on the same sweep.** Protocol of Sec. VII, full reorthogonalization, depth certified per point. Column 3 is the fraction Fig. 9 reports; column 4 the control, where this sweep—independent code, the normalization of Theorem 1, the whole line, a deep reference—first reaches relative  $L_1 < 0.05$ , the parenthesized value being the log-linear interpolation inside the bracket and not a measurement at three digits; column 5 is FR\*; columns 6–7 are where the certified value itself first falls below 0.50 and 0.20. Every bracket is delimited by two converged evaluations unless marked: \* means the upper end is marginal in depth, *n.c.* that the bracket rests only on evaluations failing the depth test and is therefore not quoted. Columns 5–7 are statements about  $A_K$  at the declared depth, not about the dense  $A_S$ . Columns 4 and 5 must be read together: FR\* alone falls with size and would suggest the certificate improves as the problem grows, whereas what the reconstruction needs falls with it and the gap widens—a gap we do not quote as a ratio, both columns being interpolations inside brackets 0.02–0.10 wide.

| $L$                                                                               | $\text{dim}(N+1)$ | published | rel- $L_1 < 0.05$    | FR*          | cert $< 0.50$ | cert $< 0.20$ |
|-----------------------------------------------------------------------------------|-------------------|-----------|----------------------|--------------|---------------|---------------|
| <i><math>\eta = 0.18t</math>, the resolution of the momentum-resolved channel</i> |                   |           |                      |              |               |               |
| 6                                                                                 | 300               | 0.82      | [0.70, 0.80]         | [0.90, 0.92] | $> 0.97$      | $> 0.97$      |
| 8                                                                                 | 3 920             | 0.56      | [0.50, 0.56]         | [0.75, 0.80] | [0.90, 0.92]  | $> 0.95$      |
| 10                                                                                | 52 920            | 0.36      | [0.30, 0.40]         | [0.60, 0.70] | [0.80, 0.85]  | [0.85, 0.90]  |
| 12                                                                                | 731 808           | 0.18      | [0.14, 0.18]         | [0.40, 0.45] | [0.60, 0.70]* | <i>n.c.</i>   |
| <i><math>\eta = 0.15t</math>, the resolution at which Fig. 9 is drawn</i>         |                   |           |                      |              |               |               |
| 6                                                                                 | 300               | 0.82      | [0.80, 0.85] (0.816) | [0.92, 0.95] | $> 0.97$      | $> 0.97$      |
| 8                                                                                 | 3 920             | 0.56      | [0.50, 0.56] (0.553) | [0.80, 0.82] | [0.92, 0.95]  | $> 0.95$      |
| 10                                                                                | 52 920            | 0.36      | [0.30, 0.40] (0.356) | [0.60, 0.70] | [0.80, 0.85]  | [0.85, 0.90]  |
| 12                                                                                | 731 808           | 0.17–0.18 | [0.14, 0.18] (0.168) | [0.45, 0.50] | [0.60, 0.70]* | <i>n.c.</i>   |

is of interest. We state it without qualification: at the accuracy this paper accepts elsewhere, Theorem 1 does not certify the operating points of Fig. 9, and the shot cost of the regime where it would exceeds the saving that figure reports.

What the table does certify at a fraction a sampler can plausibly reach is weaker, and it is not empty. The trivial bound caps the relative error at  $1 + w_S \leq 2$  for *any* subspace, a reconstruction of pure noise included; at  $L = 12$  and 60% of a 731 808-dimensional sector the certified 0.717 sits a factor 2.8 below that cap, and at 70% and  $\eta = 0.25t$  a factor 9.2 below it. Those are exclusions of gross failure made from the subspace alone, at a dimension where the exact spectrum is expensive

and, at the sizes this method is aimed at, unavailable.

#### D. Where the certificate is informative, and where it is not

One positive result survives the sweep, and it is the reason we report the sweep. Across all sixteen  $(L, \eta)$  cells the crossing of the trivial bound occurs at a true relative  $L_1$  error of  $1.2 \times 10^{-3}$  to  $7.3 \times 10^{-3}$ , median  $2.8 \times 10^{-3}$ : a spread of 5.9 over a table whose true errors span five decades, and in every cell at least 47 times the drift of our own reference.  $\hat{\Lambda}_S(\eta)/\eta$  is therefore a calibrated a posteriori risk tracker—its crossing of the trivial bound

TABLE V. **The threshold in the only currency a sampler has.**  $T_{\text{marg}}$  is the shots per snapshot at which the marginal determinant of the cut appears once in expectation—the reciprocal of its Born probability in the accumulated distribution of the 19 snapshots, from the protocol’s own cut weights; covering all of  $S$  costs a further factor  $\log|S|$ . *It is a different quantity from the measured budgets  $T$  of Table IX*, which are totals over the 19 snapshots at which the finite-shot reconstruction reaches relative  $L_1 < 0.05$ ; the two carry different names for that reason, they are not interchangeable, and only ratios within this table are used. Column 2 uses the grid point nearest the published fraction: 0.80 against a published 0.82 at  $L = 6$ , and 0.40 against 0.36 at  $L = 10$ , where the published fraction is the worse of the two. Factors are from unrounded budgets and need not reproduce from the two-figure entries. The marginal Born probability at the  $L = 12$  threshold is  $2.2 \times 10^{-9}$  at  $\text{FR} = 0.40$ ,  $1.2 \times 10^{-9}$  at 0.45.

| $L$ | published fraction $\rightarrow T_{\text{marg}}$ | $\text{FR}^* \rightarrow T_{\text{marg}}$                          | factor  |
|-----|--------------------------------------------------|--------------------------------------------------------------------|---------|
| 6   | $0.80 \rightarrow 2.2 \times 10^4$               | $[0.90, 0.92] \rightarrow 3.1\text{--}3.6 \times 10^4$             | 1.4–1.6 |
| 8   | $0.56 \rightarrow 2.1 \times 10^5$               | $[0.75, 0.80] \rightarrow 1.6\text{--}2.1 \times 10^6$             | 7.5–10  |
| 10  | $0.40 \rightarrow 5.2 \times 10^6$               | $[0.60, 0.70] \rightarrow 3.4 \times 10^7\text{--}1.1 \times 10^8$ | 6.7–21  |
| 12  | $0.18 \rightarrow 2.3 \times 10^7$               | $[0.40, 0.45] \rightarrow 4.7\text{--}8.1 \times 10^8$             | 20–35   |

TABLE VI. **The strongest certified statements available from converged data.** Each row is the tightest certified value at that size whose bracket is delimited by evaluations passing the depth test; rows that would be stronger but rest on evaluations failing it are not listed. “True error” is the measured relative  $L_1$  at the same point, in the normalization of Theorem 1, given only where it exceeds its own reference drift by more than a factor 20 (App. C, H3); at  $L = 10$  it does not, and is withheld rather than rounded. At  $L = 6$  the recursion runs to  $n_\ell = |S|$ , so the evaluation is exact and the depth indicator is not defined. The gap between the certified value and the true error is three to four decades: the object is a certificate and not an error estimate, and Sec. VD states what it is good for.

| $L$ | $\eta/t$ | fraction             | certified rel- $L_1 \leq$ | true rel- $L_1$       | $\Lambda_{\text{in}}/\Lambda_{\text{out}}$ |
|-----|----------|----------------------|---------------------------|-----------------------|--------------------------------------------|
| 6   | 0.18     | 291/300 = 0.97       | 0.502                     | $3.94 \times 10^{-4}$ | exact ( $n_\ell =  S $ )                   |
| 8   | 0.18     | 3724/3920 = 0.95     | 0.219                     | $4.33 \times 10^{-5}$ | $8 \times 10^{-4}$                         |
| 10  | 0.18     | 47628/52920 = 0.90   | 0.136                     | withheld (H3)         | $1.6 \times 10^{-3}$                       |
| 12  | 0.18     | 439085/731808 = 0.60 | 0.717                     | $1.89 \times 10^{-4}$ | $7.0 \times 10^{-3}$                       |
| 12  | 0.15     | 0.60                 | 0.878                     | $2.66 \times 10^{-4}$ | $1.4 \times 10^{-2}$                       |
| 12  | 0.25     | 512266/731808 = 0.70 | 0.218                     | $1.99 \times 10^{-5}$ | $4.7 \times 10^{-2}$                       |

locates a true error of  $\approx 3 \times 10^{-3}$  to within a factor of three over  $L = 6\text{--}12$  and  $\eta = 0.10\text{--}0.25t$ , and it is computed from  $H$ ,  $S$ ,  $\phi$  and  $\eta$  alone, without the answer. That factor of three is a drift and not scatter, and the deposited grid settles it in a few lines: the sixteen crossing errors are ordered without a single exception, falling with  $L$  at every resolution—5.65, 3.50, 2.52 and  $1.60 \times 10^{-3}$  at  $\eta = 0.18t$  for  $L = 6\text{--}12$ , a factor 2.8–3.7—and rising with  $\eta$  at every size by 1.6–2.1, the product of the two being the observed 5.9. The drift runs the conservative way in the variable one would extrapolate: the larger the system, the smaller the true error at which the certificate first says anything, so reading the crossing as  $\approx 3 \times 10^{-3}$  beyond the sizes measured overstates the error rather than understating it. In  $\eta$ , which the user chooses and which lies inside the measured range, it does not. The objection to anticipate is that this is Fig. 9 relabelled. It is not, and the difference is the content: that figure is built knowing the exact spectrum and this quantity is computed without it. That the two track each other to within a factor 5.9 across five decades of error is not a redundancy—it is the calibration, and it is what makes the quantity usable where the second curve cannot be drawn.

Four limits come with it. The crossing is one point

per cell, so the tracker locates one error level and does not estimate the error away from it: at a *fixed* subspace the looseness of the bound against the measured error is constant to within  $\pm 17\%$  ( $L = 6$ ) and  $\pm 25\%$  ( $L = 8$ ) over a decade in  $\eta$ , but across subspaces it varies by a factor 51, and at the threshold itself it is  $3 \times 10^2$  to  $2.4 \times 10^3$ —the object is a certificate, not an error estimate, and calling it the latter would be refutable in one line. Only one of its two terms is a posteriori: the comparison is  $\hat{\Lambda}_S/\eta$  against  $\tau(w_S)$ , and  $\tau$  needs  $w_S$ , which a user does not have. That dependence can be discharged conservatively, and doing so is what makes the decision itself a posteriori. Equation (15) is decreasing in  $1 - w_S$  with  $\tau(1) = 1$ , so any assumed floor on the captured weight yields a usable threshold: at the  $L = 12$  published operating point  $w_S = 0.998622$  and  $\tau = 0.9625$ , and every subspace with  $w_S \geq 0.995$  has  $\tau \geq 0.9280$ . A user willing to assume only that the sampler captured 99.5% of the probe’s Born weight can therefore run the comparison as  $\hat{\Lambda}_S(\eta)/\eta < 0.928$  and use no other ground-state information, at the cost of at most 7% in the threshold. Where not even that floor can be assumed, or where  $w_S$  is estimated against an already-truncated  $\Psi_0$ —an estimate biased optimistically—the test

is not conservative and the bound is not a posteriori. And sixteen cells over four sizes and four resolutions is the whole of the evidence: no exponent is fitted to it and no extrapolation to  $L = 14$  is made, the local log–log slopes of  $\text{FR}^*(L)$  already varying by a factor 4.1 across the four sizes measured.

The fourth limit is the one a reader holding the deposited grid will test first, and it is a real restriction on what the crossing is worth. A sampler already holds the captured Born weight, so the question is whether a stopping rule indexed on  $1 - w_S$  alone locates the same error level, in which case the crossing would be that weight relabelled. It is not: over the sixteen cells,  $1 - w_S$  at the crossing ranges over a factor 45 against the true error's 5.9, and still over 9.4 against 3.8 once the extreme cell at each end is dropped. A rule with constants fitted to the grid, however, comes close enough. Take  $1 - w_S < c \eta^p$ , fit  $c$  and  $p$  on some of the four sizes with its operating point matched to the certificate's, evaluate it on the sizes held out, and repeat over all fourteen train/test splits: out of sample it localizes the crossing error about as tightly as the parameter-free certificate does, and more tightly on ten of the fourteen—a count that moves between six and twelve as the two constants are fitted differently, so that neither object dominates. We therefore claim no superiority for the certificate as a tracker of the error against the captured weight; out of sample it has none. The difference lies elsewhere. The fitted rule fires at  $1 - w_S \sim 10^{-5} - 10^{-4}$ , one to two orders of magnitude finer than the floor  $1 - w_S \leq 5 \times 10^{-3}$  at which the threshold above still works, and it asks that precision of the one quantity this section has already said a user does not have and, estimated against an already-truncated  $\Psi_0$ , has optimistically; strip its second parameter and an  $\eta$ -blind  $1 - w_S < c$  is the looser of the two on all fourteen splits. Theorem 1, for its part, is an inequality with a proof: it holds at a size no grid has seen, and its decision survives replacing  $w_S$  by a floor. The fitted rule is a regularity with two constants and no guarantee outside the range it was fitted on, which is the only place either would be used.

Within those limits the procedure is operational and identical at every size we measured: assemble  $\hat{\Lambda}_S(\eta)/\eta$  from the Ritz vectors already in hand and compare it with  $\tau(w_S)$ . Above  $\tau$  the certificate says nothing, which is the regime Fig. 9 operates in; at  $\tau$  the true relative error is  $\approx 3 \times 10^{-3}$  within a factor of three; below  $\tau$  it is smaller than that, by an amount the bound states loosely and the calibration does not sharpen. That is the whole content of the certificate, and it is a number a user can compute before knowing whether the reconstruction is right.

### E. Why the threshold cannot be lowered: a structural obstruction

Where does the  $\approx 0.45$  of sector fraction between what the reconstruction needs and what the certificate certifies go? We measured the displacement rung by rung along the

chain of inequalities that proves Theorem 1, at  $L = 8$ , and the answer removes the one repair that looked available: the pointwise step from a scalar distance to a distance in the Hilbert space accounts for 44% of it, the operator-norm step  $\|R(z)\| \leq 1/\eta$  applied to a source that depends on  $\omega$  for 38%, the Feshbach/Minkowski split for 11%, and the Cauchy–Schwarz step in  $\omega$  for 4.6% (medians over 15 of 18 cells; App. C). The working hypothesis that Cauchy–Schwarz is what places the threshold is therefore refuted by its own measurement: its total margin for tightening is a factor 1.05 typically and 3.29 at the worst point measured, which moves the threshold by a fraction of one grid interval.

The second measurement says that no rearrangement of the subspace helps either, and it is the one that turns the result into a statement about the problem.  $\Lambda_S = 0$  holds if and only if  $\text{span}(S)$  contains the Krylov space  $\mathcal{K}(H, \phi)$ , so the coordinate support of that Krylov space is a floor under any coordinate subspace of vanishing leakage. We measured it directly: 296 of 300 determinants at  $L = 6$ , where  $\dim \mathcal{K} = 65$ , and 3920 of 3920—the entire sector—at  $L = 8$ , where  $\dim \mathcal{K} = 596$ –663. (It is the *coordinate support* of  $\mathcal{K}$  that fills the sector, not  $\mathcal{K}$  itself, whose dimension is three orders of magnitude smaller; containment, not spanning, is what  $\Lambda_S = 0$  requires.) At  $L = 10$  the same construction—a Lanczos recursion on the whole sector to depth 700 with full reorthogonalization, orthogonality defect  $1.6 \times 10^{-15}$ —leaves not one of the 52 920 rows identically zero, so there the support is the entire sector at a threshold of exactly 0.

Beyond  $L = 10$  that basis cannot be stored, and we use a lower bound instead: every  $H^k \phi$  lies in  $\mathcal{K}$ , so the union over  $k$  of the coordinate support of  $H^k \phi / \|H^k \phi\|$  is contained in the support of  $\mathcal{K}$ . A count of this kind is meaningless without its threshold, so both are given. At an amplitude threshold of  $10^{-12}$  and  $k \leq 30$  the union is the entire sector at  $L = 8, 10, 12$  and  $14$ : 3920, 52 920, 731 808 and 10 306 296 determinants. At the stricter  $10^{-10}$  and the same 30 powers it returns 3919 of 3920, 52 919 of 52 920, 731 807 of 731 808 and 10 306 063 of 10 306 296—and 52 920 of 52 920 at  $L = 10$  once the recursion is allowed 36 powers. At  $L = 6$  it returns 296 of 300, reproducing the row count above; the four determinants at issue carry amplitudes between  $10^{-14}$  and  $10^{-10}$ , which makes  $L = 6$  the one size whose answer depends on where the threshold is put. The two estimators are independent and they agree, to one determinant in 3920 and exactly in 52 920, wherever both were run.

From  $L = 8$  on the leakage is therefore strictly positive for *every* proper coordinate subspace, and no certificate of leakage type, in this paper or another, can be non-vacuous there at low fraction in the determinant basis—at every size this work reports, the largest being the  $L = 14$  sector at which Fig. 9 reports a published fraction of 0.08. That is a property of the basis in which the device measures, not of the bound and not of the ranking. Nor is the ranking the weak link: replacing the Born ordering by a Krylov-diagonal oracle—determinants ordered by the diagonal

of the projector onto  $\mathcal{K}(H, \phi)$ , an ordering no sampler could produce—gives a uniformly *worse* certificate at  $L = 6$ ,  $\eta = 0.18t$ , 12.97 against 4.27 at fraction 0.85 and 5.82 against 0.911 at 0.95. The ordering the device realizes by measuring is the better of the two at every fraction we checked, which is the opposite of what an oracle comparison is usually for.

Together the two measurements say what the threshold is made of, and neither component is a frontier of the method. At fixed size the displacement is the looseness of the proof, only a twentieth of it in the step that looked easiest to tighten; the position of the threshold near unity is geometry, the Krylov space of the probe filling the sector in the basis where a leakage certificate is by construction a statement about what a subspace fails to contain. The reconstructions themselves are untouched by either—the control column of Table IV reproduces Fig. 9 to 7% at worst and to under 2% against the scan’s own fractions, with nothing shared—and what is new, bounded and measured is the crossing of Sec. VD. Section VI closes the picture from the other side: nothing cheaper than that crossing—no invariant of the state, computable before the subspace exists—can be put in its place.

One route out of the determinant basis has been measured, and it closes this section as an outlook and not as a result, because the same measurement shows where it stops. The qualifier *in the determinant basis* above is load-bearing, and it is measurable. An orbital rotation is a Givens network and the circuit of Sec. II already contains them, so the measurement basis is a choice; for the momentum-resolved probe of Fig. 3 it is a forced one, since  $H$  commutes with lattice translation and  $\phi_k$  is an eigenvector of it, so that  $\mathcal{K}(H, \phi_k)$  lies inside a single momentum block—rigorously, at every  $L$ —and that block is exactly  $\dim(N+1)/L$  determinants. At  $L = 6$  and  $L = 8$ , the two sizes we measured, the Krylov support fills 96–100% of it and the certificate becomes non-vacuous at 48–54 of 300 determinants and at 490 of 3920—a factor 5–7 fewer than the determinant basis requires and, at matched accuracy, 10–66 fewer shots at  $L = 8$ —while on the block itself  $\Lambda_S$  vanishes identically and the bound is zero rather than vacuous. The gain, though, is  $1/L$  against an exponential: as a fraction of the fraction Fig. 9 publishes the block is 0.20 at  $L = 6$  and then 0.22, 0.28, 0.49 and 0.89 at  $L = 14$ , the two crossing just past  $L = 14$ . It is also a laptop Lanczos—60 984 determinants at  $L = 12$ , 736 164 at  $L = 14$ —so the regime it certifies is one that momentum-resolved exact diagonalization has enumerated since the 1970s. Three things are unmeasured here and would have to be measured before more is said: whether the support

fills the block at  $L \geq 10$ , where the threshold then sits, and what the  $\approx L(L-1)$  extra two-qubit gates do to the sampled distribution.

## VI. NO SHORTCUT TO THE BOUNDARY: AN OBSTRUCTION AND ITS WITNESS

Section III leaves one question open: can the a posteriori certificate be replaced by an a priori predictor computed from the state?  $\Lambda_S(\eta)/\eta$  is cheap, but it is cheap *after* the subspace exists; a quantity read off  $|\Psi_0\rangle$  before any circuit runs would be worth more. The natural candidates are the resource monotones of fermionic non-Gaussianity, efficiently measurable from two-point functions, which a large recent literature has put in exactly that role. None of them can play it, for either term of Eq. (7), and the two statements are about two different objects, which we separate here once and keep separate throughout. The first is about the determinant support the shots must populate: a *static* count on the ground state, written  $|\mathcal{S}|_{\varepsilon_w}$  below, and the quantity the cost analysis of this paper has always been about. The second is about the leakage  $\Lambda_S(\eta)/\eta$ , and its  $S$  is not that count at all—it is the coordinate subspace of Theorem 1, the one the sampler returned. Only the second is an obstruction on the theorem’s own variable, and it is the one that makes this section inseparable from Sec. III:  $\Lambda_S(\eta)$  is defined by the resolvent at resolution  $\eta$  and does not exist outside the spectral half of this work. Remove Sec. III and Proposition 1 loses its subject; remove this section and the question above stays unanswered. What replaces the predictor is not a cheaper functional but the measured map of Sec. V.

The candidates have a sharp algebraic origin. Matchgate (free-fermion) circuits are classically simulable [47–50], and what leaves that manifold is fermionic non-Gaussianity: any pure non-Gaussian state injected into a matchgate circuit promotes it to universality [51], the fermionic counterpart of qubit magic [52, 53]. A ladder of measures quantifies it—Gaussian rank and extent [58], non-Gaussian gate counts [56, 57], including the count that makes a state efficiently learnable [63], covariance-matrix monotones [59, 61], and convolution- and entropy-based measures built on the violation of Wick’s theorem [93].

### A. The collapse: one-body magic is a single 1-RDM invariant

We work with the fermionic AntiFlatness (FAF) [54],

$$\mathcal{F}_k = 2n_o - \frac{1}{2} \text{tr}[(\Gamma^\top \Gamma)^k] = 2n_o - \sum_{j=1}^{2n_o} \lambda_j^{2k}, \quad \Gamma_{mn} = -\frac{i}{2} \langle [\gamma_m, \gamma_n] \rangle, \quad (16)$$

with  $\Gamma$  the  $4n_o \times 4n_o$  Majorana covariance matrix,  $\{\lambda_j\} \in$

$[0, 1]$  the moduli of its eigenvalue pairs  $\pm i\lambda_j$ , and  $2n_o$  the



number of spin-orbital *modes* (twice the spatial-orbital count), distinct from the electron number of the  $(N \pm 1)$  sectors. Each  $\mathcal{F}_k$  vanishes if and only if the state is Gaussian and is invariant under Gaussian (free-fermion) unitaries by construction [54]; the index  $k$  is a hierarchy of correlator orders, efficiently measurable from two-point functions [54, 59, 62].

$$\mathcal{F}_1 = \sum_i [1 - (2g_i - 1)^2] = 4 \sum_i g_i(1 - g_i) = 4 \text{tr}[\gamma(1 - \gamma)], \quad (17)$$

exact on any number-conserving state and verified state by state to  $10^{-13}$  across the dissociation of  $\text{N}_2$  and the suite of Table XIV, that is on the 38 molecular ground states of this work. For spin-restricted natural orbitals ( $g_{a\uparrow} = g_{a\downarrow} = n_a/2$ ) it equals  $2N_u$ , the quadratic effectively-unpaired-electron index  $N_u = \sum_a n_a(2 - n_a)$  of Takatsuka–Fueno–Yamaguchi and of Staroverov–Davidson [89, 90], and  $\text{tr}[\gamma(1 - \gamma)]$  is the Gigena–Rossignoli one-body-entanglement linear entropy [92].<sup>1</sup>

We claim no novelty for the algebra. Equation (17) is an immediate number-conserving specialization of the antiflatness of Ref. [54];  $N_u$  has been the standard diagnostic of unpaired density since 1978 [89]; and the ensemble form was noted independently and contemporaneously, for interacting-integrable eigenstates, by Świątek *et al.* [55]. What we add is the exact, state-by-state *dictionary*: three literatures—fermionic magic, effectively-unpaired electrons, one-body entanglement—name one quantity. The verification is the 38 molecular states just quoted; the further equality to  $2N_u$  is checked on the closed-shell members, the five spin-polarized ones being excluded for the reason given in the footnote. The 36 lattice ground states of this work are number conserving, so Eq. (17) applies to them by the same algebra, but they were not part of that state-by-state check and we do not count them in it. The consequence we carry forward is structural and equally exact: being a function of the occupation spectrum alone,  $\mathcal{F}_1$  inherits the orbital-rotation invariance of the eigenvalues of  $\gamma$ . It is a genuine, cheap,

On the number-conserving states of electronic structure the anomalous correlators  $\langle c_i c_j \rangle$  vanish, so  $\Gamma$  is fixed by the one-particle reduced density matrix  $\gamma$  and its singular values are  $\lambda_i = |2g_i - 1|$ , with  $g_i \in [0, 1]$  the natural spin-orbital occupations. The  $k = 1$  member then reduces to a single 1-RDM invariant,

exactly computed invariant of the state, and it is that invariance the two following subsections use.

### B. The geminal witness, and what an invariant cannot certify

The obstruction is not a continuity artifact. For the *raw* determinant support—the unthresholded Slater rank—no continuous functional of a fixed set of reduced density matrices can lower-bound the count at all, rank being coefficient-insensitive and not lower-semicontinuous, like the CP tensor rank [94]; that argument (App. E) does *not* transfer to the operative, coefficient-*sensitive*  $|\mathcal{S}|_{\varepsilon_u}$ , which collapses to 1 along the same path as the functionals. What obstructs the operative quantities is a symmetry, exhibited by a family whose amplitudes are  $\Theta(1)$  and therefore threshold-robust: an antisymmetrized product of  $K$  strongly orthogonal singlet geminals (perfect-pairing / generalized-valence-bond),

$$|\Psi_K\rangle = \bigotimes_{i=1}^K \left( \cos \theta \hat{b}_{i\uparrow}^\dagger \hat{b}_{i\downarrow}^\dagger + \sin \theta \hat{a}_{i\uparrow}^\dagger \hat{a}_{i\downarrow}^\dagger \right) |0\rangle, \quad (18)$$

on  $2K$  spatial orbitals ( $4K$  modes,  $N = 2K$  electrons), each geminal occupying a disjoint bonding/antibonding pair  $(b_i, a_i)$ .

**Theorem 3** (Geminal orbit witness). *At maximal pairing  $\theta = \pi/4$  the family (18) satisfies, exactly,*

$$\mathcal{F}_1 = 2N_u = 4K, \quad \mathcal{F}_2 = 4K, \quad |\mathcal{S}|_{\text{pair}} = 2^K, \quad \chi = 2, \quad (19)$$

with  $\mathcal{F}_{1,2}$  the order-1 and order-2, antiflatness of Eq. (16),  $|\mathcal{S}|_{\text{pair}}$  the determinant support in the pairing basis, and

$\chi$  the minimal matrix-product-state bond dimension. One-body magic, higher-order magic and determinant support

<sup>1</sup>  $2N_u$  is the quadratic index, distinct from the linear  $\sum_a \min(n_a, 2 - n_a)$  of Head-Gordon [91]. A cleanly singly-occupied spin orbital contributes 0 to  $\mathcal{F}_1$  but 2 to  $2N_u$ , so on the five spin-polarized members of Table XIV ( $\text{O}_2$ ,  $\text{NO}$ ,  $\text{CN}$ ,  $\text{OH}$ ,  $\text{BeH}$ ) the two differ and we report  $\mathcal{F}_1 = 4 \text{tr}[\gamma(1 - \gamma)]$  directly. Outside

number conservation the collapse fails: a paired Gaussian state hides fractional occupations in anomalous correlators and keeps  $\mathcal{F}_1 = 0$ .

therefore all diverge with  $K$  while, in the localized (pairing) basis, the state remains a bond-dimension-2 tensor network—representable, contractible and Born-samplable in  $O(K)$ : magic of every order is unbounded at fixed  $O(1)$  simulation cost.

The proof is four short block-diagonal computations (App. D); the step worth flagging here is that at  $\theta = \pi/4$  a single maximally-paired geminal is the two-determinant cat, whose Majorana covariance *vanishes identically*, so every order of antiflatness saturates its maximum at once. All four values are reproduced by exact diagonalization for  $K = 1, \dots, 4$ .

*The orbit is the argument.* The witness is not the trivial “a product state is easy” (Fig. 5). Because  $\mathcal{F}_1$  and  $\mathcal{F}_2$  are invariant under number-conserving orbital rotations [54], the whole orbit  $\{U|\Psi_K\rangle\}$  carries the same  $(\mathcal{F}_1, \mathcal{F}_2) = (4K, 4K)$  while the cost in the fixed computational basis—bond dimension and Born-sampling time—runs from  $O(K)$  at the localized representative up to exponential. Two conclusions follow, each in the only direction the construction supports: a Gaussian-invariant fermionic magic is not a sufficient certificate of hardness and does not lower-bound the cost (there  $\chi = 2$  while  $\mathcal{F}_k = 4K \rightarrow \infty$ ); and  $\mathcal{F}_k$  does not determine the cost, being constant on an orbit along which the cost varies exponentially. In a generic rotated basis the Born sampling of  $|\Psi_K\rangle$  is plausibly classically hard—the free-fermionic-linear-optics-with-magic-input regime of the Fermion Sampling advantage scheme [100]—while  $\mathcal{F}_k$  is unchanged: any hardness that appears belongs to the sampler’s fixed frame. We are silent on whether magic is *necessary* for hardness, and we anchor the easy end to an external result rather than our own, since for this paired class the additive-error estimation of amplitudes, overlaps and number correlators under free-fermionic dynamics is classically efficient [60].

*The same orbit reaches the leakage.* The orbit argument was written for  $|\mathcal{S}|$ ; it extends to the second budget of Eq. (7), with its statement corrected. An invariant upper bound on  $\Lambda_S(\eta)/\eta$  *does* exist—it is exhibited inside the proposition below, and we tested it on 1600 hostile cases (GOE/GUE, near-degenerate, banded,  $\|H\| \sim 300$ ) with zero violations and with its first link attained, minimum ratio 1.0000000000. The correct statement is therefore one of non-vacuity, not of non-existence: the same logical form the  $|\mathcal{S}|$  no-go always had, where the sector dimension is likewise an invariant upper bound that says nothing.

**Proposition 1** (Fibers on which every orbital-rotation invariant gives a vacuous leakage certificate). *Let  $\mathcal{U}_U$  be the Gaussian unitary implementing  $U \in \mathbf{U}(n_o)$ , and let the same physical problem written in the rotated orbital basis be  $(H_U, \Psi_U, \phi_U) = (\mathcal{U}_U^\dagger H \mathcal{U}_U, \mathcal{U}_U^\dagger \Psi, \mathcal{U}_U^\dagger \phi)$ , the determinant basis—in which the device measures, and of which  $S$  is a coordinate subspace—held fixed. Along this orbit the spectrum of  $H$ ,  $E_0$ ,  $\|\phi\|^2$ , every Lehmann weight and hence  $A(\omega)$  pointwise are constant, as is every orbital-rotation invariant of the state, in particular  $\mathcal{F}_1 = 4 \text{tr}[\gamma(1 - \gamma)]$  and all  $\mathcal{F}_k$ .  $\Lambda_S(\eta)/\eta$  is*

*not. An upper bound on  $\Lambda_S(\eta)/\eta$  is available from invariants alone—combining  $\Lambda_S \leq \|P_S H Q_S\| \|\phi_S\|$  with  $\|Q_S H P_S\| = \|Q_S(H - c)P_S\| \leq \min_c \|H - c\|$  gives  $\Lambda_S(\eta)/\eta \leq (W/2)\|\phi\|\sqrt{w_S}/\eta$ ,  $W$  the spectral width of the  $(N \pm 1)$  sector—but it is vacuous by a factor of at least  $W/2\eta$ , which grows extensively with system size. That this is the best such bound can be seen at fixed invariants: any  $B(\text{invariants}, |\mathcal{S}|, w_S)$  valid for  $\Lambda_S(\eta)/\eta$  is constant on a fiber of fixed arguments and must therefore exceed the fiber maximum. In the geminal family APSG( $\pi/4$ ) on  $2K$  orbitals with  $H = \varepsilon \hat{N} - g \sum_i A_i^\dagger A_i + \delta h$  we exhibit fibers ( $K = 2, 3, 4$ ;  $\delta = 0.01$ ;  $\eta = 0.10$ ;  $|\mathcal{S}| = 2^{K-1}$ ; spectrum, Lehmann weights and all  $\mathcal{F}_k$  matched to  $10^{-14}$ ,  $w_S$  to  $2.2 \times 10^{-16}$ ) whose endpoints carry  $\hat{\Lambda}_S(\eta)/\eta = 0.237$  and  $2.326, 0.323$  and  $2.336, 0.391$  and  $2.346$ . Since the larger value exceeds  $\tau(w_S) = (1 + w_S)/(1 + \sqrt{w_S}) - \sqrt{1 - w_S} = 0.795$ , the resolution above which the leakage branch of Theorem 1 no longer improves on the trivial bound  $\|\phi\|^2(1 + w_S)$ , every valid invariant bound is vacuous on these fibers—while the subspace-aware certificate is non-vacuous at the aligned endpoint (0.856 against a trivial 1.962). The statement is one of non-determination and non-vacuity, not of non-existence: invariants do not fail to bound the leakage, they fail to bound it usefully.*

Table VII is the exhibit. The two endpoints of each fiber are the *same* rotation angle applied within one geminal block and across two blocks, which is why the captured weight agrees bit for bit. An exhaustive search over all 276 coordinate subspaces of that size at  $K = 2$  settles both ends. At the aligned endpoint the global minimum of the certificate over the 276 is 0.856 and it is attained by the protocol’s own subspace: the subspace the sampler would return is the best available one. At the rotated endpoint the minimum over the same 276 is exactly the trivial bound, attained only by subspaces of vanishing captured weight: *no* determinant subspace of that size certifies anything there.

*What this exhibit is not.* Four limits are declared with it, and they are why we call it a fiber and not a sweep of the orbit. (i) The evidence is a matched pair at  $s = \pi/16$  for  $K = 2, 3, 4$ . At the maximal-pairing angle  $s = \pi/4$  the Born distribution of  $\phi$  is exactly flat over  $2^K$  determinants, so “the top  $|\mathcal{S}|$  of the Born distribution” does not define a subspace at all (6, 70 and 12870 equally valid outputs at  $K = 2, 3, 4$ ); any contrast quoted at that angle is a tie-breaking artifact and not a property of the fiber, and the sweep of App. D stops at  $s = 0.654$  for that reason. (ii) Nothing here grows with the system, and two separate quantities shrink. The ratio of the two leakages falls,  $9.80 \rightarrow 7.23 \rightarrow 6.01$  from  $K = 2$  to  $K = 4$ , and it dies as the perturbation is turned up, from 28.7 at  $\delta = 0.0025$  to 1.03 at  $\delta = 0.3$ . More to the point—because it is the quantity the proposition actually rests on—so does the margin of the certificate over the trivial bound:  $1.962/0.856 = 2.29$ ,  $1.962/1.026 = 1.91$ ,  $1.962/1.161 = 1.69$ , a fall of 26% across the only three sizes available. Where that fall would reach unity we do

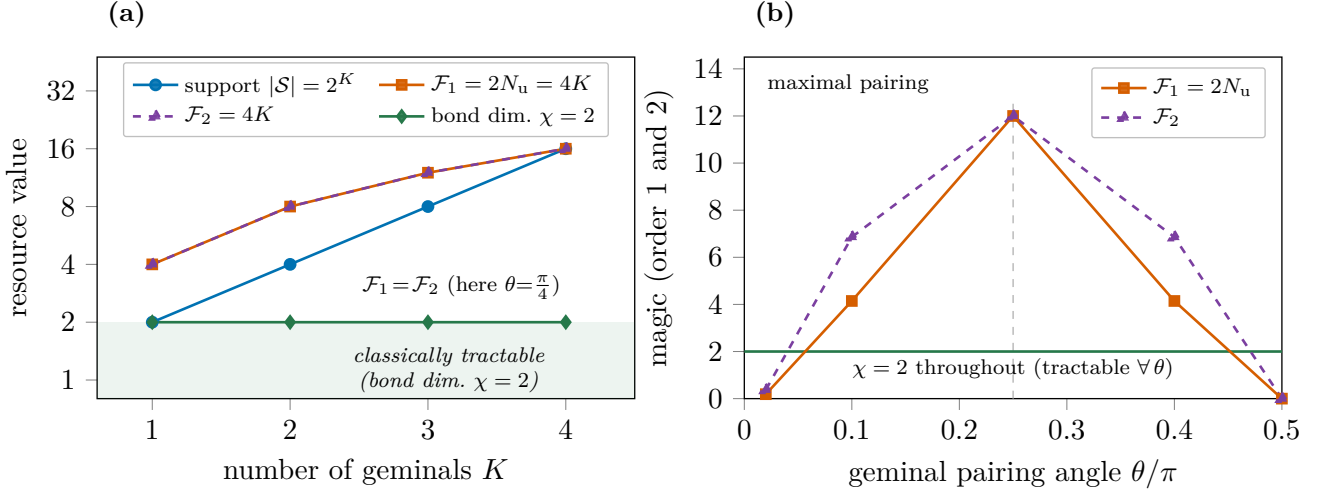


FIG. 5. **The geminal orbit witness (Theorem 3).** Antisymmetrized products of strongly orthogonal geminals in their localized (pairing) basis, where  $\chi = 2$  and the state is a classically tractable tensor network. **(a)** As  $K$  grows,  $\mathcal{F}_1 = 2N_u = 4K$ ,  $\mathcal{F}_2 = 4K$  and  $|\mathcal{S}| = 2^K$  all diverge while the bond dimension never exceeds 2. **(b)** At  $K = 3$ , sweeping the pairing angle drives both  $\mathcal{F}_1$  and  $\mathcal{F}_2$  to their maximum at  $\theta = \pi/4$ —where the covariance vanishes and all orders saturate—with  $\mathcal{F}_2 > \mathcal{F}_1$  off the peak and  $\chi = 2$  throughout. Because  $\mathcal{F}_k$  is orbital-rotation invariant the same values persist on the whole orbit, along which the cost runs from  $O(K)$  to exponential.

TABLE VII. **Fibers of fixed invariants with moving leakage**, at  $s = \pi/16$ ,  $\delta = 0.01$ ,  $\eta = 0.10$ ,  $|\mathcal{S}| = 2^{K-1}$ ,  $w_S = 0.961940$ ,  $\tau(w_S) = 0.795$ . Along each row the spectrum, every Lehmann weight,  $\|\phi\|^2$ ,  $|\mathcal{S}|$  and all  $\mathcal{F}_k$  are identical between the two endpoints ( $w_S$  to  $2.2 \times 10^{-16}$ ), so the first term of Eq. (7) is fixed and only the second moves. Quantities are relative ( $\hat{\Lambda}_S = \Lambda_S/\|\phi\|$ ) and “certificate” is  $\min\{(1 + \sqrt{w_S})(\sqrt{1 - w_S} + \hat{\Lambda}_S/\eta), 1 + w_S\}$ , the relative form of Theorem 1; “al.” and “rot.” label the aligned and rotated endpoints of the fiber. A continuous angle sweep is in App. D.

| $K$ (dim( $N+1$ )) | $\mathcal{F}_1=\mathcal{F}_2=\mathcal{F}_3$ | $\hat{\Lambda}/\eta$ al. | $\hat{\Lambda}/\eta$ rot. | cert. al.    | cert. rot.      |
|--------------------|---------------------------------------------|--------------------------|---------------------------|--------------|-----------------|
| 2 (24)             | 8                                           | 0.237                    | 2.326                     | <b>0.856</b> | 1.962 (trivial) |
| 3 (300)            | 12                                          | 0.323                    | 2.336                     | <b>1.026</b> | 1.962 (trivial) |
| 4 (3920)           | 16                                          | 0.391                    | 2.346                     | <b>1.161</b> | 1.962 (trivial) |

not say: a linear extrapolation in  $K$ , a log-linear one and the last step alone place it between  $K = 6$  and  $K = 8$ , the decrements are themselves decelerating, and three points do not support a number—which is exactly the objection this paper makes to its own fraction headline in Sec. VII. A statement of non-vacuity needs one fiber and not a magnitude, and the fiber at  $K = 4$  is still non-vacuous by 69%; but “the effect grows with  $K$ ” is false here, and the trend is the one a referee would find first, so we state it ourselves. (iii) The small-leakage endpoint is realized only by the aligned basis, which is the stabilizer of the state ( $|\langle \Psi(0) | \Psi(U_{\text{in}}) \rangle| = 1.000000000000$ ); at the fiber angle  $s = \pi/16$ , every generator that actually moves the state gives  $\hat{\Lambda}/\eta \in [2.34, 5.61]$ . That range is a statement about generators at one angle and not about the angle sweep of App. D, whose rotated endpoint descends to 1.61 as  $s \rightarrow 0$  because the rotation itself is being switched off. The rhetorical parallel with the  $|\mathcal{S}|$  witness—where the state genuinely runs from  $O(K)$  to  $2^K$ —is therefore lost, and Sec. VID states that asymmetry in the form a referee

would. (iv) The scale is a toy:  $2K \leq 8$  orbitals, sector dimension  $\leq 3920$ , no instance at the  $L = 12$  Hubbard scale of Sec. V.

What the exhibit delivers survives all four limits and is measured: on fibers where every invariant is fixed to  $10^{-14}$  and the captured Born weight to  $2.2 \times 10^{-16}$ , the certificate of Theorem 1 separates a non-vacuous value from the trivial bound in every row of Table VII, and a quantity constant on those fibers cannot make that distinction. The a priori predictor asked for at the head of this section does not exist in the invariant class; what stands in its place is the subspace-aware certificate, whose non-vacuous region is mapped, not asserted, in Sec. V.

### C. The obstruction is bilateral, and its sign is not stable

The same symmetry closes the cost side, in both directions (Fig. 6). The cost statistic of this section is the *static* determinant support  $|\mathcal{S}|_{\varepsilon_w}$ : the number of fixed-

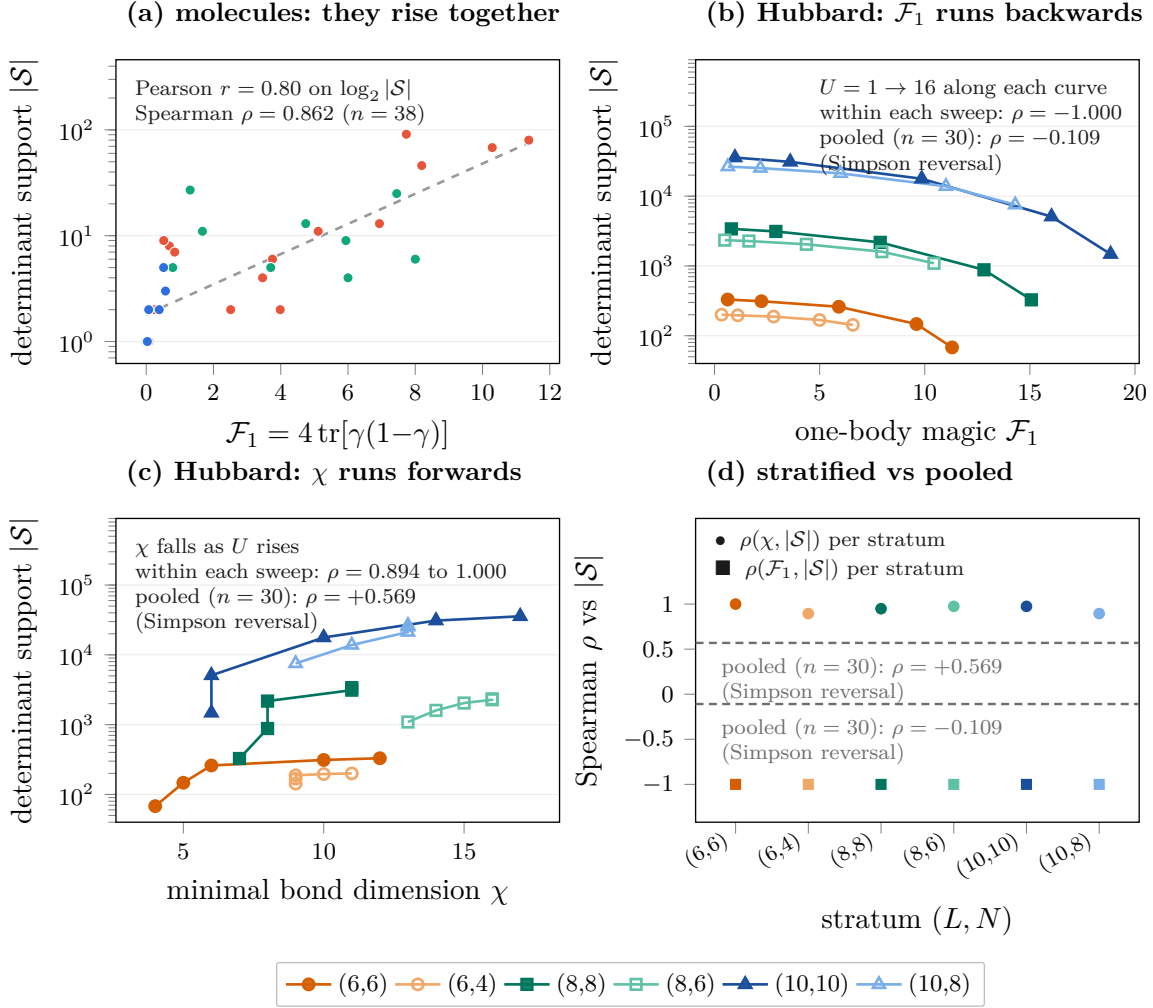
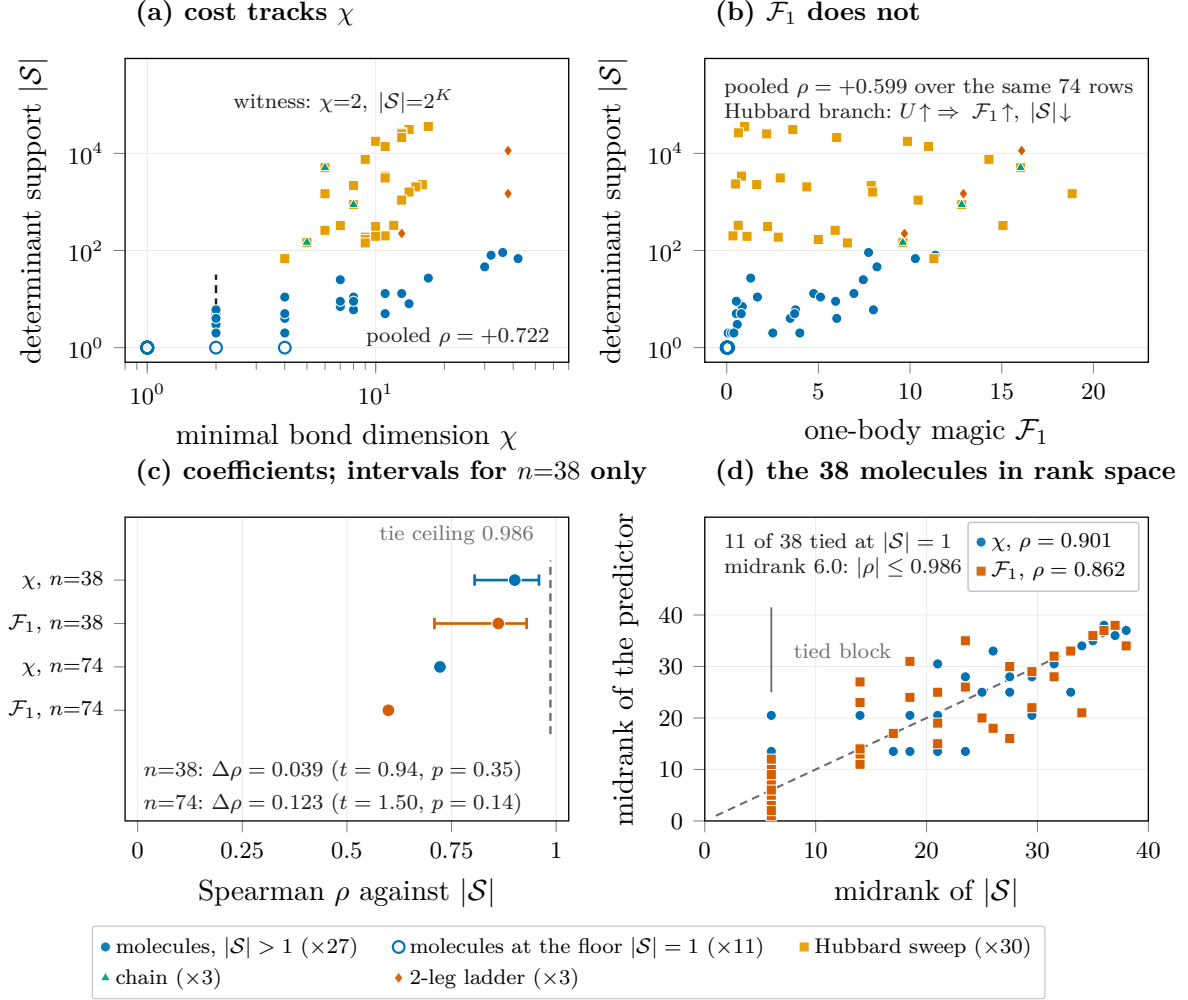


FIG. 6. **Inside every stratum one-body magic  $\mathcal{F}_1$  orders the static determinant support  $|\mathcal{S}|$  perfectly and with the sign inverted; what fails is the *pooled* coefficient, not  $\mathcal{F}_1$ .** (a) Nineteen molecular active spaces at equilibrium and at dissociation (38 points): in the static-correlation regime  $\mathcal{F}_1$  and  $|\mathcal{S}|$  rise together (Pearson  $r = 0.80$  on  $\log_2 |\mathcal{S}|$ ; Spearman  $\rho = 0.862$ ). The grey dashed line is the least-squares fit  $\log_2 |\mathcal{S}| = 0.474 \mathcal{F}_1 + 0.850$  over those same points, a guide to that correlation and not a model of the cost. (b) The Hubbard interaction sweep is not one cloud of 30 points but six strata  $(L, N)$  of five on-site repulsions  $U = 1, 2, 4, 8, 16$ ; one curve per stratum. Inside every one of the six, raising  $U$  raises  $\mathcal{F}_1$  and lowers  $|\mathcal{S}|$ , and does so with a perfect rank reversal:  $\rho(\mathcal{F}_1, |\mathcal{S}|) = -1.000$  in all six. Pooling across strata collapses the coefficient to  $\rho = -0.109$ ; that collapse is a Simpson reversal, not an absence of signal. (c) The same six strata against the minimal bond dimension:  $\rho(\chi, |\mathcal{S}|)$  runs from  $+0.894$  to  $+1.000$  within strata, with a mean of  $0.948$ , and does not change sign between them. Its pooled coefficient, like  $\mathcal{F}_1$ 's, is a Simpson reversal and is not used: pooling six deterministic five-point sweeps destroys within-stratum structure whichever way the result falls, and it fell in our favour on one of the two. (d) The six within-stratum coefficients against the two pooled ones. Stratified, the  $\mathcal{F}_1$  evidence is one-sided and exact: Kendall  $\tau = -1.000$  with 0 concordant of 60 within-stratum pairs and no ties, one-sided exact permutation  $p = 3.3 \times 10^{-13}$  over the  $(5!)^6 = 2.99 \times 10^{12}$  within-stratum relabellings. The reading “ $\rho = -0.11$  ( $n = 30$ ,  $p \approx 0.56$ , not significant)” is the pooled number and should be replaced by the stratified test. Two limits of this evidence, which the stratified test does not remove. First, the five points of a stratum are a deterministic monotone sweep in  $U$ , so the effective number of independent observations is 6, not 30; the exact  $p$  counts relabellings, not independent samples. Second, this is a statement about rank direction only: inside a fixed sweep  $\mathcal{F}_1$  orders the five costs *perfectly*, so the claim is not that  $\mathcal{F}_1$  is uninformative about  $|\mathcal{S}|$  but that its sign is not stable—inverted here, positive over the molecular set of panel (a)—which is exactly what an orbital invariant compared against a basis-dependent cost must do.

basis Slater determinants of the *ground state* carrying all but  $\varepsilon_w$  of its weight. Three tolerances called 0.05 or 0.01 appear in this paper and they are three different objects, so we subscript all of them.  $\varepsilon_w$  cuts the support:  $\varepsilon_w = 10^{-2}$  for the molecular suite (`n19_suite.py`, 99%

of the weight) and  $\varepsilon_w = 2.5 \times 10^{-3}$  for the lattice families (`cost_vs_entanglement.py`, 99.75%, written there as  $1 - \varepsilon^2$  with  $\varepsilon = 0.05$ ).  $\varepsilon_\chi = 0.05$  cuts the Schmidt spectrum for  $\chi$ , by discarded tail  $\leq \varepsilon_\chi^2$ , in both families. Neither is the third 0.05 of this work, which is a thresh-





**FIG. 7. The resource thesis over the 74 deposited rows (71 distinct exact ground states).** Nineteen molecules at equilibrium and dissociation, a 30-point Hubbard interaction sweep, and a matched chain/two-leg-ladder pair; all by exact diagonalization. **(a)** The determinant support tracks the minimal bond dimension in every family (pooled Spearman  $\rho = +0.722$  over all 74 rows,  $+0.746$  over the 71 distinct states—see *Provenance* below), with the geminal witness  $\chi=2$ ,  $|\mathcal{S}|=2^K$  the deliberately loose lower bound. Open circles are the 11 of 38 molecular points pinned at the floor  $|\mathcal{S}| = 1$ —29% of the molecular set—where the support cannot resolve anything and from which both molecular coefficients draw much of their leverage. **(b)** One-body magic does not ( $\rho = +0.599$  over the same rows,  $+0.592$  over the 71 distinct states): the Hubbard branch runs the wrong way ( $U \uparrow$  raises  $\mathcal{F}_1$  and lowers  $|\mathcal{S}|$ ) and the chain/ladder pair sits at  $\mathcal{F}_1$  values agreeing to within 0.9% while  $|\mathcal{S}|$  differs by 1.52, 1.68 and  $2.24\times$ . **(c)** The two coefficients with the intervals the deposited data actually support. Bars are 95% cluster bootstrap intervals over the 19 molecules ( $B = 20\,000$ , cluster = molecule):  $\chi$  gives 0.901 [0.805, 0.959],  $\mathcal{F}_1$  gives 0.862 [0.709, 0.929]. *The molecular comparison does not resolve the two resources:*  $\Delta\rho = 0.039$ , Williams  $t = 0.94$  on 35 df,  $p = 0.35$  (Steiger  $z = 0.94$ ), bootstrap 95% CI of the difference  $[-0.070, +0.189]$ ; neither does the pooled comparison ( $\Delta\rho = 0.123$ ,  $t = 1.50$ ,  $p = 0.14$ ). The  $n = 74$  entries are point estimates—only the difference was resampled. The dashed line is the ceiling  $|\rho| \leq 0.986$  imposed by the ties in  $|\mathcal{S}|$ , against which  $\chi$  reaches 91% and  $\mathcal{F}_1$  87% of what is attainable. What discriminates between the two candidate resources is therefore not this figure but the stratified Hubbard sweep of Fig. 6(b,d). **(d)** The same 38 molecular points in the rank space in which a Spearman coefficient is actually computed, against the identity line. The 11 floor points share midrank 6.0—the vertical block at the left—which is the origin of the ceiling quoted in (c). *Provenance and caveats.* The three “chain” rows are the same states as the  $U = 8$  half-filled entries of the Hubbard sweep (identical  $|\mathcal{S}|$  and  $\chi$ , identical  $\mathcal{F}_1$  to the precision at which they are stored), re-plotted as the ladder’s matched partner; the 74 rows are therefore 71 distinct states and three of them enter the pooled coefficients twice. Recomputed over the 71 distinct states the pooled coefficients are  $\rho(\chi, |\mathcal{S}|) = +0.746$  and  $\rho(\mathcal{F}_1, |\mathcal{S}|) = +0.592$ . The 38 molecular points are 19 molecules  $\times$  2 geometries, i.e. dependent pairs (effective  $n \approx 31.3$ ), which is why every molecular interval here is clustered by molecule.

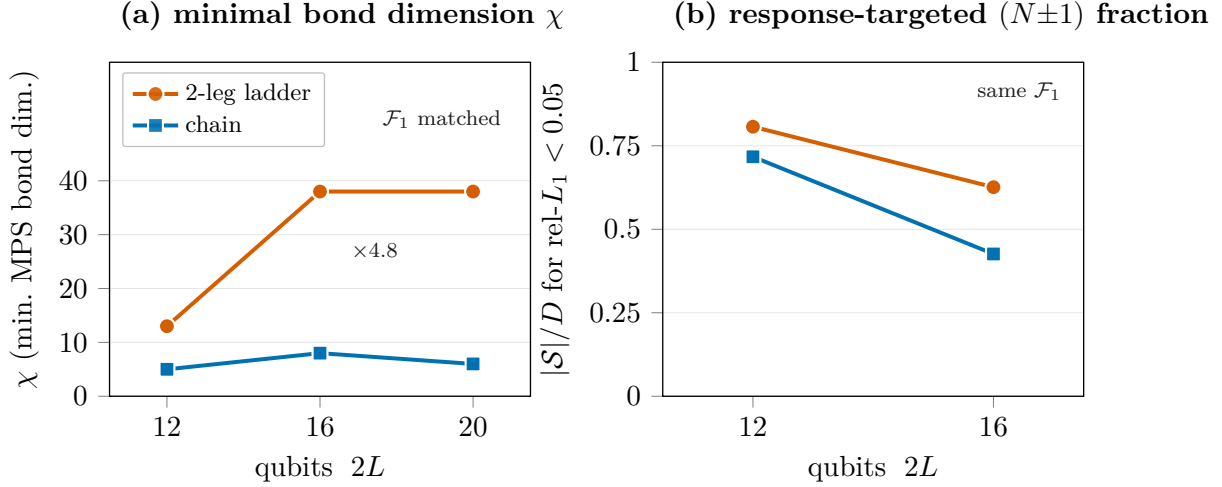


FIG. 8. **The cost is tracked by entanglement, not one-body magic—a controlled chain-versus-ladder contrast.** A two-leg Hubbard ladder versus a chain of the *same* site count (half filling,  $U/t = 8$ ; exact diagonalization). At matched fermionic magic  $\mathcal{F}_1$  (open chain vs. ladder: 9.60/9.69, 12.82/12.91, 16.03/16.09 at 12/16/20 qubits— $\mathcal{F}_1$  is a geometry-blind one-body invariant) the ladder’s perpendicular cut has an area-law boundary of two rather than one. **(a)** Its minimal matrix-product-state bond dimension  $\chi$  is far larger; the chain-to-ladder ratio grows over  $12 \rightarrow 20$  qubits ( $\times 2.6$ ,  $\times 4.8$ ,  $\times 6.3$ ), the ladder  $\chi$  saturating at 38 while the chain  $\chi$  stays small and does *not* grow (5, 8, 6 at 12/16/20 qubits—non-monotone, with no systematic increase over this range, consistent with the at most logarithmic entanglement growth of the spin sector of the one-dimensional chain)—the reported  $\chi$  are threshold-truncated Schmidt ranks. **(b)** The response-targeted  $(N\pm 1)$  fraction needed for spectral relative- $L_1 < 0.05$  is correspondingly higher for the ladder (these union  $(N\pm 1)$  response fractions differ from the single-sector  $(N+1)$   $A(\omega)$  fractions of Fig. 9, a different observable). Same one-body magic, higher cost: the determinant support is lower-bounded in the raw count and empirically tracked by the basis-dependent bond dimension  $\chi$  (Sec. VIE), not by  $\mathcal{F}_1$ . A ladder of this width ( $\chi \approx 38$ ) remains classically near-exact for dynamical DMRG; it is a resource contrast and no claim of advantage is attached to it.

old on a relative  $L_1$  spectral error at a stated resolution and defines the *dynamical* support  $|S|_\eta$  of Sec. VII; and neither is the  $S$  of Theorem 1, which is whatever the sampler returned. This subsection is about  $|S|_{\varepsilon_w}$  and about nothing else. Where the *raw* support—determinants of nonzero amplitude—is meant we say so, because the two do not obey the same bounds. Unlike  $\mathcal{F}_1$ ,  $|S|$  is basis dependent: a single Slater determinant has  $|S| = 1$  in its own natural orbitals and exponentially many terms in a generic basis. An orbital rotation is a free operation that leaves the occupation spectrum, hence  $\mathcal{F}_1$ , exactly fixed while changing  $|S|$ ; a quantity invariant under operations that change  $|S|$  at fixed value cannot determine it.

Both directions are realized and neither is a limiting case. A free-fermion ground state at  $U = 0$ —one Slater determinant,  $\mathcal{F}_1 = 0$ , the minimum—has site-basis support growing exponentially,  $|S| = 35, 336, 3496, 37361$  for  $L = 4, 6, 8, 10$ : minimal invariant, divergent cost. Along a Hubbard chain at fixed filling, raising  $U$  raises  $\mathcal{F}_1$  while lowering  $|S|$  as the ground state localizes: rising invariant, falling cost. Across the nineteen molecules of Table XIV the two rise *together* (Pearson  $r = 0.80$  on  $\log_2 |S|$ ), both merely tracking static correlation. The relationship is two-sided and its sign is not stable—the signature of comparing an orbital invariant to a basis variant.

We state what this is and what it is not. It is not a refutation of anyone’s claim: Refs. [54, 61] propose  $\mathcal{F}_k$  as a magic monotone for matchgate circuits, which is

what it is, and no one proposed it as a predictor of the determinant support. It is our own no-go, about our own cost, and a corollary consistent with—not a competitor to—the invariant/basis-dependent dichotomy of Tarabunga *et al.* [59], who six weeks before this work set out the same division on general grounds: occupation entropies are Gaussian invariants and lower-bound the non-Gaussian gate count, while participation entropies in the natural-orbital basis are basis dependent and *upper-bound the classical simulation cost*. That is the quantity on the useful side of the division, and the priority is theirs. The same acknowledgement is owed to Leone and Bittel [112], whose fermionic entropy—equal to the squared Frobenius norm of the correlation matrix, with an unbiased  $O(\varepsilon^{-2})$  estimator—is a Gaussian monotone of the same class and appeared seventeen days before v1 of this work. What our construction adds to that division is the constructive instance it does not contain: a family (Theorem 3) on which every order of the invariant is maximal while the cost is  $O(K)$ , and fibers (Proposition 1) on which the invariants are frozen to  $10^{-14}$  while the certificate of Theorem 1 still separates.

#### D. Two blindnesses, and why we do not sell them as one symmetry

The two halves of this section are not one statement wearing two hats, and the difference is better said by us than by a referee. On the cost side the invariance is a genuine physical group: the Gaussian orbital rotations  $U(n_o)$  are free operations, they act on the whole problem, and along one orbit the cost sweeps an *extensive* range,  $O(K)$  to  $2^K$ , at fixed invariant—a symmetry argument in the ordinary sense. On the precision side it is weaker and of a different kind:  $\Lambda_S(\eta)$  reads the off-diagonal block  $P_S H Q_S$  in the working basis, and an invariant is blind to that block—informational blindness, not a gauge. The orbit there does not sweep an extensive range; as declared above, the small-leakage endpoint is realized only by the stabilizer of the state, and the contrast shrinks with  $K$  rather than growing.

What joins the two is therefore not a symmetry at all, and nothing stronger is claimed for it than what it is: the “+” of Eq. (7), one inequality whose two summands are the two halves of this paper, one bought by shots and one not, each carrying its own obstruction. What makes that join a measurement rather than an analogy is that both summands have been evaluated on the same object,  $\mathcal{S}^* = \text{supp}(\phi)$ , where the first summand is identically zero while the second still leaves a relative  $L_1$  error of 0.43 at  $L = 6$  and 0.33 at  $L = 8$  on the periodic rings at  $\eta = 0.18t$ , and 0.28 on the open chain (Secs. III and IV). The two halves are joined by those numbers.

#### E. What does track the cost: $\chi$ , and the statistics the right way up

The computable lower bounds on  $|\mathcal{S}|$  are themselves basis dependent. The orbital-Schmidt rank across a bipartition—the tensor-network bond dimension  $\chi$  [102]—is a rigorous floor on the *raw* support,<sup>2</sup> and, empirically, a tracker of the measured  $|\mathcal{S}|_{\varepsilon_w}$ . It is not the winner of a contest:  $\chi$  is measured rather than predicted, and it has no purchase on the leakage at all. On the fibers of Table VII it is *identical* at the two endpoints—2 and 2 at  $K = 3$ , 1 and 1 at  $K = 4$ , at equal raw support—while  $\hat{\Lambda}/\eta$  moves by a factor 6–7; in a control at  $K = 3$  it jumps

from 2 to 16 with  $\hat{\Lambda}/\eta = 2.336$  and relative  $L_1 = 0.238$  unchanged to three figures; and across a subfamily of 58 points on which  $\chi$  and the raw support are *exactly* equal,  $\hat{\Lambda}/\eta$  still varies by 2.8 and the measured error by 8.3. The only tracker of the leakage is the boundary coupling  $\|P_S H Q_S\|$  of Sec. III. What follows is a statement about  $|\mathcal{S}|_{\varepsilon_w}$  and about nothing else.

*The Hubbard sweep, stratified.* The 30 lattice points are not 30 independent observations. They are six  $(L, N)$  strata—(6, 6), (6, 4), (8, 8), (8, 6), (10, 10), (10, 8)—each swept over the same five values  $U = 1, 2, 4, 8, 16$ . Within every stratum the rank correlation of  $\mathcal{F}_1$  with  $|\mathcal{S}|_{\varepsilon_w}$  is  $-1.0000$ , exactly, in all six; the corresponding  $\rho(\chi, |\mathcal{S}|_{\varepsilon_w})$  are 1.000, 0.894, 0.949, 0.975, 0.975, 0.894, mean 0.948. The stratified Kendall statistics are  $\tau = -1.000$  for  $\mathcal{F}_1$ , with 0 concordant pairs out of 60, and  $\tau = +1.000$  for  $\chi$ , with 50 concordant, 0 discordant and 10 tied. Because each  $\mathcal{F}_1$  stratum is a perfect reversal of five points, the exact one-sided stratified permutation  $p$ -value is  $(1/120)^6 = 3.35 \times 10^{-13}$ , against  $1.93 \times 10^{-10}$  for  $\chi$ , which has ties. Both take a product across strata and therefore assume the six independent. They are not six independent experiments: they are three lengths crossed with two fillings, on one code path and one  $U$  grid. The per-stratum coefficients, Fisher’s more conservative combination and the tie structure are in App. F.

Two pooled coefficients are available over these same 30 points:  $\rho = 0.57$  for  $\chi$ , which invites the reading that the bond dimension tracks the cost, and  $\rho = -0.11$  for  $\mathcal{F}_1$ , which invites the reading that it carries no evidence. Neither reading is used here, and for one reason. The 30 points are six deterministic five-point sweeps; the effective  $n$  is  $\approx 6$ , not 30; and pooling strata with different offsets destroys within-stratum structure. Here it destroys it in our favour on one coefficient and against us on the other, which is precisely why neither may be kept. What is used instead is the within-stratum evidence above, which is stronger on both sides and symmetric between them:  $\mathcal{F}_1$  anti-orders the support perfectly in all six strata, and  $\chi$  orders it at a mean  $\rho$  of 0.948.

That the evidence is stronger does not make it what we were about to call it. A rank correlation of exactly  $-1$  in all six strata is the strongest evidence a rank statistic can give of a *deterministic monotone* relation between  $\mathcal{F}_1$  and the support inside a stratum—with the wrong sign. In this family  $\mathcal{F}_1$  is not decoupled from the cost; it is an excellent rank predictor of it whose *sign* reverses from family to family:  $+1.000$  in the chain and ladder sweeps below,  $+0.86$  across the molecules,  $-1.000$  within every Hubbard stratum. That is the failure mode the no-go of Secs. VIB and VIC predicts for an orbital invariant compared against a basis variant, and “unstable in sign”, not “decoupled”, is the phrase the measurement supports. The physical content is one mechanism observed six times—raising  $U$  localizes the ground state, so  $|\mathcal{S}|_{\varepsilon_w}$  falls and  $\chi$  falls while  $\mathcal{F}_1$  rises—and a  $p$ -value of  $3.35 \times 10^{-13}$  is the probability of six coincident orderings under that one mechanism, not evidence that it generalizes beyond the

<sup>2</sup> For any state and any bipartition into  $(L, R)$ ,  $\chi \leq |\mathcal{S}|$  with  $|\mathcal{S}|$  the number of determinants of nonzero amplitude: each determinant factorizes across the cut, so the matricization  $A_{D_L, D_R} = c_D$  has at most  $|\mathcal{S}|$  nonzero entries and rank at most  $|\mathcal{S}|$ . It is elementary, which is why we record it as an observation, and it is proved for the raw support only: once amplitudes are thresholded the inequality need not survive—in the molecular suite  $|\mathcal{S}|$  is cut at 99% of the weight while  $\chi$  is cut at a discarded tail of  $\varepsilon_\chi^2 = 2.5 \times 10^{-3}$ , i.e. at 99.75%, so the two are not truncated at the same place—and the measured  $|\mathcal{S}|_{\varepsilon_w}$  falls below  $\chi$  in 7 of the 38 molecular points. It is also one-sided and can be exponentially loose: Theorem 3 has  $\chi = 2$  at  $|\mathcal{S}|_{\text{pair}} = 2^K$ .

one-dimensional Hubbard family in which it was measured.

*The molecular set does not discriminate, and now we can show it.* The  $n = 38$  molecular points are 19 molecules at two geometries—dependent pairs, not 38 draws—and 76% of them lie in some tie in  $|\mathcal{S}|_{\varepsilon_w}$ , 11 pinned at the floor  $|\mathcal{S}| = 1$ . The comparison the molecular set is asked to settle is not resolvable by any of three procedures:  $\rho = 0.90$  for  $\chi$  against 0.86 for  $\mathcal{F}_1$  differ by 0.039, a molecule-level cluster bootstrap puts a 95% interval of  $[-0.070, +0.189]$  on that difference, and Williams’s and Steiger’s tests for dependent correlations give  $p = 0.35$ . The pooled pair over all 74 states, 0.72 against 0.60, is likewise unresolved. The text already said that the molecular set does not separate the two candidate resources; it now has the formal statement (App. F), and the whole weight of the separation rests on the stratified Hubbard sweep above.

*The two families where  $\mathcal{F}_1$  wins, written out.* In the matched chain and two-leg ladder families of Fig. 8 ( $n = 3$  each),  $\rho(\mathcal{F}_1, |\mathcal{S}|) = +1.000$  exactly, against  $\rho(\chi, |\mathcal{S}|) = 0.50$  and 0.87. We report it and then dismantle it. The parameter swept there is  $L$ , not  $U$ : for the open chains  $\mathcal{F}_1/L = 1.6008, 1.6020, 1.6031$  at  $L = 6, 8, 10$ , that is  $\mathcal{F}_1 = 1.60 L$  to three figures, so  $\rho = +1$  restates “both quantities grow with system size” and nothing else, and conditioning on  $L$  leaves zero degrees of freedom. The three chain rows are moreover literal duplicates of three of the 30 Hubbard rows—the  $U = 8$ , half-filled entries at  $L = 6, 8, 10$ , identical to the last digit in  $\mathcal{F}_1$ ,  $|\mathcal{S}|$  and  $\chi$ —so they are not independent evidence. And with  $n = 3$  the smallest attainable permutation  $p$ -value is  $1/6$ : less evidence than three coin flips. What the chain/ladder pair does establish is the controlled contrast it was built for, which needs no correlation coefficient to read: at  $\mathcal{F}_1$  equal to within 1%, the ladder’s support exceeds the chain’s by up to  $\sim 2.2\times$ , while  $\chi$  moves from 5 to 13 at  $L = 6$  and from 8 to 38 at  $L = 8$ .

*What stands, and where the tracker fails too.* Figure 7 consolidates the picture over all 74 exact ground states: the determinant support is bounded below by the bond dimension in the raw count, and the bond dimension is the better empirical tracker of the thresholded one. It is not a monotone tracker, and the exception is in the same file as the rule. Within every Hubbard stratum  $\chi$  orders the support at  $\rho = 0.89$ –1.00. Along the chain family, where the swept parameter is  $L$  and not  $U$ , it does not:  $\chi = 5, 8, 6$  at  $L = 6, 8, 10$  while  $|\mathcal{S}|_{\varepsilon_w} = 147, 880, 5084$ , so  $\chi$  falls across the last step while the cost grows by 5.8, and  $\rho(\chi, |\mathcal{S}|_{\varepsilon_w})$  there is 0.50; in the ladder family  $\chi$  is flat at 38 over that same step while the cost grows by 7.7. That non-monotonicity of the chain  $\chi$  is stated here explicitly, and it is a limit on  $\chi$  itself. What  $\mathcal{F}_1$  does instead is fail in two *additional* and visibly distinct ways—the Hubbard branch runs the wrong way at a stratified  $\tau = -1.000$ , and the chain/ladder points sit at identical  $\mathcal{F}_1$  with  $|\mathcal{S}|_{\varepsilon_w}$  differing by up to  $\sim 2.2\times$ . Neither statement supplies the predictor this section set out to find,  $\chi$  being itself basis depen-

dent, measured rather than predicted, and blind to the leakage. What  $\mathcal{F}_1$  is remains intact and useful: a faithful, efficiently measurable detector of static (multireference) correlation, generalizing to fermionic non-Gaussianity, and to the nineteen molecules of Table XIV, the qubit non-stabilizerness peak of molecular bonding [64] and its reported proportionality to Hartree–Fock weight, itself reported to break down beyond the Coulson–Fischer point [97]; that a single-particle invariant can be maximal while bond entanglement stays  $O(1)$  is the fermionic counterpart of the separation between magic and entanglement reported for qubit matrix-product states [103]. The cost is a property of the sampler’s frame, read after the fact by  $\chi$  and priced in Sec. VII; the error is a property of the subspace’s boundary, read after the fact by  $\Lambda_S(\eta)$  and mapped in Sec. V. Per-species  $|\mathcal{S}|_{\varepsilon_w}$ ,  $D$  and  $\chi$  are in App. G; the full stratified analysis—the six per-stratum coefficients, the exact one-sided permutation values, the tie structure, the cluster bootstrap and the dependent-correlation tests—is in App. F.

## VII. THREE PRICES: SUPPORT, RESOLUTION, AND SHOTS

A sample-based reconstruction is paid for on three axes, not one: the determinant support the sampler must populate, the resolution  $\eta$  at which the response is asked for, and the shot budget  $T$  that populates the support. It is easy to price the first two and to quote the second as though it were a property of the model. It is not. This section reports what each axis costs, states which of the three exponents belongs to the Hubbard chain and which belongs to the question we put to it, and prices the third, which is usually left unpriced.

*One symbol, three counts.* Because the same letter carries three different objects in this paper, we fix the notation before using it.  $|\mathcal{S}|_\eta$ , the subject of this section, is *dynamical*: the number of determinants needed for the reconstruction of  $A(\omega)$  to reach relative  $L_1 < \text{thr}$  at resolution  $\eta$  and recursion depth  $n_\ell$ , so it depends on all three of  $\text{thr}$ ,  $\eta$  and  $n_\ell$  (`scaling_lanczos_mf.py`, `stage_frac`:  $\text{thr} = 0.05$ ,  $\eta = 0.15t$ ,  $n_\ell = 150$ ).  $|\mathcal{S}|_{\varepsilon_w}$ , the subject of Sec. VI, is *static*: the number of determinants carrying a fixed fraction  $1 - \varepsilon_w$  of the weight of the *ground state*, with no dynamics, no resolution and no threshold on a spectral error. And  $S$ , the subject of Theorem 1, is neither: it is whatever set of bitstrings the sampler actually returned. The three coincide only by accident. Where this section writes  $|\mathcal{S}|_\eta$  it means the first, and no statement here transfers to the other two without saying so.



### A. The support exponent, and what does and does not survive a change of resolution

Across  $L = 4\text{--}14$  (8 to 28 qubits, half-filled *periodic* rings at  $U/t = 8$ ; the three largest sizes reached by matrix-free Lanczos/Haydock and validated against dense diagonalization at  $L \leq 8$ ), the determinant support needed to reconstruct  $A(\omega)$  to relative  $L_1 < 0.05$  at  $\eta = 0.15t$  and  $n_\ell = 150$  grows exponentially,

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$$|\mathcal{S}|_{\eta=0.15} = 22, 246, 2180, 19183, 124407, 824503 \implies |\mathcal{S}|_\eta \sim e^{1.05 L} \quad (L = 4\text{--}14, R^2 = 0.998), \quad (20)$$


---

more slowly than the  $(N+1)$  sector itself,  $\dim = 24, 300, 3920, 52920, 731808, 10306296 \sim e^{1.30 L}$ . Restricting the same fit to  $L = 4\text{--}12$ , the five sizes Table VIII covers, gives 1.082 at  $R^2 = 0.998$ ; we quote the fit range with the exponent throughout, because it matters at the second decimal. The same caution applies to the object fitted: 1.082 is the fit to the published points, while the 1.090 in the bottom block of Table VIII is the fit to that table’s own coarser replication grid at the same resolution. They are two fits to two datasets and the difference is theirs, not a rounding.

*The two exponents are one exponent.* It would be convenient to report that the support exponent, unlike the fraction exponent, is resolution-independent. It is not, and the arithmetic is immediate: the fraction *is* the support divided by the sector dimension — exactly, to the last digit, at all six sizes — and the sector dimension does not depend on  $\eta$ . The two exponents therefore differ by a constant,  $b_{|\mathcal{S}|}(\eta) - b_{\text{frac}}(\eta) = b_{\text{dim}} = 1.291$  on  $L = 4\text{--}12$ , and they move together: across the decade  $\eta = 0.05 \rightarrow 0.50t$  both shift by 0.148 in absolute value (Table VIII). There is one resolution-dependent exponent in this analysis, and quoting the fraction and the support separately reports it twice.

What differs is the base it is measured against, and that is not cosmetic. On the fraction exponent 0.148 is three quarters of the quantity itself:  $e^{-0.13 L}$  becomes  $e^{-0.27 L}$ , and the headline changes qualitatively with a choice the model does not make. On the support exponent the same 0.148 carries it from 1.165 to 1.017—a seventh of its own value, 13% measured against the larger end and 15% against the smaller—and “exponential in  $L$  with an exponent near one” survives at every resolution in the decade. We therefore report the support, and we report it for two reasons, neither of which is stability: it is the absolute resource a device has to populate, and it is not divided by a sector dimension that grows for a reason—the combinatorics of the Hilbert space—that has nothing to do with the method. The fraction is reported as the derived, resolution-dependent quantity that it is, with its full grid attached.

### B. The sector fraction is a property of the question, not of the model

The headline these numbers offer—that the fraction needed for a fixed spectral accuracy falls from 0.92 at  $L = 4$  to 0.08 at  $L = 14$ , as  $e^{-0.25 L}$ —is tunable, and no headline of that form is claimed here. Over the decade of  $\eta$  the fraction exponent moves by a factor 2.2, from  $-0.1266$  at  $\eta = 0.05t$  to  $-0.2743$  at  $\eta = 0.50t$ . At  $\eta = 0.50t$  the fraction at  $L = 12$  is already 0.10, a quarter above the  $L = 14$  value at the resolution used throughout this work and reached two sizes earlier: any headline of this form is reachable by choosing the broadening.

Nor is the fraction exponential even at fixed  $\eta$ . The local logarithmic slopes between consecutive published points are  $-0.056, -0.194, -0.214, -0.379, -0.377$ —a factor 6.8 across five intervals, steepening monotonically over the first four. The quoted  $R^2 = 0.94$  is a straight line drawn through a curve of increasing slope, and the fitted exponent depends on where the fit stops:  $L = 4\text{--}10$  gives  $-0.159$ ,  $L = 4\text{--}12$  gives  $-0.209$ , and the published  $-0.2477$  requires reaching  $L = 14$ .

The mechanism that makes the fraction move with  $\eta$  is available without derivation, and Sec. IV states it as Conjecture 1 rather than as a theorem: the polynomial-approximation route to a geometric rate in the Krylov order places the kernel poles at  $\omega \pm i\eta$ , so any such rate approaches 1 as  $\eta \rightarrow 0^+$  and any prefactor diverges (App. B3). At finer resolution more Krylov order—hence more of the sector—is needed for the same  $L_1$  error. We use that as a mechanism and not as a bound: no  $C(\eta)$  or  $\rho(\eta)$  is exhibited anywhere in this work, and none is needed to read Table VIII, which is a measurement.

One further constraint has to be declared rather than inherited. The reference against which the threshold is applied is itself a finite-depth Lanczos computation, recomputed at the same depth as the subspaces so that the comparison is internally consistent ( $n_\ell = 150$ ; `scaling_lanczos_mf.py`). Two consequences follow, and both belong in the open. First, the published relative  $L_1$  is a distance between two depth-150 objects, not between the reconstruction and the exact  $A$ . Second, that reference

TABLE VIII. **Sector fraction required for relative  $L_1 < 0.05$ , versus system size and resolution.** Same ground state, same time-evolution ranking (protocol  $K = 18$ ,  $dt = 0.5$ , 16 substeps, 19 snapshots; `scaling_lanczos_mf.py`), same threshold, same depth  $n_\ell = 150$ ; only  $\eta$  changes. The scan over subspace size in this replication is coarser than the bisection that produced the published points, and returns the smallest scanned fraction meeting the threshold, so the  $\eta = 0.15t$  column is an upper bound on them: against the published 0.9167, 0.8200, 0.5561, 0.3625, 0.1700 it reads 0.917, 0.820, 0.560, 0.380, 0.180, agreeing at the two smallest sizes and standing 0.7%, 4.8% and 5.9% high at the three largest. The same granularity is why the fraction exponent of this column,  $-0.2012$ , is not identical to the  $-0.2093$  obtained by fitting the published points themselves over  $L = 4$ –12. Bottom block: least-squares exponents. *Both* fit windows are given for the support, because the choice moves the third digit and this work criticizes exactly that elsewhere in this section. The fraction exponent moves by a factor 2.2 over the decade, from  $-0.1266$  to  $-0.2743$ , while the support exponent moves from 1.165 to 1.017; in absolute value both move by 0.148, which is the whole content of the comparison.

| $L$                          | $\eta/t$  |       |           |           |       |           |       |           |
|------------------------------|-----------|-------|-----------|-----------|-------|-----------|-------|-----------|
|                              | 0.05      | 0.075 | 0.10      | 0.15      | 0.18  | 0.25      | 0.35  | 0.50      |
| 4                            | 0.917     | 0.917 | 0.917     | 0.917     | 0.917 | 0.917     | 0.917 | 0.875     |
| 6                            | 0.900     | 0.890 | 0.870     | 0.820     | 0.800 | 0.730     | 0.680 | 0.630     |
| 8                            | 0.740     | 0.680 | 0.640     | 0.560     | 0.540 | 0.480     | 0.420 | 0.380     |
| 10                           | 0.520     | 0.480 | 0.440     | 0.380     | 0.340 | 0.260     | 0.220 | 0.200     |
| 12                           | 0.340     | 0.280 | 0.240     | 0.180     | 0.160 | 0.140     | 0.120 | 0.100     |
| fraction exponent, $L=4$ –12 | $-0.1266$ | —     | $-0.1681$ | $-0.2012$ | —     | $-0.2395$ | —     | $-0.2743$ |
| $R^2$                        | 0.904     | —     | 0.918     | 0.929     | —     | 0.970     | —     | 0.982     |
| support exponent, $L=4$ –12  | 1.165     | —     | 1.123     | 1.090     | —     | 1.052     | —     | 1.017     |
| $R^2$                        | 0.999     | —     | 0.999     | 0.998     | —     | 0.999     | —     | 0.999     |
| support exponent, $L=6$ –12  | 1.136     | —     | 1.088     | 1.053     | —     | 1.022     | —     | 0.992     |

drifts with depth, and the drift is resolution-dependent: between  $n_\ell = 100$  and  $n_\ell = 200$  at  $L = 12$  the relative  $L_1$  change of the reference *alone* is 11.1% at  $\eta = 0.05t$ , 3.0% at  $\eta = 0.10t$ , 0.93% at  $\eta = 0.15t$  and 0.05% at  $\eta = 0.25t$ . At the depth used here the protocol is self-consistent only for  $\eta \gtrsim 0.15t$ —which is exactly where the published column sits, at the margin, and it is *below* that margin, at  $\eta = 0.10t$ , that Fig. 12 is drawn.

The replacement is therefore stated as a declared window rather than as a number. The fraction is reported only for  $\eta \geq 0.15t$  at  $n_\ell = 150$ ; the full  $\text{frac}(L, \eta)$  grid of Table VIII is published with it, so that the resolution dependence is visible rather than absorbed into a single exponent; and the quantitative scaling claim of this work is carried by the support, Eq. (20), whose exponent is quoted with its fit window attached.

### C. The third price: shots

The two axes usually priced are the fraction and the absolute support, and that pair is easily called the honest two-sided picture. There is a third, it is the one that binds, and it is the one priced here.

Populating a subspace by measurement is a coupon-collector problem on the Born distribution of the time-evolved seed, so the budget exceeds the support it buys. Table IX reports that budget *measured*, by multinomial sampling of the declared protocol, at the five sizes where the measurement is affordable; only  $L = 14$  carries no

measurement. Two independent runs contribute, and we name them as two rather than merging them: run A draws eight seeds at  $L = 4, 6$  and 8, run B draws four seeds at  $L = 4, 6, 8$  and 12 and six at  $L = 10$ , with different seeds and a different budget grid. Where they overlap they agree—2.43 against  $2.44 \times 10^4$  at  $L = 6$  and 2.70 against  $2.61 \times 10^5$  at  $L = 8$ , that is to 0.2% and 3.2%—which is why the two larger rows are reported as measurements and not as extrapolations of the smaller ones. Every budget is for *one* orbital seed and *one* branch; the momentum-resolved observable consumes the full set of orbital seeds in both  $(N \pm 1)$  sectors, sixteen channels at  $L = 8$ .

Four statements are supported by the measured rows, and a fifth is not. Supported: the budget is two orders of magnitude larger than the support it buys, already at  $L = 6$ ; the budget itself grows exponentially,  $7.8 \times 10^2$  to  $1.9 \times 10^7$  over  $L = 4$ –12, a log-linear fit over the four sizes at which it is measured giving  $T \sim e^{1.12L}$  at  $R^2 = 0.996$ ; at  $L = 8$  the full momentum-resolved observable costs  $16 \times 2.7 \times 10^5 \approx 4 \times 10^6$  shots, which is a device experiment, while the  $L = 14$  floor of  $\sim 10^8$  for a *single* seed and branch is neither a laptop experiment nor, at present readout rates, a device one; and the overhead per determinant is a level and not a growth law—35, 99, 124, 161, 154 against the published support across  $L = 4$ –12, rising by a factor 4.4 over the first three sizes and flat within the seed spread over the last two. Not supported, and not written: an exponent for that overhead. Fitted on the four sizes where  $T$  is measured it is 0.054 per site with a

TABLE IX. **The third price, measured at every size but the largest.**  $T$  is the total shot budget, split evenly over the 19 snapshots of the protocol, for a single orbital seed and a single branch; it is the budget at which the finite-shot reconstruction itself first reaches relative  $L_1 < 0.05$ , and  $\pm$  is the sample standard deviation across seeds, not the standard error of the mean. Run A is `sampled_honest.json` (8 seeds,  $L \leq 8$ ); run B is `honest_sampling.json` (4 seeds, 6 at  $L = 10$ ), and the two are separate runs with different seeds and different budget grids, listed separately at  $L = 6$  and  $L = 8$  so that their agreement can be read rather than asserted. At  $L = 4$  run B is censored by its own cap of  $10^8$  shots per snapshot, the 24-state sector being degenerate for this purpose, and run A is quoted there. The last column divides  $T$  by the *published* support; dividing instead by the support each run actually populated gives 95, 107, 141 and 124 at  $L = 6$ –12. Only  $L = 14$  is a floor: the overhead held at its  $L = 12$  value, which is a lower bound and not an estimate.

| $L$ | $\dim(N+1)$ | $ \mathcal{S} _\eta$ | $T$                           | $T/ \mathcal{S} _\eta$ | status                   |
|-----|-------------|----------------------|-------------------------------|------------------------|--------------------------|
| 4   | 24          | 22                   | $(7.8 \pm 2.0) \times 10^2$   | 35                     | measured, run A, 8 seeds |
| 6   | 300         | 246                  | $(2.43 \pm 0.34) \times 10^4$ | 99                     | measured, run A, 8 seeds |
| 6   | 300         | 246                  | $(2.44 \pm 0.23) \times 10^4$ | 99                     | measured, run B, 4 seeds |
| 8   | 3 920       | 2 180                | $(2.70 \pm 0.13) \times 10^5$ | 124                    | measured, run A, 8 seeds |
| 8   | 3 920       | 2 180                | $(2.61 \pm 0.08) \times 10^5$ | 120                    | measured, run B, 4 seeds |
| 10  | 52 920      | 19 183               | $(3.09 \pm 0.04) \times 10^6$ | 161                    | measured, run B, 6 seeds |
| 12  | 731 808     | 124 407              | $(1.91 \pm 0.01) \times 10^7$ | 154                    | measured, run B, 4 seeds |
| 14  | 10 306 296  | 824 503              | $\gtrsim 1.3 \times 10^8$     | 154                    | floor, not measured      |

95% interval  $[-0.065, 0.172]$  that contains zero, so *these data do not establish that the shot budget grows with a larger exponent than the support*, only that it is larger in absolute terms at every size measured. That distinction is the same one this section enforces two subsections above, applied to our own new axis.

#### D. What the published curve is, and what a declared budget costs

Precision about the label matters more here than anything else. The ranking that produced the published fractions is `argsort` applied to the Born distribution of the time-evolved state accumulated over the 19 snapshots of the protocol (`scaling_lanczos_mf.py`, `stage_rank`)—that is, to the distribution the device would itself measure, in the limit of infinitely many shots. It is *not* a clairvoyant ordering by exact eigenvector amplitudes; the repository contains such an oracle elsewhere (`amortized_recovery.py`) and labels it as one. But it is also not a finite-shot reconstruction, and to describe it as *sampled* without a shot count is to mislabel it. The published curve is the  $T \rightarrow \infty$  limit of the declared protocol, and it is labelled that way from here on.

The published curve is therefore a strict lower bound on the finite-shot cost. At  $L = 4$  the two agree because the sampler exhausts the sector; at  $L = 6$  and  $L = 8$  the finite-shot fraction exceeds the published one by 0.037 and 0.071. That is a direction across three sizes, one of which is degenerate, and we state it as a direction and do not fit an exponent to it—which is the same objection this section raises against the fraction headline, applied to our own replacement. At the published subspace itself,  $L = 8$  and  $|\mathcal{S}|_\eta = 2180$ , the infinite-shot ordering gives relative  $L_1 = 0.0490$ , just inside the threshold, against

$0.1006 \pm 0.0036$  for finite shots at  $T = 1.47 \times 10^5$ : a factor  $2.05 \pm 0.07$ , and the reason the finite-shot fraction has to rise to 0.63.

The finding that runs the other way is the more useful one, and it is what this subsection closes on: the penalty is budgetary, not structural, and it extinguishes with a budget that is affordable at this size. On the same pair as the last row of Table X— $L = 8$ ,  $|\mathcal{S}|_\eta = 2331$ —raising the budget from  $T = 2.04 \times 10^5$  to  $T = 2.60 \times 10^6$ , a factor 12.7, takes the ratio of the finite-shot error to the infinite-shot-ordering error from  $1.96 \pm 0.12$  to  $1.06 \pm 0.01$ , through 1.51, 1.18 and 1.09 at the intermediate budgets. No sampling penalty in this family can be quoted without its budget, because the penalty is a statement about  $T$  and not about the method. We therefore report finite-shot fractions only with  $T$  and the seed count attached, and we report the infinite-shot ordering for what it is—the floor that any budget approaches from above, at the price Table IX now states.

### VIII. PHYSICAL VALIDATION: THE TWO GAPS SCALE IN OPPOSITE DIRECTIONS

The two collective channels of the half-filled Hubbard chain are the strongest physical check available to this method, because their answer is known exactly and independently of it. This section reports that check as a scaling statement across  $L = 6, 8, 10, 12$  rather than as a picture at a single size—which is what it was—withdraws three words that a single size could not support, and separates what is a validation of the method from what is a result.

TABLE X. **Finite shots, with the budget attached.** Multinomial sampling of the declared protocol, eight independent seeds,  $\eta = 0.15t$ ;  $\pm$  is the sample standard deviation across seeds, not the standard error of the mean. Top block: the sector fraction the finite-shot reconstruction needs to reach relative  $L_1 < 0.05$ , and the budget at which it reaches it. Bottom block: at *matched* subspace size, the error of the infinite-shot ordering against the error of finite shots at the stated budget, and the ratio between them.

|                                                     | $L = 4$                     | $L = 6$                       | $L = 8$                       |
|-----------------------------------------------------|-----------------------------|-------------------------------|-------------------------------|
| fraction, $T \rightarrow \infty$ (published)        | 0.9167                      | 0.8200                        | 0.5561                        |
| fraction, finite shots                              | $0.9167 \pm 0.0000$         | $0.8571 \pm 0.0135$           | $0.6275 \pm 0.0031$           |
| budget $T$ at which it is reached                   | $(7.8 \pm 2.0) \times 10^2$ | $(2.43 \pm 0.34) \times 10^4$ | $(2.70 \pm 0.13) \times 10^5$ |
| matched $ \mathcal{S} $ (size the sampler returned) | 22                          | 247                           | 2331                          |
| budget $T$ at that size                             | $7.8 \times 10^2$           | $1.74 \times 10^4$            | $2.04 \times 10^5$            |
| relative $L_1$ , $T \rightarrow \infty$ ordering    | $1.4 \times 10^{-14}$       | 0.0469                        | 0.0362                        |
| relative $L_1$ , finite shots                       | $1.4 \times 10^{-14}$       | $0.0702 \pm 0.0115$           | $0.0709 \pm 0.0042$           |
| ratio                                               | 1.00                        | $1.50 \pm 0.25$               | $1.96 \pm 0.12$               |

**A. Charge converges to a finite value, spin closes as  $1/L$**

*Charge.*  $\Delta(L)$  decreases monotonically and stays above the Lieb–Wu thermodynamic value  $4.67952t$  at every size, the residual falling from  $0.679t$  at  $L = 6$  to  $0.289t$  at  $L = 12$ —that is, to 6.2% of the thermodynamic value (Table XI).  $\Delta(L = 12) = 4.96876t$  reproduces the  $4.97t$  this work has quoted throughout, and is the same number that appears as the Hubbard-band separation of Fig. 4 and as the low-frequency scale of Fig. 11. We fit no extrapolation exponent: a straight line in  $1/L$  through the four points extrapolates to  $4.561 \pm 0.044t$ , which under-shoots Lieb–Wu by  $0.12t$  and is therefore evidence that the functional form is not  $1/L$ , not evidence about the limit. The statement supported is that the charge scale converges to a finite thermodynamic value and is already within 6.2% of it at the largest size reached here.

*Spin.* The lowest pole of  $S^{zz}(q = \pi, \omega)$  gives  $L\Delta_s = 2.091, 1.917, 1.996, 1.996$ : a  $1/L$  closing with a coefficient of about 2.1. Two fits make that precise and bound it honestly. Fitting  $\Delta_s = c_0 + c_1/L$  over these four sizes gives an intercept  $c_0 = -0.020 \pm 0.020$ —consistent with zero at one standard error—against  $c_0 = 4.561 \pm 0.044$  for the charge gap: one intercept is consistent with zero and the other is not, and that separation, not either intercept on its own, is the result. It has to be read with the control that limits it, which we state here rather than leave to the figure: the pure Heisenberg ring, whose spin gap is known to vanish in the thermodynamic limit, gives a linear-in- $1/L$  intercept of  $0.0142 \pm 0.0014t$ —above zero, not consistent with it—so at these sizes a vanishing intercept is not by itself evidence that a gap closes. The statement that survives without a fitted model is the flatness of  $L\Delta_s$  below. Fitting instead  $L\Delta_s = d_0 + d_1/L$  gives  $d_0 = 2.09 \pm 0.17$  with  $d_1 = -0.010 \pm 0.018$ , a slope consistent with zero: over  $L = 6$ – $12$  the product  $L\Delta_s$  is flat to the resolution of four points. What is *not* supported is a third significant figure on the coefficient. The four

values span 8% and are not monotone, and we quote  $L\Delta_s \approx 2.1$  with the spread attached rather than 2.00.

The independent control is the effective spin model. The pure Heisenberg ring at  $J = 4t^2/U = 0.5t$ —no charge degree of freedom, no double occupancy, a different Hamiltonian on a different Hilbert space, solved by dense diagonalization on a different code path—gives  $\Delta_{\text{ST}}(N = 12) = 0.35585J = 0.17792t$  and reproduces the same  $1/L$  law across the same four sizes, with  $L\Delta_{\text{ST}} = 2.054, 2.091, 2.116, 2.135$ . Two things have to be said about that agreement rather than left for a referee. First, the two numbers at  $L = 12$  differ by 6.5% of the Heisenberg value, equivalently 7.0% of the Hubbard one; we name the denominator because the two differ in the third figure. Second, the residual is not monotone and changes sign—the Hubbard gap is 1.8% *above* the Heisenberg one at  $L = 6$  and 8.3%, 5.7% and 6.5% *below* it at  $L = 8, 10, 12$ —so it cannot be attributed to a smooth  $O(t^4/U^3)$  correction to the effective exchange, which would not change sign. A correction of that order is the expected *scale* of the difference at  $U/t = 8$ , and it is the right scale; its size dependence is not resolved here and we claim nothing about it. The control confirms the  $1/L$  scaling and the magnitude of the coefficient to within 10%; it does not pin the coefficient, and its own  $L\Delta_{\text{ST}}$  drifts upward by  $0.0134 \pm 0.0014$  per site over the same range.

The two channels therefore behave *qualitatively oppositely* under a change of system size—one converging to a finite thermodynamic value, the other extrapolating to zero as  $1/L$ —with a factor 29.9 between them at  $L = 12$ , and with  $1/L$  intercepts that separate. That opposition, measured at four sizes with an independent control on the smaller scale, is the physical statement this work makes. It is a statement about pole positions, not about lineshapes, so it does not depend on the broadening at which either response is plotted.



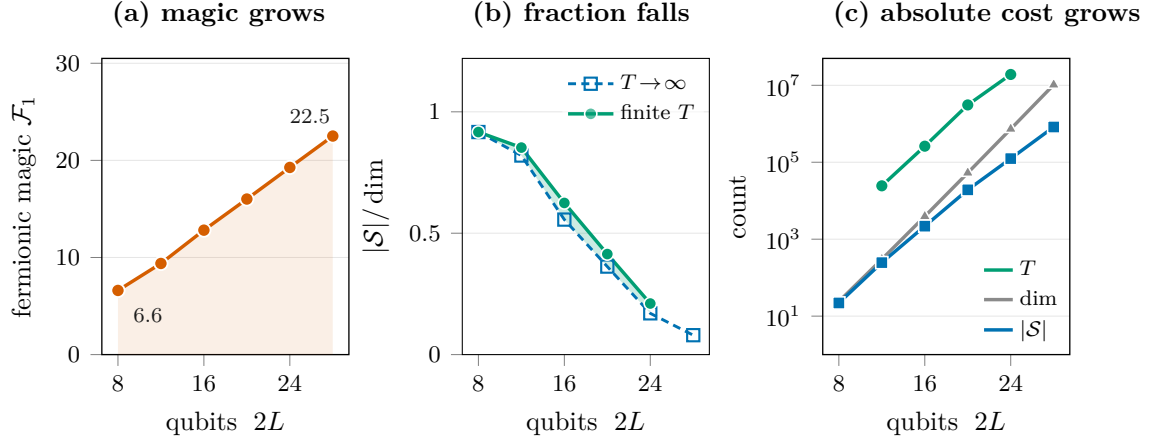


FIG. 9. **Three faces of the cost, measured with shots.** One-dimensional Hubbard ring,  $U/t = 8$ , half filling, periodic boundary conditions; the subspace is built in the  $(N+1, S_z+1)$  sector seeded by  $c_{0\uparrow}^\dagger|\Psi_0\rangle$  from the  $K+1 = 19$  time snapshots of the declared protocol ( $K = 18$ ,  $\delta t = 0.5$ ), and a subspace is accepted when its Haydock reconstruction reaches relative- $L_1 < 0.05$  against the exact spectrum at broadening  $\eta = 0.15t$ . All fractions below are at that single  $\eta$ . **(a)** One-body fermionic magic of the ground state, 6.6074 at  $L = 4$  to 22.5050 at  $L = 14$  (periodic ring; the open-chain families elsewhere in this paper carry different values and must not be substituted here). **(b)** The required fraction of the sector, twice. Open squares, dashed: the infinite-shot limit of the same protocol — the subspace obtained by ranking the exact Born weights of the snapshots — 0.9167, 0.8200, 0.5561, 0.3625, 0.1700, 0.0800 for  $L = 4$ –14, the sequence previously reported. Filled circles, solid: the same acceptance criterion reached by actually drawing shots,  $0.9167 \pm 0.0000$ ,  $0.8527 \pm 0.0096$ ,  $0.6246 \pm 0.0033$ ,  $0.4141 \pm 0.0012$ ,  $0.2104 \pm 0.0004$  (mean  $\pm$  s.d. over 4 independent shot seeds; 6 seeds at  $L = 10$ ); the error bars are drawn and are smaller than the markers. Finite sampling needs +4.0%, +12.3%, +14.2% and +23.7% more of the sector than the infinite-shot limit at  $L = 6, 8, 10, 12$ : the penalty does not cancel, it grows with size, and the shaded strip is that difference. An independent 8-seed run of a separately written sampler reproduces  $0.8571 \pm 0.0135$  ( $L = 6$ ) and  $0.6275 \pm 0.0031$  ( $L = 8$ ), agreeing with the curve to within one standard deviation. No finite-shot point is shown at  $L = 14$ : its Born weights were not computed, and they are not extrapolated. **(c)** The absolute counts that panel (b) normalises away, plus the third one. Sector dimension  $\dim$  (grey) and the support  $|S|$  of the reported subspace (blue) both grow exponentially; the measured number of shots  $T$  that reaches the finite-shot operating point of panel (b) (green) is  $2.4 \times 10^4$ ,  $2.6 \times 10^5$ ,  $3.1 \times 10^6$ ,  $1.9 \times 10^7$  at  $L = 6, 8, 10, 12$  (s.d. over seeds  $2.3 \times 10^3$ ,  $8.1 \times 10^3$ ,  $3.8 \times 10^4$ ,  $1.2 \times 10^5$ ).  $L = 4$  carries no shot count: its 24-state sector is degenerate for this purpose and its budget is censored by the  $10^8$ -per-snapshot cap. Log-linear fits over the six sizes drawn in (c) (four for  $T$ , which has no  $L = 4$  and no  $L = 14$  point) give  $\dim \sim e^{1.30L}$  ( $R^2 = 0.9999$ ),  $|S| \sim e^{1.05L}$  ( $R^2 = 0.998$ ) and  $T \sim e^{1.12L}$  ( $R^2 = 0.996$ , four sizes). Those exponents are fits to the points this panel plots, not to the five-size fits stored in `honest_sampling.json` (1.108 and 1.082). The exponent of  $T$  exceeds that of  $|S|$  by 0.054 per site, with a 95% interval  $[-0.065, 0.172]$ : these data do *not* establish that the shot budget grows with a larger exponent than the support, only that it is larger in absolute terms at every size measured (Table IX). **Resolution dependence.** Over  $\eta/t \in [0.05, 0.50]$  the exponent of  $|S|$  moves from 1.165 to 1.017 and that of the fraction in panel (b) by a factor 2.2, from  $-0.1266$  to  $-0.2743$  (Table VIII,  $L = 4$ –12). In absolute value both move by the same 0.148: they are one exponent, and neither is stable.

## B. What a single size at $\eta = 0.10t$ could not support

The previous version made the spin-charge statement at one size and read it off a lineshape. Two things were wrong with that, and we retract both.

First, resolution. At  $L = 12$  the spin scale is  $0.16635t$  against the  $\eta = 0.10t$  used for Fig. 12: the gap is  $1.66\eta$ , i.e. 0.83 of the Lorentzian full width at half maximum. At that resolution a spin response with a  $0.17t$  gap and one with none are not distinguishable by eye in a single-size plot. The figure could *illustrate* the physics; it could not establish it. Three terms applied across the  $L = 12$  channels of the previous version—“continuum”, “gapless”, and “edge of the continuum”—were therefore not licensed by it, and they are withdrawn here, once, for every channel of this work at this size.

Second, provenance. The numerical value that stood behind the sentence asserting that the two-spinon band touches  $\omega = 0$  was  $-0.100$ , and it is not a measurement: it is the first entry of the frequency window `np.linspace(-0.1, 3.5, nw)` declared in `spin_lanczos.py`, propagated into the edge table by a minimum over the computed signal. It is the lower end of the computation window, censored at that end, and it never reaches the plotted axis.

What replaces both is measurement rather than illustration: the four-size scaling of Table XI, plotted in Fig. 10, in which the spin scale is a converged pole position at every size and closes as  $\approx 2.1/L$  with an independent effective-Heisenberg control, and the pole census of Table XII, which states what is actually inside the window at each size instead of naming it.

TABLE XI. **The two collective scales, across four system sizes.** Charge: the Mott gap  $\Delta = \mu^+ - \mu^-$  from exact ( $N \pm 1$ ) ground-state energies (half-filled Hubbard ring,  $U/t = 8$ ; particle-hole symmetry  $\mu^+ + \mu^- = U$  verified to  $10^{-8}$  at every size, ARPACK checked against dense `eigh` on every sector of dimension  $\leq 5000$ ).  $\Delta_\infty = 4.67952t$  is the Lieb–Wu thermodynamic value [104], evaluated here from its exact integral representation to an absolute quadrature error of  $2 \times 10^{-9}$  and cross-checked against a second exact representation, the two agreeing to  $10^{-13}$ . Spin: the lowest pole of  $S^{zz}(q = \pi, \omega)$ , a pole position of the tridiagonal recursion and therefore independent of the broadening used to display it. Right-hand block: the same quantity in the pure Heisenberg ring at the effective exchange  $J = 4t^2/U = 0.5t$ , a model with no charge degree of freedom at all. The residual column is quoted to four decimals so that it is the difference of the two columns to its left; at five it would not be.

| $L$ | charge     |                              | spin (Hubbard) |                | spin (Heisenberg, $J = 0.5t$ ) |                   |
|-----|------------|------------------------------|----------------|----------------|--------------------------------|-------------------|
|     | $\Delta/t$ | $(\Delta - \Delta_\infty)/t$ | $\Delta_s/t$   | $L \Delta_s/t$ | $\Delta_{ST}/t$                | $L \Delta_{ST}/t$ |
| 6   | 5.35813    | 0.6786                       | 0.34857        | 2.091          | 0.34237                        | 2.054             |
| 8   | 5.18584    | 0.5063                       | 0.23958        | 1.917          | 0.26134                        | 2.091             |
| 10  | 5.02224    | 0.3427                       | 0.19956        | 1.996          | 0.21162                        | 2.116             |
| 12  | 4.96876    | 0.2892                       | 0.16635        | 1.996          | 0.17792                        | 2.135             |

### C. A pole census replaces the word

At  $L = 12$  and the broadenings used here, none of the three momentum-resolved channels is a dense spectrum inside the plotted window; each momentum carries a small number of discrete poles that the broadening merges into a band. Counting poles that carry more than 1% of the channel maximum—Lanczos with full reorthogonalization at  $n_\ell = 120$ , checked for depth stability at  $n_\ell = 60, 120$  and  $240$  at  $q = \pi$  in every channel, and validated pole by pole against dense diagonalization at  $L = 6$  and  $L = 8$  [40, 41]—gives Table XII. The census is the deposited `pole_structure.json` and is taken on the momentum grid the figures of this section plot, seven values of  $k$  and six of  $q$ ; a count over a coarser grid is a count of different momenta and is not comparable entry by entry. The census covers the three channels the figures of this section plot; the  $k$ -integrated  $A(\omega)$ , which is a sum over the same branch, and the removal branch are not counted, and no statement is made about them here.

The count does grow with size, and the size of the growth is the point. Between  $L = 6$  and  $L = 12$  the  $(N+1)$  sector grows by a factor 2440, while the largest per-momentum pole count grows from 3 to 7 in the addition branch, from 4 to 9 in the charge channel and from 2 to 3 in the spin channel. A line count that roughly doubles while the Hilbert space grows by three orders of magnitude is a discrete spectrum whose resolvable lines are being broadened together, not a dense one forming. The spin channel is the extreme case: at every momentum a single pole carries between 80% and 99% of the weight, at separations of at least 1.08 full widths, and at  $q = \pi$  that single pole *is* the  $0.16635t$  scale of Table XI. What Fig. 12

shows at  $L = 12$  is one dominant antiferromagnetic mode plus at most two satellites.

This has a consequence for how the reference itself must be described, and we state it rather than leave it for a referee. The previous version describes  $A(k, \omega)$  as computed exactly, equivalent to the Lehmann sum. With  $n_\ell = 150$  and no reorthogonalization in a sector of dimension 731 808, what is computed is a 150-node Gauss quadrature of the spectral measure [40, 41]. It is accurate enough here—and it is accurate *precisely because* the measure it integrates is a handful of poles; a dense spectrum at this dimension would not be reproduced by 150 nodes. It is not, however, converged: the same depth moves the reference by 0.93% at  $\eta = 0.15t$  and by 3.0% at the  $\eta = 0.10t$  of Fig. 12 when it is taken from  $n_\ell = 100$  to  $n_\ell = 200$  (Sec. VII). The reference is therefore labelled for what it is: a finite-depth continued fraction, checked against dense diagonalization at  $L \leq 8$  and quoted with its depth. The pole positions of Table XI are a separate and stronger object—they are computed at full reorthogonalization and are stable across  $n_\ell = 60, 120, 240$ —which is why the physical claim of Sec. VIII A rests on them and not on a lineshape.

### D. The sum rule, as measured

The analytic identity  $\int A^+ d\omega = 1 - \langle \hat{n}_p \rangle = 0.5$  is exact and grid-independent, but it is an identity of the *ground state*: evaluating it costs one ground-state expectation value,  $\langle \Psi_0 | \hat{c}_p \hat{c}_p^\dagger | \Psi_0 \rangle$ , and never touches the reconstruction. It is therefore not an a posteriori check of  $A_S$ , and the previous caption of Fig. 2 was wrong to present it as one.

Integrating the plotted curves instead, over the 600-point window at  $\eta = 0.15t$ :

$$\int_{\text{window}} A_{\text{exact}} d\omega = 0.486588, \quad \int_{\text{window}} A_S d\omega = 0.486471. \quad (21)$$

TABLE XII. **Pole census, across three sizes.** Poles carrying more than 1% of the channel maximum, counted per momentum; each entry is the range over the momenta plotted (7 values of  $k$  for  $A(k, \omega)$ , 6 values of  $q$  for the collective channels). The  $L = 6$  and  $L = 8$  columns are dense diagonalizations, where the count needs no recursion, on the subset of momenta available at those sizes; the  $L = 12$  column is the reorthogonalized recursion at  $n_\ell = 120$ , which reproduces the dense counts pole by pole at the two smaller sizes. “Leading weight” is the fraction carried by the single largest pole at  $L = 12$ . The addition branch of  $A(k, \omega)$  and the charge channel are at  $\eta = 0.18t$ , the spin channel at  $\eta = 0.10t$ . The last column is the smallest separation between retained poles in units of the full width at half maximum, which is why the figures show bands: at the tightest momenta the lines are genuinely merged, and the census says how many of them there are.

| channel                    | poles above 1% |       |        | dim ratio | leading   | $\delta\omega_{\min}/$ |
|----------------------------|----------------|-------|--------|-----------|-----------|------------------------|
|                            | $L=6$          | $L=8$ | $L=12$ |           | weight    | FWHM                   |
| $A(k, \omega)$ , addition  | 2–3            | 2–4   | 3–7    | 2440      | 0.32–0.69 | 0.12–2.04              |
| $S(q, \omega)$ , charge    | 3–4            | 3–6   | 6–9    | 2440      | 0.38–0.68 | 0.24–2.20              |
| $S^{zz}(q, \omega)$ , spin | 1–2            | 1–3   | 1–3    | 2134      | 0.80–0.99 | 1.08–2.37              |

Both fall 2.7% below the analytic 0.5 because Lorentzians of width  $\eta$  leak outside any finite window: the analytic in-window mass is 0.486589, the tail outside it is 0.013411, and the residue of the trapezoidal quadrature itself is  $3 \times 10^{-6}$  relative—so the deficit is entirely the window and not the reconstruction. The a posteriori statement the reconstruction does support is the comparison of the two integrated curves, which differ by  $1.16 \times 10^{-4}$  absolute, that is  $2.4 \times 10^{-4}$  relative. That is the number we report, and it brings this caption into line with the  $\sim 2\%$  window effect that Fig. 4 already conceded.

#### E. Size: the convention, and what it does and does not license

The system size is not hidden and is not the weak point. The exact-diagonalization convention for this observable is  $L = 12$ –16 at  $\eta \approx 0.1$ – $0.2t$ : Xu, Lu, Tohyama and Shao [121] compute this spectral function by Lanczos at  $L = 12$  and  $\eta = 0.2t$  at half filling, and at  $L = 16$  at quarter filling. Their model is the extended Peierls–Hubbard chain, and their  $L = 16$  run is a dimerized lattice away from half filling; it is their half-filled  $L = 12$ , and the plain-Hubbard  $L = 12, 8, 6$  of their Appendix A, that sit alongside ours. Our  $L = 12$  at  $\eta = 0.10$ – $0.18t$  is that convention, not an outlier within it.

What the convention does not license is the vocabulary of the thermodynamic limit, and the criterion that says so is in the dynamical-DMRG literature this paper already cites. Jeckelmann’s rule is explicit [71]: “ $\eta_0(N)$  must be larger than the distance  $\delta\omega$  between two consecutive excited states.” At  $L = 12$  the largest spacing between consecutive excited-state energies in the window is  $\approx 0.48t$ , a factor 2.7 above the  $\eta = 0.18t$  used for  $A(k, \omega)$  and 4.8 above the  $\eta = 0.10t$  used for the spin channel. The rule is not satisfied here. Satisfying it would demand  $\eta > 0.48t$ . That is only a tenth of the Mott scale, so the Hubbard bands would survive; but it is  $0.96J$  and 2.9 times the  $L = 12$  spin gap itself, so the entire low-frequency structure of

Fig. 12 would merge into one feature. The number the rule demands is set by the spacing,  $0.48t$ ; we take it from the spacing and not from a rounded-up interval. The correct response is not a larger  $\eta$ : it is to stop using words that require the rule to hold. The dynamical-DMRG standard fixes the product  $\eta L$  and lets  $\eta$  fall with the size: Benthien and Jeckelmann use  $\eta = 5t/L$  in one place, the  $L = 100$  spin calculation, and declare  $\eta L = 12t$  for the finite-size scaling that carries the charge channel to  $L = 120$  [117]; Jeckelmann’s own prescription is  $\eta(N) = c/N$  with  $c = 6.4$  and  $12.8t$ , on chains of up to  $N = 256$  [71]. This work is at  $L = 12$  with  $\eta L = 1.2$ – $2.2t$ , a factor of two to ten below every constant in that range, and it is the words, not the size, that we withdraw. Section IX E states the same concession in the list of what this work does not claim, so that it is not read only here.

What the convention does license is exactly the claim of Sec. VIII A: pole positions at four sizes, each stable in recursion depth, whose ratios and  $1/L$  law are properties of the spectrum and not of the broadening used to display them. Table XII is of the same kind: it counts poles, and the counting does not depend on  $\eta$  either—only the picture does. That is the form in which the physics is stated from here on.

#### F. Spin–charge separation is a check of the method, not a result

That the spin and charge responses of the one-dimensional Hubbard model live on separate energy scales has been known exactly since Lieb and Wu [104], is textbook [118], was seen in angle-resolved photoemission thirty years ago [81, 119], is already legible in the spectral weight of twelve-site clusters embedded by cluster perturbation theory [120], and has been computed by dynamical DMRG since 2007 [117] in all three of the momentum-resolved channels reported here—in that reference’s notation the charge factor  $N(q, \omega)$  at  $L = 120$ , the spin factor  $S(q, \omega)$  at  $L = 100$  and the one-particle

function  $A(q, \omega)$  at  $L = 90$ , which are respectively our  $S(q, \omega)$ , our  $S^{zz}(q, \omega)$  and our  $A(k, \omega)$ , for the extended Hubbard model of which ours is the  $V = 0$  line; the same literature treats the Hubbard chain at  $L = 48$  with  $\eta = 0.1t$  by correction-vector DMRG, and reaches  $L = 64$  in the Heisenberg chain [73]. We present it here as what it is: a validation of one sampling primitive against a known answer, in four channels, with the des Cloizeaux–Pearson lower boundary [105] and the Müller upper bound [106] recovered as thermodynamic-limit references rather than as findings. No part of it is offered as new physics.

What is offered as a result is the size dependence. On the same lattice, in the same calculation, from the same primitive, the charge scale converges to a finite thermodynamic value while the spin scale closes as  $\approx 2.1/L$ , with an independent effective-Heisenberg control reproducing the latter to within 10% and a factor 29.9 separating the two at  $L = 12$ . Opposite finite-size behaviour in two channels of one Hamiltonian is a statement that no single-size picture can make, and it is the statement the four diagonalizations of Table XI support.

## IX. HARDWARE, SCOPE, AND THE REGION WHERE THE METHOD HAS NO ADVANTAGE

Three claims of the previous version were larger than the quantity that fixes them: the hardware claim is fixed by a subspace size, the chemistry claim by a set of sector dimensions, and the advantage claim by the same bond dimension this paper puts forward as its own cost tracker. This section states each with its fixing quantity attached, and derives—rather than concedes—the region in which this method has no advantage over a classical one.

### A. The two hardware runs, and what “exact by coverage” means

Both device runs are preserved in full and they are the entirety of the quantum hardware used in this work. The first was executed on the IBM Heron processor `ibm_fez` (job `d9s16avpemt73ct6g8g`): the  $L = 6$  Hubbard chain at  $U/t = 4$ ,  $\eta = 0.15t$ , 12 qubits,  $3.5 \times 10^5$  computational-basis shots— $5 \times 10^4$  on each of seven circuits  $k = 0, \dots, 6$ , of which  $k = 0$  is the  $t = 0$  reference—pooled into one subspace, under TREX readout twirling, Pauli twirling and dynamical decoupling and with no zero-noise extrapolation (Fig. 1 is the primitive, Fig. 14(a) the result). Self-consistent configuration recovery returned  $|\mathcal{S}| = 300$  determinants of an  $(N+1)$  sector of dimension 300; the recorded relative  $L_1$  is 0.0, and the deposited provenance string says why in its own words—the reconstructed  $A_{\text{hw}}$  equals the exact  $A$  byte for byte because the recovered subspace spans the whole sector.

That number is exact *by coverage*, and the distinction is not rhetorical: at  $|\mathcal{S}| = \dim$ , Rayleigh–Ritz on  $\text{span}(\mathcal{S})$  is diagonalization of the full sector,  $P_{\mathcal{S}} H P_{\mathcal{S}} = H$ , and a

vanishing error is an identity rather than a measurement of device fidelity. The run establishes that the pipeline executes end to end on a processor—transpiled circuits, device counts, recovery to the correct particle-number sector, and the classical Lehmann assembly of Sec. II—which is what a proof of principle is, and it establishes nothing about accuracy at incomplete coverage, the regime the rest of this paper is about. The caption of Fig. 14 already said “exact by coverage”, while the abstract of v1–v2 said the method “runs on the IBM Heron processor” and the conclusion said “demonstrated on” it. Both claimed more than  $|\mathcal{S}| = \dim$  can support and both are withdrawn: the runs are *executed on* the named processors, and the coverage qualification now governs the abstract and the conclusion as well as the caption.

*The second run, and a comparison we withdraw.* Stretched  $\text{N}_2$  ( $R = 2.0$  Å, CAS(10e,12o)/cc-pVDZ, 24 qubits) was executed on `ibm_marrakesh` (job `da125f2ein7c73bcsqs0`,  $10^4$  shots, double-factorized Trotter circuit). Averaged over 8 recovery seeds on the cached device counts it reaches  $0.59 \pm 0.14$  mHa of the exact active-space value (best seed 0.40 mHa), while the noiseless statevector simulation of the same shallow circuit stops at 29.5 mHa (Fig. 14(b)). Versions 1–2 read that contrast as device noise *helping*, and we withdraw the reading, because the mechanism named for it is a statement about a quantity the record does not contain: recovery is said to turn corrupted bitstrings into valid configurations and thereby *enlarge* the variational subspace—a valid-shot economy, not a fidelity gain—so the comparison is decided by the two recovered subspace sizes, and neither is stored. The same gap runs through the controlled sweep, whose script saves the size of the naively post-selected subspace and not of the recovered one, so the two places where an error *below* the noiseless value is reported—CO from 1.73 to 0.63 mHa and  $\text{C}_2$  from 0.184 to 0.026 mHa, between  $\varepsilon = 0$  and  $\varepsilon = 20$ –30%—rest on the same unrecorded variable, at the price of one line in a script that already builds the subspace.

The mechanism itself is not in dispute, and the group that established it belongs in the paragraph that withdraws our measurement of it, not in a referee’s report. On the dissociation of  $\text{N}_2$  in a large active space, with the same LUCJ ansatz used for the run above, Vaquero-Sabater, Carreras, Broers, Shirakawa, Yunoki and Casanova report that “sampling noise can enhance Hilbert-space exploration by generating additional configurations beyond those supported by the ideal ansatz,” and that “when combined with configuration recovery, this leads to systematically improved energies” [46]: the same molecule, the same ansatz and the same mechanism as the reading withdrawn above, and there it is carried by a controlled comparison. What this work withdraws is therefore its own measurement of the effect and not the effect. Our two arms differ in the one quantity neither of them recorded, so they cannot be evidence either way, whatever the literature independently establishes.

What *is* controlled is the comparison in which both



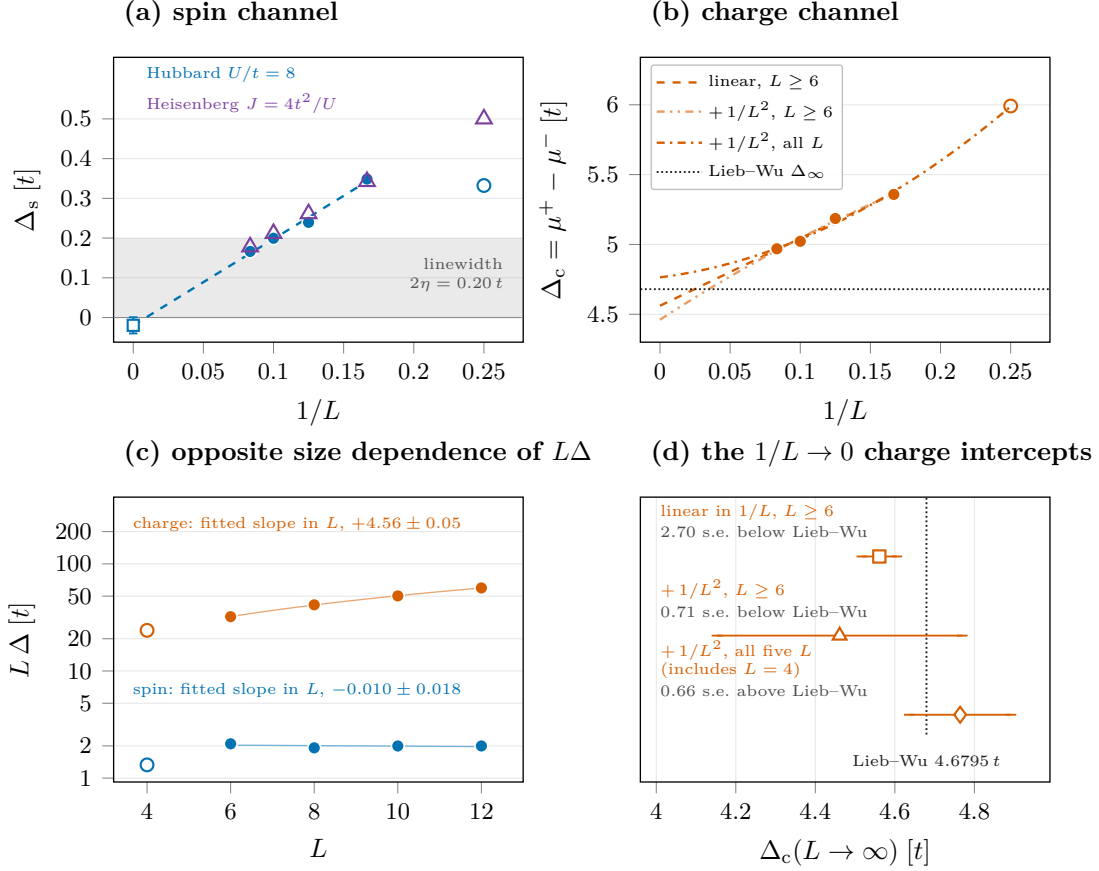


FIG. 10. **The two gaps of the half-filled Hubbard ring scale in opposite directions.** Periodic ring,  $U/t = 8$ ,  $L = 4, 6, 8, 10, 12$ ; in (a) and (b) the abscissa is  $1/L$ , so the sizes run  $L = 12, 10, 8, 6, 4$  right to left, and open symbols are  $L = 4$ , which the  $L \geq 6$  fits exclude. Every gap is an exact Lanczos pole, full reorthogonalisation, *no* broadening. **(a)** Spin gap, lowest pole of  $S^{zz}(q = \pi, \omega)$ :  $0.34857, 0.23958, 0.19956, 0.16635 t$  at  $L = 6, 8, 10, 12$  ( $0.33232 t$  at  $L = 4$ ). Dashed: linear fit in  $1/L$ ,  $L \geq 6$  ( $R^2 = 0.988$ ); the open square is its intercept  $-0.0196 \pm 0.0204 t$ ,  $0.96$  s.e. from zero. Triangles: the pure Heisenberg ring  $J = 4t^2/U = 0.50$ , an independent control on a different code path, whose lowest  $S^{zz}(q = \pi)$  pole reproduces its singlet-triplet gap to  $8 \times 10^{-15}$ . For  $L \geq 6$  the two channels agree to within  $8.3\%$  (ratios  $1.018, 0.917, 0.943, 0.935$ ); at  $L = 4$  they do not ( $0.665$ ), which is why that size is excluded. Grey band: the linewidth actually used for the published spin figure,  $\text{FWHM} = 2\eta = 0.20 t$  at  $\eta = 0.10 t$ . The  $L = 12$  gap lies inside it ( $0.16635 t = 0.83 \text{ FWHM}$ ), and broadened by this  $\eta$  it still carries  $26.5\%$  of its peak height at  $\omega = 0$ . One size alone therefore cannot separate a closing gap from no gap. **(b)** Charge gap  $\Delta_c = \mu^+ - \mu^-$  from the  $N \pm 1$  sectors:  $5.3581, 5.1858, 5.0222, 4.9688 t$  at  $L = 6, 8, 10, 12$  ( $5.9914 t$  at  $L = 4$ ). At  $L = 12$  that is  $27.6 \eta = 13.8 \text{ FWHM}$  for the  $\eta = 0.18 t$  of the one-particle channel,  $16.6$  times further from  $\omega = 0$  in units of its own linewidth than the spin gap. Three extrapolations are drawn and they disagree. The ordinate omits zero because the question here is which finite value the gap tends to, not whether it vanishes. **(c)** The same data with no extrapolation model, logarithmic ordinate:  $L\Delta_s$  is flat over  $L \geq 6$  (mean  $2.0 t$ , the four values spanning  $8\%$  and not monotone, so no third figure is quoted; fitted slope in  $L$   $-0.010 \pm 0.018$ ,  $0.57$  s.e. from zero) while  $L\Delta_c$  grows linearly ( $+4.56 \pm 0.05$ ,  $101$  s.e.). The thin lines are those two linear fits, curved here only because the ordinate is logarithmic. **(d)** Those three intercepts on a scale where their standard errors are legible; on the scale of (b) the  $\pm 0.0439 t$  bar is thinner than its marker. Linear,  $L \geq 6$ :  $4.5609 \pm 0.0439 t$ , undershooting the exact Lieb-Wu gap  $4.6795 t$  by  $0.1186 t$ ,  $2.70$  s.e. Same protocol plus a  $1/L^2$  term:  $4.4611 \pm 0.3084 t$ , undershooting by  $0.2184 t$  ( $0.71$  s.e., standard error  $2.4$  times larger). Only the  $1/L^2$  fit over all five sizes brackets Lieb-Wu ( $4.7641 \pm 0.1278 t$ ,  $0.66$  s.e. above), and it is the one fit here that includes  $L = 4$ , the size (a) and (b) mark anomalous. These sizes therefore fix the  $L \rightarrow \infty$  charge gap only to about  $\pm 0.1$ – $0.3 t$ : compatible with Lieb-Wu once a  $1/L^2$  term is allowed, not evidence for it. What all three exclude is zero, the weakest by  $14.5$  s.e. and the linear one by  $104$  s.e. *Caveat, because it limits (a).* A vanishing intercept is not by itself the evidence: the Heisenberg control, whose gap is known to vanish in the thermodynamic limit, has a linear-in- $1/L$  intercept  $0.0142 \pm 0.0014 t$ ,  $10.0$  s.e. above zero, and its own  $L\Delta$  drifts  $+6.8\%$  from  $L = 4$  to  $L = 12$ , both from its logarithmic corrections. The model-free statement is (c).

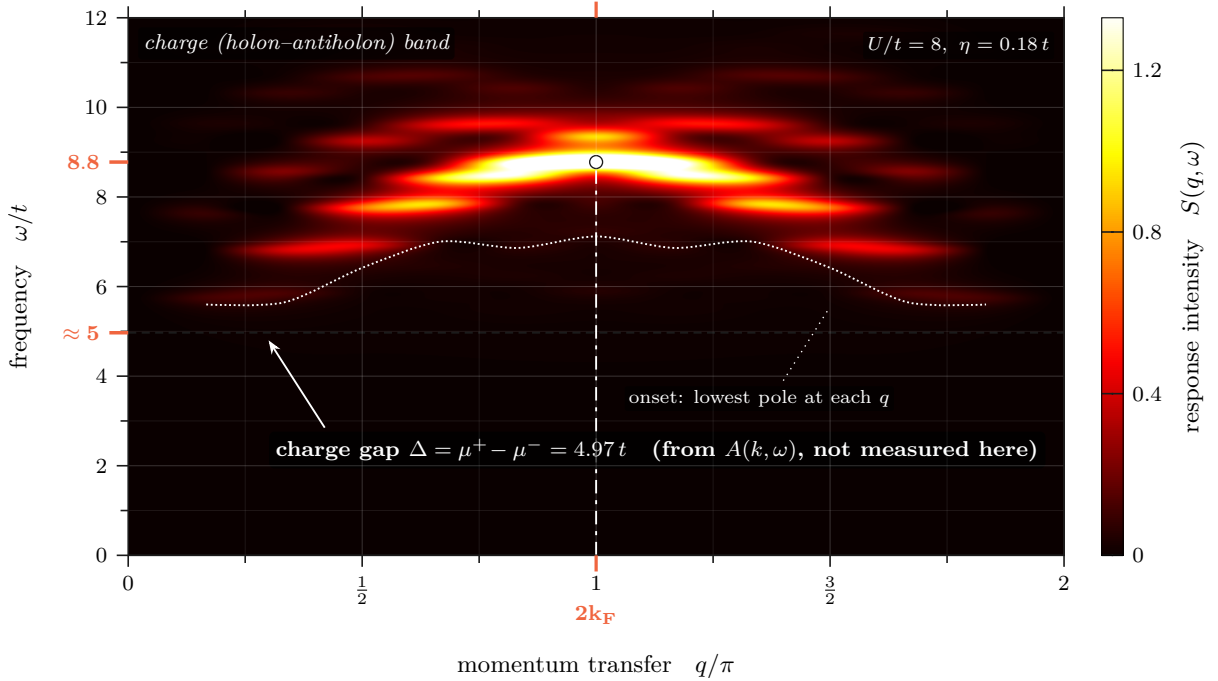


FIG. 11. **Dynamical structure factor: the density response.**  $S(q, \omega)$  of the half-filled Hubbard chain ( $U/t = 8$ ,  $L = 12$ , Lorentzian broadening  $\eta = 0.18t$ )—the density-density response, a number-conserving (neutral) excitation, computed by a Haydock continued fraction of depth  $n_\ell = 150$  in the  $N$ -particle sector—a finite-depth quadrature of the spectral measure, not the Lehmann sum (Sec. VIII). Colour is the *response intensity*  $S(q, \omega)$  (units  $1/t$ ): bright means the system responds strongly at that  $(q, \omega)$ , dark means it does not. The weight forms the two-particle charge (holon-antiholon) band, whose dotted curve is the onset, the lowest pole resolved at each  $q$ , and sets the low-frequency scale of the band and traces its lower-boundary dispersion; the diffuse upper boundary is left unmarked. At  $L = 12$  this band is six to nine discrete poles per momentum merged by the broadening, not a dense spectrum (Table XII). Two features are decisive. (i) The band is gapped on the scale of the charge gap: every resolved pole lies above  $\Delta = \mu^+ - \mu^- = 4.97t$  (exact  $(N \pm 1)$  ground-state energies), the lowest onset reached here being  $5.60t$  at  $q/\pi = 1/6$ . The limit  $q \rightarrow 0$  is inaccessible to this observable,  $S(q=0, \omega>0)$  vanishing identically by density conservation, so  $\Delta$  is drawn as a horizontal reference level imported from the single-particle channel and nothing is claimed about that limit. It is the *same* Mott gap that separates the Hubbard bands in the single-particle channel (Fig. 4): two independent dynamical observables, one correlation gap. (ii) The band peaks at  $q = 2k_F = \pi$  at  $\omega = 8.78t$  (coral tick, at the measured peak), within 10% of the bare Hubbard scale  $U = 8t$  of a doubly-occupied site; the tick marks the peak, not  $U$ . The reflection  $S(q) = S(2\pi - q)$  holds to  $\sim 10^{-4}$  (Lanczos precision), and the analytic f-sum rule  $\int \omega S(q, \omega) d\omega \propto (1 - \cos q)$  is recovered by the plotted (finite-window) grid to  $\sim 0.3\%$ . This is the collective observable the same bitstring-sampled primitive targets; as for  $A(k, \omega)$ , faithfully sampling the *full* response carries the sector-fraction cost of Sec. VII. This channel carries no shot budget: it is a full-sector reference (Table I).

arms consume the same corrupted bitstrings at the same rate with the same seeds and differ only in the recovery step, and there the gain from recovery is measured and grows with the noise: at  $\varepsilon = 2\%$  eighteen of nineteen molecules reach chemical accuracy with recovery against seventeen without it, at  $\varepsilon = 30\%$  recovery roughly halves the error on  $N_2$  ( $31.9 \rightarrow 13.2$  mHa, Fig. 16), and on the Hubbard chain it holds the spectral relative  $L_1$  near 0.05 out to  $\varepsilon \approx 16\%$  (Fig. 15)—seed-averaged, with  $\pm 1$  s.d. bands, at matched noise. That is the noise result of this work.

*Scale.* These runs are 12 and 24 qubits, and the same class of observable has been executed elsewhere at larger scale—retarded spin Green’s functions of the one-dimensional Fermi–Hubbard model on IBM Heron r2

at up to 30 qubits [108], an ARPES-type  $A(k, \omega)$  on a 27-site chain at 54 qubits [109], dynamical structure factors on trapped ions against scattering data [26]—so our device record is smaller than every entry in that class and is not offered as a scale result. It is not the smallest such record either, and we name the one below it: a spectral function was reconstructed on a trapped-ion device from a four-qubit Hubbard dimer [36]. What is new in our run is the object assembled, the  $(N \pm 1)$ -sector Lehmann construction from a bitstring-sampled subspace, at a size where the coverage question is degenerate, and Sec. VII says how degenerate in shots: the  $3.5 \times 10^5$  spent—seven circuits at  $5 \times 10^4$  shots each, pooled into one subspace—are  $14.4$  times the  $(2.43 \pm 0.34) \times 10^4$  that Table IX measures for one orbital seed and one branch at  $L = 6$ , so complete

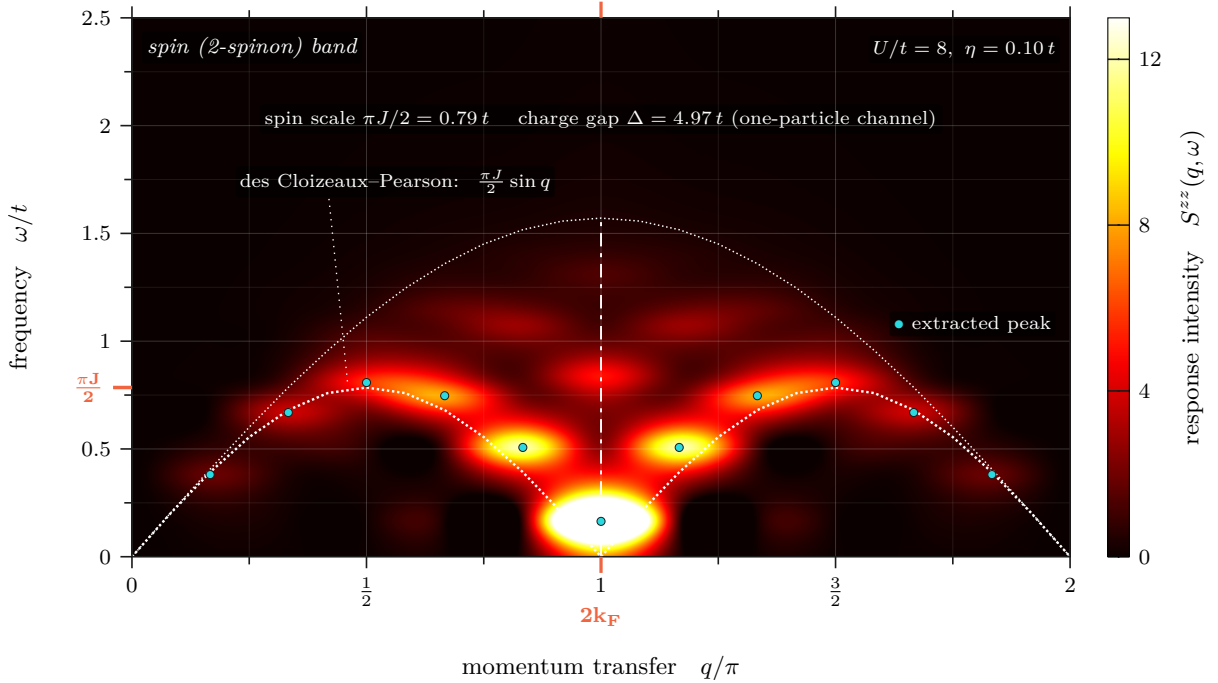


FIG. 12. **Spin structure factor: the low-frequency companion of the charge response.** The dynamical spin structure factor  $S^{zz}(q, \omega)$  of the same half-filled Hubbard chain ( $U/t = 8$ ,  $L = 12$ ; here  $\eta = 0.10 t$ , finer because the spin scale is small), computed by a Haydock continued fraction of depth  $n_\ell = 150$  from a spin seed  $S_q^z|\psi_0\rangle$  in the  $N$ -particle sector; at this  $\eta$  that depth moves the reference by 3.0% between  $n_\ell = 100$  and 200, so the lineshape is quoted with its depth and not as exact (Sec. VIII). *Contrast Fig. 11:* the charge scale stays at the Mott gap  $\Delta \approx 5 t$  at every size measured, while the spin scale is  $0.16635 t$  at  $L = 12$  and closes as  $\approx 2.1/L$  (Table XI). At this size the spin response is one dominant pole plus at most two satellites per momentum (Table XII), not a dense band, and what the figure shows is a set of broadened lines; whether the band touches  $\omega = 0$  in the thermodynamic limit is not established here and is not claimed. This is spin-charge separation in the collective (two-particle) channel: charge and spin live in energetically separate worlds, the spin set by the superexchange  $J = 4t^2/U = 0.5 t$ , some  $16\times$  smaller than  $U$ . The spectral weight piles up on the des Cloizeaux-Pearson [105] lower boundary  $\frac{\pi J}{2}|\sin q|$  (dotted), bounded above by  $\pi J|\sin \frac{q}{2}|$  [106]; the numerically-extracted peak dispersion (cyan markers) tracks this lower boundary to  $\sim 3\%$ ; the antiferromagnetic soft mode at  $q = \pi$  carries the dominant weight. The reflection  $S^{zz}(q) = S^{zz}(2\pi - q)$  holds to double-precision round-off and  $S^{zz}(q \rightarrow 0) \rightarrow 0$  (total- $S^z$  conservation). The same bitstring-sampled primitive that yields  $A(k, \omega)$  and  $S(q, \omega)$  targets this collective response as well.

coverage of a 300-dimensional sector is what the budget predicts rather than a surprise.<sup>3</sup> The first size at which a device run would test the reconstruction rather than the pipeline is  $L = 8$ : sector 3920, at  $(2.70 \pm 0.13) \times 10^5$  shots per orbital seed and branch—a factor 1.3 *below* what was already spent here, so what confined this run to  $L = 6$  was not the shot budget. The momentum-resolved observable at that size is a different matter: sixteen channels,  $\approx 4 \times 10^6$  shots, an order of magnitude above this job.

<sup>3</sup> The seven circuits are indexed  $k = 0, \dots, 6$ , and the  $k = 0$  circuit carries no evolution: it is the  $t = 0$  reference, so six of the seven are evolution circuits. Table IX instead replicates the repository protocol,  $K = 18$  with 19 snapshots, at  $U/t = 8$  rather than the  $U/t = 4$  of the device run; this is a consistency statement between two budgets, not an identity.

## B. The molecular suite, and the size of its sectors

The nineteen-molecule suite reconstructs the orbital-projected addition spectral function  $A(\omega)$  at dissociation in a valence bond-breaking active space, against a subspace grown by multinomial sampling of the time-evolved probe: 8000 draws at each of 27 evolution times,  $2.16 \times 10^5$  per molecule, at  $\eta = 0.10$  Ha, from a single fixed random seed and with no spread reported across seeds. What decides the reach of that test is not the shot budget but the sector it is poured into, and Table XIII gives it.

Twelve of the nineteen reconstructions are exact by construction rather than by approximation, and the dimensions say why: seven sectors have dimension six or less, and five more are exhausted by six to twelve determinants. The sentence of v1-v2 concluding from this suite that the reconstruction is “both exact and efficient across chemistry” is withdrawn, because on twelve of the

TABLE XIII. **What the nineteen-molecule reconstruction tests.** The suite grouped by whether the reported error is an approximation or an identity.  $D$  is the dimension of the  $(N+1)$  sector the reconstruction runs in—all nineteen species take the addition branch—and  $|\mathcal{S}|$  the absolute subspace size at the end of the protocol. Top row: five sectors of dimension 2, one of 3 and one of 6, where the reported error is identically zero and where in three cases the sampled subspace is the entire two-dimensional sector. Middle row: five further cases whose subspace already contains the whole support of the probe, so what is reported is floating-point round-off. Bottom row: the seven cases in which a truncation costs something. The largest sector in the suite is 300, the dimension of the  $L = 6$  Hubbard sector of Sec. IX A, against 731 808 at  $L = 12$  on the lattice side of this work.

|                                  | count | $D$    | $ \mathcal{S} $ | rel- $L_1$                                  |
|----------------------------------|-------|--------|-----------------|---------------------------------------------|
| sector of dimension $\leq 6$     | 7     | 2–6    | 1–2             | 0 exactly                                   |
| subspace spans the probe support | 5     | 24–90  | 6–12            | $\leq 1.4 \times 10^{-14}$                  |
| genuine truncation               | 7     | 20–300 | 5–149           | $2.2 \times 10^{-6}$ – $1.0 \times 10^{-4}$ |

TABLE XIV. **The nineteen-molecule fermionic-magic suite**, computed in a cc-pVDZ active space; the active-space CASCI energies are converged to  $1.1 \times 10^{-13}$  Ha, which is a convergence check and not a validation against an independent method (Sec. IX B). The fermionic AntiFlatness  $\mathcal{F}_1$  ( $k=1$  Majorana antiflatness;  $\mathcal{F}_1=0$  iff Gaussian/mean-field, maximum  $\mathcal{F}_1^{\max}=2n_o$ ) is listed near equilibrium and at dissociation, in a valence bond-breaking active space CAS( $n_e, n_o$ ). Fermionic magic switches on for covalent bond-breaking, is already large at equilibrium for inherently multireference species ( $\text{C}_2$ ,  $\text{O}_2$ ), and stays near-Gaussian for ionic and weakly-correlated bonds—so  $\mathcal{F}_1$  marks the multireference, strongly-correlated regime rather than bond length *per se*. The last two columns validate the *method*: at the dissociation geometry the reconstruction of the (orbital-projected) spectral function  $A(\omega)$  from a subspace drawn in  $2.16 \times 10^5$  shots matches the exact Lehmann result to relative- $L_1$  error, on a subspace that is a fraction  $|\mathcal{S}|/D$  of the full  $(N+1)$  sector. Twelve of the nineteen sectors are exhausted by that subspace and their agreement is exact by construction; see Table XIII. The listed  $\mathcal{F}_1 = 4\text{tr}[\gamma(1-\gamma)]$  are directly computed spin-orbital values (for the five open-shell species they are *not*  $2N_u$ ; see Sec. VI). Per-molecule equilibrium and dissociation geometries, active-space orbital indices, FCI energies and residuals, and the absolute  $|\mathcal{S}|$ , sector dimension  $D$  and minimal bond dimension  $\chi$  are listed in App. G.  $\dagger$ : near-degenerate ground state.

| Molecule                           | Bonding                   | CAS( $n_e, n_o$ ) | $\mathcal{F}_1^{\text{eq}}$ | $\mathcal{F}_1^{\text{diss}}$ | $\mathcal{F}_1^{\text{diss}}/\mathcal{F}_1^{\text{max}}$ | rel- $L_1$         | $ \mathcal{S} /D$ |
|------------------------------------|---------------------------|-------------------|-----------------------------|-------------------------------|----------------------------------------------------------|--------------------|-------------------|
| <i>Covalent bond-breaking</i>      |                           |                   |                             |                               |                                                          |                    |                   |
| F <sub>2</sub>                     | covalent single           | (1+1,2)           | 0.068                       | 3.985                         | 1.00                                                     | $< 10^{-13}$       | 0.50              |
| N <sub>2</sub>                     | covalent triple           | (3+3,6)           | 0.847                       | 11.377                        | 0.95                                                     | $2 \times 10^{-5}$ | 0.21              |
| H <sub>2</sub> O                   | multicentre               | (2+2,4)           | 0.007                       | 6.936                         | 0.87                                                     | $< 10^{-13}$       | 0.50              |
| HF                                 | polar single              | (1+1,2)           | 0.001                       | 3.460                         | 0.87                                                     | $< 10^{-13}$       | 1.00              |
| NH <sub>3</sub>                    | multicentre               | (3+3,6)           | 0.080                       | 10.289                        | 0.86                                                     | $1 \times 10^{-5}$ | 0.47              |
| CO                                 | polar triple              | (3+3,6)           | 0.688                       | 8.192                         | 0.68                                                     | $1 \times 10^{-4}$ | 0.41              |
| H <sub>6</sub>                     | H-chain                   | (3+3,6)           | 0.521                       | 7.735                         | 0.64                                                     | $2 \times 10^{-5}$ | 0.50              |
| H <sub>4</sub>                     | H-chain                   | (2+2,4)           | 0.252                       | 5.113                         | 0.64                                                     | $< 10^{-13}$       | 0.50              |
| H <sub>2</sub>                     | covalent single           | (1+1,2)           | 0.038                       | 2.511                         | 0.63                                                     | $< 10^{-13}$       | 0.50              |
| BeH <sub>2</sub>                   | multicentre               | (2+2,4)           | 0.039                       | 3.754                         | 0.47                                                     | $< 10^{-13}$       | 0.25              |
| <i>Multireference / open-shell</i> |                           |                   |                             |                               |                                                          |                    |                   |
| C <sub>2</sub> <sup>†</sup>        | covalent double           | (4+4,6)           | 4.743                       | 8.001                         | 0.67                                                     | $< 10^{-13}$       | 0.10              |
| NO                                 | polar (radical)           | (4+3,6)           | 0.790                       | 7.449                         | 0.62                                                     | $2 \times 10^{-6}$ | 0.25              |
| O <sub>2</sub>                     | covalent double (triplet) | (5+3,6)           | 1.673                       | 6.000                         | 0.50                                                     | $1 \times 10^{-5}$ | 0.25              |
| CN                                 | polar (radical)           | (4+3,6)           | 1.307                       | 5.938                         | 0.49                                                     | $1 \times 10^{-5}$ | 0.24              |
| OH <sup>†</sup>                    | polar (radical)           | (3+2,4)           | 0.013                       | 3.703                         | 0.46                                                     | $< 10^{-13}$       | 0.33              |
| <i>Ionic / weakly correlated</i>   |                           |                   |                             |                               |                                                          |                    |                   |
| BH                                 | metal hydride             | (2+2,4)           | 0.518                       | 0.573                         | 0.07                                                     | $< 10^{-13}$       | 0.25              |
| BeH                                | metal hydride (radical)   | (2+1,3)           | 0.035                       | 0.386                         | 0.06                                                     | $< 10^{-13}$       | 0.67              |
| LiH                                | metal hydride             | (1+1,2)           | 0.002                       | 0.075                         | 0.02                                                     | $< 10^{-13}$       | 1.00              |
| LiF                                | ionic                     | (1+1,2)           | 0.000                       | 0.001                         | 0.00                                                     | $< 10^{-13}$       | 1.00              |

nineteen there is nothing for efficiency to mean. A second label goes with it: the suite was described as “FCI-verified” to better than  $10^{-5}$  Ha, but the comparison behind that phrase is an active-space CASCI energy against itself, and

the residual the appendix records is  $1.1 \times 10^{-13}$  Ha across all 38 points—a convergence check on a deterministic diagonalization, worth keeping in the reproducibility appendix at its true magnitude, but not a validation against



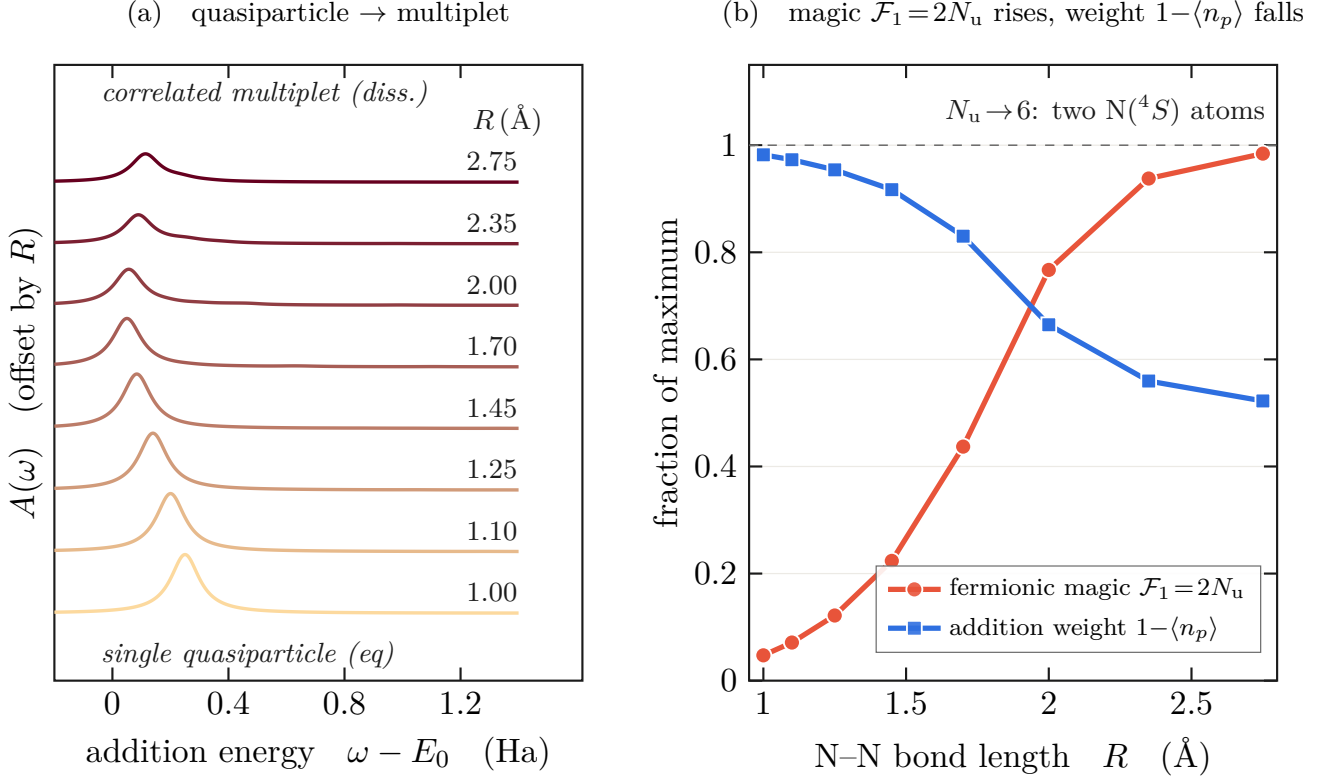


FIG. 13. **Magic and multireference character in  $N_2$  dissociation.** (a) The exact addition spectral function  $A(\omega)$  (orbital-projected, CAS(6,6), cc-pVDZ) along the triple-bond stretch, offset by  $R$ : the sharp coherent quasiparticle peak at equilibrium ( $R = 1.0$  Å) broadens into a correlated multiplet on dissociation as its spectral weight collapses. (b) Two physically independent quantities along the same coordinate. The fermionic magic obeys the *exact* identity  $\mathcal{F}_1 = 2N_u$  [Eq. (17), verified to  $< 10^{-13}$ ], so a single curve carries both: it rises with the orbital-rotation-invariant multireference count—the effective number of unpaired electrons  $N_u = \sum_i n_i(2 - n_i)$  from the natural-orbital occupations  $n_i$ , approaching its atomic limit of 6 (the two triplet  $N(^4S)$  atoms) as the bond fully breaks. The integrated addition weight  $\int A^+ d\omega = 1 - \langle \hat{n}_p \rangle$  (from panel a) falls from 0.98 to 0.52 over the same stretch; the two cross as the bond breaks. The active-space CASCI energies are converged to  $1.1 \times 10^{-13}$  Ha, which is a convergence check and not a validation against an independent method (Sec. IX B).

an independent method.

What the suite does test is the bottom row of Table XIII, and it is a result. On seven Hamiltonians— $O_2$ , CN, NO, CO,  $N_2$ ,  $NH_3$  and  $H_6$ , in sectors of dimension 20 to 300—the same primitive reproduces the orbital-projected  $A(\omega)$  to a relative  $L_1$  between  $2.2 \times 10^{-6}$  and  $1.0 \times 10^{-4}$  from 21% to 50% of the sector. Those seven are independent chemical Hamiltonians rather than one molecule repeated, and their agreement is a consistency check of the primitive outside the Hubbard family. It is not a scaling statement and is not on a common axis with Sec. VII, since the Hamiltonians, the broadening and the error level all differ. Per-species dimensions, subspace sizes and bond dimensions are in App. G, whose column  $D$  is the one whose absence from Table XIV—which prints  $|\mathcal{S}|/D$  and never  $D$ —made the scope invisible.

### C. The boomerang: a region of no advantage, derived from our own resource statement

The resource statement of Sec. VI has a consequence for the standing of this method, and we derive it here rather than wait to have it put to us. What tracks the determinant support is  $\chi$ , the minimal bond dimension across a bipartition in the working basis (Sec. VI E). But  $\chi$  is, by definition, the cost parameter of a matrix-product representation of the same state. A cost claim built on a basis-dependent boundary quantity therefore cannot select for the quantum side of the comparison: wherever our own diagnostic says the determinant support is small it says a matrix-product state is small too, and both methods are charged the same number. The thesis of Sec. VI implies its own region of no advantage, and implies it more sharply than an external objection would.

The instance is in the literature and it is not marginal. Ouyang, Chi and Chan reproduce the 7 260 trajectories

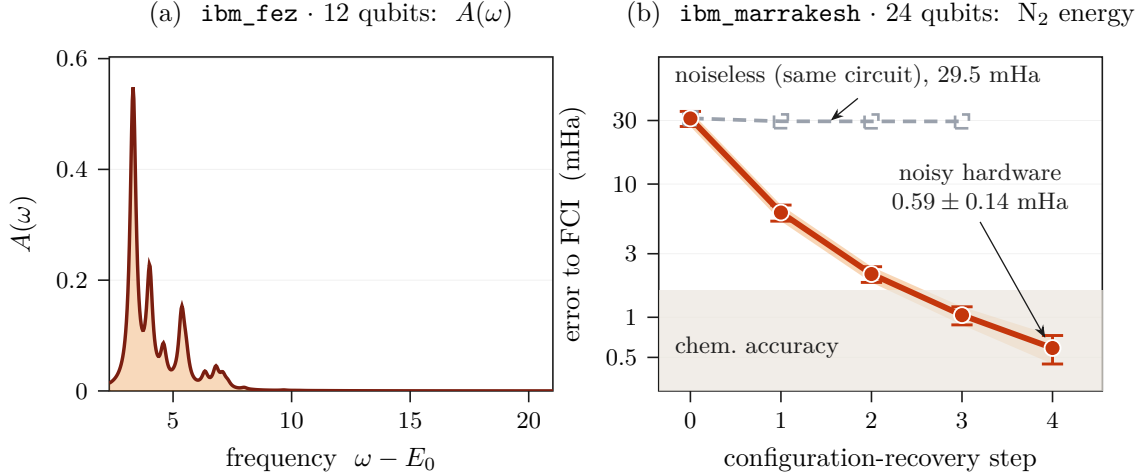


FIG. 14. **The complete quantum-hardware record of this work: two runs on IBM Heron processors.** (a) Single-particle spectral function  $A(\omega)$  of the  $L = 6$  Hubbard model at  $U/t = 4$ ,  $\eta = 0.15t$  (12 qubits; the classical validation of Fig. 2 is a different instance,  $U/t = 8$ ), reconstructed from 350,000 computational-basis bitstrings—50,000 from each of seven circuits—sampled on `ibm_fez` (TREX readout twirling, Pauli twirling, and dynamical decoupling). The device histogram covered the complete  $(N+1)$  sector ( $|S| = 300/300$ ), so the reconstruction is exact by coverage—a proof of principle, and neither a fidelity measurement nor a claim of advantage. (b) Ground-state energy error of stretched  $N_2$  ( $R = 2.0$  Å, CAS(10e,12o)/cc-pVDZ, 24 qubits) versus self-consistent configuration-recovery step, on `ibm_marrakesh` (10,000 shots, double-factorized Trotter circuit). The noisy device (solid) reaches  $0.59 \pm 0.14$  mHa of the exact FCI value—well below chemical accuracy—whereas the noiseless statevector simulation of the *same* shallow circuit (dashed) is stranded at 29.5 mHa. That contrast is not read here as device noise *helping*: the mechanism such a reading needs is decided by the two recovered subspace sizes, and the record stores neither (Sec. IX A). The solid curve is the mean over 8 recovery seeds on the cached device counts; the shaded envelope and the capped whiskers at every step both show the  $\pm 1$  s.d. spread across those 8 seeds (widest at step 0,  $31.1 \pm 4.0$  mHa, where it starts together with the noiseless curve, and tapering as recovery converges to  $0.59 \pm 0.14$  mHa; best seed 0.40 mHa). The systematic version across nineteen molecules under a controlled noise model is Fig. 16. These two runs are the entirety of the quantum hardware used here; every other result in this work is an exact classical simulation.

of the Fermi–Hubbard dynamics experiment of Hartnett *et al.*—120 qubits, up to 90 Trotter steps [114]—by tensor contraction at bond dimension 32, faster and to higher accuracy than the device run [113]: the same pattern in which tensor-network and Pauli-propagation methods have overtaken earlier hardware demonstrations, the first “quantum utility” claim [22] among them [23, 24, 65], and in which unitary cluster-Jastrow sampling is itself now classically reproduced, including by an extended match-gate simulator [15, 16]—against which the standing worst-case hardness result for that circuit family, its embedding of instantaneous-quantum-polynomial sampling [67, 68], is not a contradiction but a statement about a different regime. Two architecture-independent results point the same way: polynomial-time classical estimation of expectation values in noisy circuits [115], and classical estimation in noiseless chaotic circuits across geometries [116]. On the ground-state energy the strong classical selectors [76–78] already win on cost [19, 20, 95], and the case for a generic exponential advantage in ground-state chemistry was found wanting before that [14]; v1–v2 conceded this without reservation and we concede it again.

One framing does not survive, and it is ours. Versions 1–2 opened by arguing that the energy is where classical methods have caught up, and moved the tar-

get to dynamics; in one dimension that inverts, because dynamical DMRG computed all three of the momentum-resolved channels reported here nineteen years ago [117]—at  $L = 120$  in the charge channel,  $L = 100$  in the spin channel and  $L = 90$  in the single-particle channel—and the correction-vector and optical branches of the same literature reach  $L = 64$  and  $N = 256$  [71, 73], at the sizes Sec. VIII tabulates. Dynamics in one dimension is where classical methods arrived first, and no framing of this work says otherwise. Within the region this work measures—one-dimensional Hubbard rings at  $L \leq 14$ , the nineteen active spaces, and a largest minimal bond dimension of 42 over the 74 deposited exact ground states, attained on a molecular active space, the largest on the lattice being 38 in the two-leg ladder—there is no computation performed here that a classical method cannot perform, and we make no advantage claim at any size reached. Deriving one’s own no-advantage region is a stronger position than conceding it; the credit does not change the fact, and the fact is the one just stated.

The same derivation says where the boomerang stops, and that is the measurement this subsection closes on. The controlled contrast of Sec. VI E is a search for a target on which the two costs come apart: at one-body magic matched to within 1%, the two-leg ladder carries

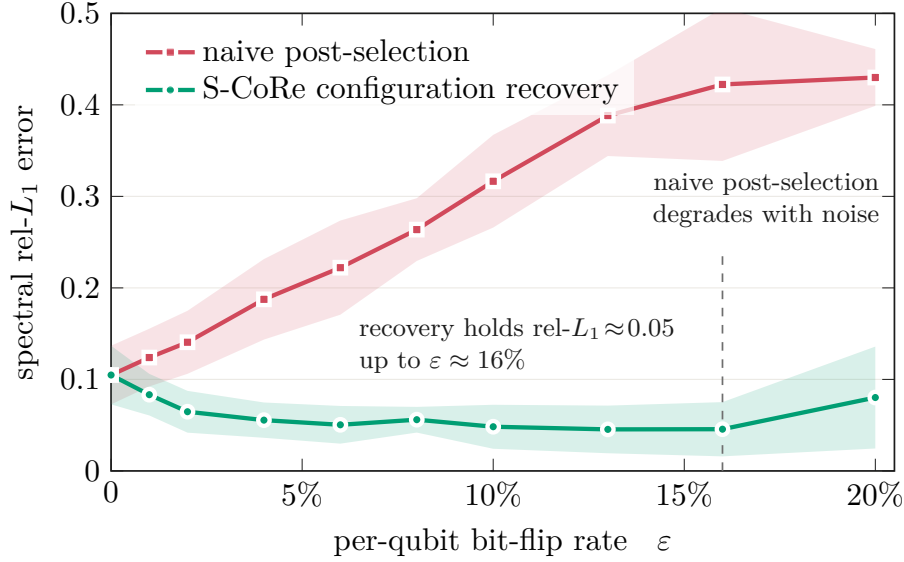


FIG. 15. **Noise robustness (synthetic per-qubit bit-flip channel,  $L = 6$ ).** Spectral relative- $L_1$  error vs the per-qubit bit-flip rate  $\varepsilon$ , applied to bitstrings Born-sampled from the exact distribution (this is a controlled classical simulation, not a device run). Naive post-selection degrades steadily, whereas S-CoRe self-consistent configuration recovery holds the error near 0.05 out to  $\varepsilon \approx 16\%$  (seed-averaged over 6 seeds at 6,000 shots each; bands =  $\pm 1$  s.d. across seeds).

$\chi = 13$  against the chain’s 5 at  $L = 6$  and  $\chi = 38$  against 8 at  $L = 8$ , while its determinant support exceeds the chain’s by 1.52, 1.68 and  $2.24\times$ . The quantity to look for is a target whose minimal bond dimension is large in *every* ordering while the sampled support stays compact—a property of geometry and orbital connectivity rather than of dimensionality, so two-dimensional lattices and molecular orbital graphs are where it would be measured, and nothing here measures it. Section V supplies the second coordinate of that search: in the determinant basis the Krylov space of the probe already has full coordinate support at  $L = 8$ , so a non-vacuous leakage certificate at small sector fraction requires a different working basis, not a better ranking within this one.

#### D. The dequantization test, and a resolution floor that five seeds cannot cross

The one place where a learned model could remove the quantum sampling from this pipeline is the generation of the dominant determinants, and we tested it. On an under-sampled device head for  $N_2$  (1 000 shots, five seeds), a cheap classical Epstein–Nesbet perturbative tail [43] applied to the *same* histogram—no additional shots, no additional quantum processing, one application of  $H$  to the sampled wavefunction—cuts the energy error from  $39.24 \pm 3.95$  to  $26.98 \pm 0.66$  mHa, a 31% reduction. A generative flow network [83] trained as a tail completer and fused onto that corrected prior then moves the point estimate a further 4.04 mHa, to  $22.94 \pm 3.30$  mHa.

Whether that further 4.04 mHa is resolved cannot be

decided from the deposited record, and we say so rather than settle it by choosing a convention. The deposit holds five summary rows and a note that the per-seed values are available on request: the seeds are paired across conditions, but the pairing is not stored. A paired  $t$  statistic for a difference of 4.04 mHa between arms of standard deviation 0.66 and 3.30 over five seeds is then bounded only by the correlation between the arms, the standard deviation of the differences lying anywhere in  $[|3.30 - 0.66|, 3.30 + 0.66] = [2.64, 3.96]$  and the statistic accordingly in  $t \in [2.281, 3.422]$ , against a two-sided critical value of 2.776 at four degrees of freedom. The interval brackets the critical value, so no reading of the deposited numbers decides the question; and the two ratios quoted in v1–v2, “3.1 s.d.” and “1.2 s.d.”, are the difference divided by one arm’s standard deviation, which is not a test statistic for a difference at all, and are withdrawn with the reading they supported.

A second limit is not repaired by repairing the deposit, because it belongs to the design. With five paired seeds the exact two-sided sign test cannot return a  $p$ -value below  $2/2^5 = 0.0625$ , and the signed-rank test shares that floor: *no* five-seed comparison in this family can be declared significant by a distribution-free test, whatever its effect size. Six paired seeds bring the floor to 0.031 and are the minimum design that could resolve anything. What survives every admissible pairing is the classical step: for its 12.26 mHa improvement the same construction gives  $t \geq 5.95$  and  $p \leq 0.004$  at four degrees of freedom across the whole range of correlation between the arms.

Both figures of this test are therefore withdrawn and the surviving statement is made in prose. The generative

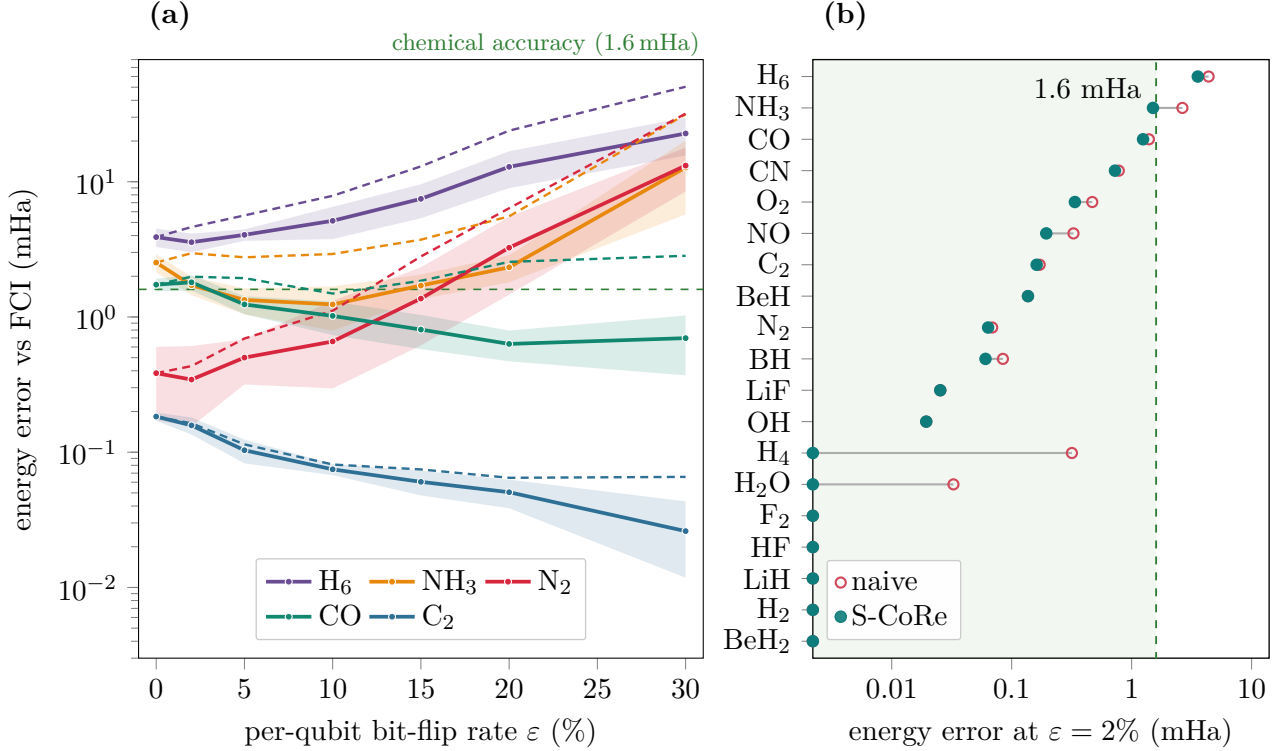


FIG. 16. **SQD ground-state energy under a controlled bit-flip model across the molecular suite.** (a) Energy error vs the per-qubit bit-flip rate  $\varepsilon$  for five representative molecules ( $H_6$ ,  $NH_3$ ,  $N_2$ ,  $CO$ ,  $C_2$ ; seed-averaged over 8 seeds at 4,000 shots each, bands =  $\pm 1$  s.d. across seeds; log scale). Solid = S-CoRe recovery, dashed = naive post-selection. The advantage of recovery over naive post-selection, both arms consuming the same corrupted bitstrings at the same rate with the same seeds, grows with noise. For  $CO$  and  $C_2$  the error at finite  $\varepsilon$  also falls below its own  $\varepsilon = 0$  value; that second comparison is not used here (Sec. IX A): it is decided by the size of the recovered subspace, which this sweep does not record. (b) The full nineteen-molecule suite at a realistic  $\varepsilon = 2\%$  (log scale, sorted): for each molecule the naive post-selection error (open circle) is linked to the S-CoRe value (filled), so the leftward pull is the recovery gain; the shaded band is chemical accuracy ( $< 1.6$  mHa). Eighteen of nineteen land inside it;  $H_6$ , the most multireference case, is the residual challenge. Panels (a) and (b) are independent seed ensembles (panel (a): 8 seeds; panel (b): 5 seeds, 4000 shots), so a given molecule's  $\varepsilon = 2\%$  value can differ by up to a factor of a few between them (for  $N_2$ , 0.34 vs 0.06 mHa; both far below chemical accuracy).

comparison goes because no script in the deposited source produces its data file—the figure script carries the five numbers as literals copied from a file belonging to a different project, and the deposited note records a sector dimension that does not match the active space named beside it—and a figure the deposit cannot regenerate does not belong in a version that asks for credit for its deposit. The amortized companion result goes for a reason internal to its own numbers: across six leave-one-out folds both arms sit at relative  $L_1$  between 0.62 and 0.88, individual folds reaching 1.25, against the 0.05 acceptance threshold used everywhere else here, so it compares two reconstructions that both fail it; and its oracle arm is worse than the untrained classical selector in two of six folds at three of the four subspace fractions, which a well-defined oracle cannot be.

What the test establishes is the statement we keep. On this instance the dominant determinants are supplied by the device sampling and recovered by drawing more shots, not manufactured by a learned generator—the class

of construction is by now several: physics-informed generative priors, neural quantum states built on selected configurations, autoregressive and dual-channel samplers for selected configuration interaction, and neural-network classifiers [17, 18, 84, 98, 99]; the largest single improvement available to an under-sampled sample-based head costs no quantum resources at all and should be the default post-processing step, since a demonstration that omits it is measured against a baseline that is too weak; and the defensible role for a learned model is the one that consumes the device histogram or schedules device work [25, 82, 85–88], the boundary the companion study maps from the ground-state side [21].

### E. Scope, declared

Collected in one place, so that no reader has to assemble it from five sections. The lattice results are one-dimensional Hubbard rings at  $L \leq 12$  with  $\eta = 0.10$ –



$0.18t$ , which is the exact-diagonalization convention for this observable—Xu, Lu, Tohyama and Shao compute it at  $L = 12$  and  $\eta = 0.2t$  at half filling, and at  $L = 16$  at quarter filling, in the extended Peierls–Hubbard chain [121]—but the resolution criterion the dynamical-DMRG literature applies at those sizes is not met here, and Sec. VIII quotes that criterion in its original words and withdraws the three terms it did not license instead of enlarging  $\eta$ ; the size was never the indefensible part, the words were. The published subspace curves are the infinite-shot limit of the declared ranking protocol, and finite-shot fractions are reported only where Table X attaches a measured budget and a seed count. The certificate of Theorem 1 is vacuous at every fraction this work publishes, by factors of 1.66 to 8.61 at the five sizes measured, so the reconstructions reported here are at present uncertified and the non-vacuity threshold is given as a bracket rather than as a number (Sec. V). The device record is the two runs of Sec. IX A, at 12 and 24 qubits, the first exact by coverage; the chemistry record is Table XIII. No result in this work is offered as evidence of an advantage over a classical method, at any size reached here.

What the work does claim is four things, each with its measurement attached: one sampling primitive returning four frequency-resolved channels from a single measurement protocol (Fig. 1); a two-sided a posteriori bound on the error of the reconstruction, with its non-vacuity threshold measured at five sizes up to a sector of dimension 10 306 296 and its crossing calibrated to a true relative error of  $\approx 3 \times 10^{-3}$  within a factor of three (Sec. V); an obstruction theorem showing that no orbital-rotation invariant supplies a cheaper a priori predictor of that bound (Sec. VI); and three measured prices—support, resolution and shots—of which the third had no number at all before this version (Sec. VII).

## F. Data availability statement

The data that underlie the figures and the quoted numbers of this work are openly available in the reproduction repository [134] at <https://github.com/nicolasbonilla/dynamical-spectral-functions-sqd>, the code under the MIT licence and the manuscript, the deposited datasets and the reproduction documentation under CC BY 4.0. Every figure’s plotted values are deposited—as a JSON table in `data/` or as a plain-text table in `paper/figs/`—and `docs/FIGURE_PROVENANCE.md` names the file that is authoritative for each; the one exception is Fig. 1, a circuit schematic that plots no data. The classical data—the exact-diagonalization spectral functions of the one-dimensional Hubbard model, the molecular suite and the resource statistics—are additionally *regenerable*: the computation scripts that produced them are deposited in `src/`, and `docs/REPRODUCE.md` gives the reproduction commands together with a measured table of what a clean clone does and does not reach. An automated

guardian (`python src/verify.py`) recomputes the underlying physics by exact diagonalization and checks the deposited artefacts against it—831 checks over 9 843 numeric assertions, exiting non-zero on any discrepancy. It covers the deposited data files and the numbers printed in the figure sources, in the tables and in the abstract; it is not a proof that every sentence of this manuscript is verified, and five known open defects are printed by name on every run.

The quantum-hardware results are deposited as the arrays that enter the figures, together with the backend name, the shot count and the IBM Quantum job identifier for each of the two runs (`data/heron_spectral.json`, `data/hw_lucj_n2_result.json`). *The raw measured bitstrings are not part of the deposit*: the notebooks in `notebooks/` retrieve the counts from the IBM Quantum service by job identifier, which requires an account and a job that is still retrievable, so the classical post-processing of the raw counts cannot be re-run from the deposited files alone. The hardware runs themselves cannot be regenerated in any case—the jobs are closed, and a device executed today has a different calibration, so new counts would be new data rather than a reproduction. For the generative-model control of Sec. IX D only the five summary rows are deposited, in `data/gflow.json`; no script in the repository regenerates them, which is the reason given there for withdrawing the two figures that were built on them, and the per-seed values are available from the author on reasonable request. Known gaps between the deposited code and the deposited artefacts—six of the seventeen figures enter as PDF with no standalone L<sup>A</sup>T<sub>E</sub>X source in the deposit, the tables plotted in Fig. 17 are deposited but the script that builds them is not, three deposited computation scripts have no deposited output, and parts of the pipeline require `pyscf` or an IBM Quantum account—are enumerated in `docs/KNOWN_DISCREPANCIES.md` rather than left for the reader to discover.

## X. CONCLUSION

The error of a spectral function reconstructed on a subspace of sampled configurations splits, by Theorem 1, into the Born weight the shots failed to capture and the Hamiltonian coupling that leaks across the boundary of that subspace at the resolution asked for; only the first of the two is bought by drawing more bitstrings, and this paper measures—subspace by subspace, at five system sizes, in a sector of dimension up to 10 306 296—where the resulting certificate says something and where it does not.

Four things were needed to make that statement, and all four are measurements rather than expectations. The bound itself stands where a weight-only bound cannot: no inequality indexed on the captured weight alone can hold, by an analytic counterexample and by counting (Sec. III D); what a Krylov-adequate subspace does guarantee is moment exactness through order  $2K + 1$ , stated

with the proof the code realizes (Theorem 2). Evaluated, the new certificate is vacuous at every published fraction, at each of the five sizes evaluated, by factors of 1.66 to 8.61, converged in Krylov depth at  $L = 12$ —and the same sweep yields the one positive property that survived four independent attacks: the crossing of the trivial bound occurs at a true relative  $L_1$  error between  $1.2 \times 10^{-3}$  and  $7.3 \times 10^{-3}$ , median  $2.8 \times 10^{-3}$ , over  $L = 6$ –12 and  $\eta = 0.10$ – $0.25t$ —a drift, not scatter: ordered without exception in both arguments and conservative in  $L$ —and is computed without the answer (Sec. V). No orbital-rotation invariant substitutes for it: one-body fermionic magic is constant along an orbit on which the determinant support runs from  $O(K)$  to  $2^K$  at bond dimension 2, and constant on fibers on which the subspace-aware certificate still separates (Theorem 3, Proposition 1), the general invariant/basis-dependent division being due to Tarabunga *et al.* [59]. And the method is priced on three axes rather than one, the third—shots—measured here for the first time (Sec. VII), while the primitive itself is validated against the one answer known exactly and independently of it: on the same lattice and from the same calculation, the charge scale converges to a finite thermodynamic value while the spin scale closes as  $\approx 2.1/L$ , with an independent effective-Heisenberg control (Table XI).

Three limits bound all of it, and we state them as flatly as the results. *First, the certificate does not certify the reconstructions this paper reports.* It is vacuous at every published fraction, at each of the five sizes evaluated, and the obstruction is structural rather than incidental: the coordinate support of the Krylov space  $\mathcal{K}(H, \phi)$  is 296 of 300 determinants at  $L = 6$  and all 3920 of 3920 at  $L = 8$ , so at  $L = 8$  the leakage is strictly positive for every proper coordinate subspace and no leakage-type certificate in the determinant basis—this one or another—can be non-vacuous there at low fraction. That support is the whole sector at  $L = 8, 10, 12$  and 14—every size this work reports. *Second, the largest momentum-resolved spectrum is  $L = 12$ , by exact diagonalization.* The three momentum-resolved responses reported here have been computed classically by dynamical DMRG since 2007 [117]—at 120, 100 and 90 sites in the charge, spin and single-particle channels respectively—and at  $L = 48$  in the Hubbard chain,  $L = 64$  in the Heisenberg chain, with correction vectors [73]; the wider classical toolbox for one-dimensional spectra is broader still—correction-vector and time-dependent DMRG, Chebyshev matrix-product states and kernel-polynomial expansions [66, 70, 72, 74, 75], and the real-time propagation of selected configuration-interaction wavefunctions [79]; one dimension is where tensor networks, since White’s formulation [69], are near-optimal [102], and nothing in this work competes with that standard; the vocabulary of the thermodynamic limit is accordingly absent from it (Sec. VIII). *Third, the certified regime costs more shots than the method saves.* Moving from the published fraction to the non-vacuous band multiplies the shots per snapshot needed for the marginal configuration by 7.5 to 10 at  $L = 8$  and by 20 to 35 at  $L =$

12, where that configuration carries a Born probability of 1 to  $2 \times 10^{-9}$ ; a regime that a device can certify is, on these systems, a regime in which diagonalizing the whole sector is already the cheaper thing to do.

Two questions follow, and only one of them is a matter of more computing. Whether a certified regime exists at low sector fraction is now a well-posed measurement rather than an expectation, and the calibration above is the instrument for making it; whether it exists at all is decided by a prior question that this work can pose but not answer—whether there is any single-particle basis in which the coordinate support of  $\mathcal{K}(H, \phi)$  is not the entire sector, since it is that support, and not the quality of the sampler, that pins the threshold measured here.

What this paper leaves behind is therefore narrow and, we think, useful: a bound on the error of a sample-based spectral function that can be evaluated without the answer, a map of where it is informative, and the negative half of that map stated in the same voice as the positive one.

## ACKNOWLEDGMENTS

We acknowledge the use of IBM Quantum services for this work; the two device runs reported in Sec. IX A were executed on `ibm_fez` and `ibm_marrakesh`. The views expressed are those of the author and do not reflect the official policy or position of IBM or the IBM Quantum team.

## Appendix A: Counterexamples to the weight-only bound, and the pooled sweep

*The three-level family.* Section IIID exhibits a two-level instance forcing the constant of the weight-only bound (9) to its trivial value at  $w_S = 1$ . With  $w_S < 1$  strictly the failure is unbounded. Take  $H = \text{diag}(\begin{pmatrix} 0 & g \\ g & 0 \end{pmatrix}, \delta)$ ,  $\phi = e_0 + \varepsilon e_2$  and  $S = \{0\}$ , so that  $1 - w_S = \varepsilon^2/(1 + \varepsilon^2)$  may be made as small as wished while the weight displaced to  $\pm g$  stays  $O(1)$ . The ratio of the measured error to the weight-only bound then diverges as  $\delta^{-2}$  in the displacement  $\delta$  of the discarded component; we measured it to  $7.6 \times 10^7$  before double precision limits the family.

*The pooled sweep behind Fig. 17.* Five machine-readable scans, written by two independent implementations, contribute 621 subspaces: 243 from the original sweep, and  $162 + 112 + 104$  from a re-implementation deposited as `cert_stress.json`, `cert_akw.json` and `cert_teqsci.json`. Fifteen reproduce the target to double-precision round-off (relative  $L_1 \leq 10^{-10}$ ) and define no ratio, leaving 606 live subspaces over four Hamiltonian families — Hubbard chains, XXZ,  $t$ - $V$ , Anderson and sparse random matrices — and eight construction rules, of which 156 are finite-shot realizations at 4000 shots per snapshot and the remainder deterministic weight-ordered or structured selections. On those 606: the weight-only bound fails on 468; among the 363 with  $w_S \geq 0.99$  — the regime a sampling experiment aims for — it fails on all 363, the mildest by a factor 2.10, with 50 of them carrying  $w_S = 1$  exactly, so that the bound is identically zero and the ratio infinite; below  $w_S = 0.99$  it survives on 138 of 243, in every case where the relative error is already of order unity and the bound excludes nothing. Theorem 1 is violated by none of the 606; the largest ratio of error to bound is 0.9999999997, attained on four subspaces with  $w_S = 0$  exactly, where  $A_S \equiv 0$  and the trivial branch is an identity rather than an inequality. The leakage branch is the smaller of the two on only 61 of the 606: the sweep shows that the bound *holds*, not that it is tight.

*Independent re-verification.* The 378 rows of the three deposited scans were re-audited for this version by a script sharing no code with the one that produced them. It confirms 378 of 378 for each side of Eq. (7); a largest error-to-bound ratio of 0.9999999997 over the 376 rows that define one; the exclusion lemma (10) on 378 of 378, with 357 rows at  $K_S = -1$  and 21 at  $w_S = 1$  exactly; and violations of the weight-only bound on 219 of the 219 finite-ratio rows with  $w_S \geq 0.99$ , the mildest by a factor 2.00.

## Appendix B: Moment exactness: statement, proof, sharpness, and the shot count

We make precise the two things a Krylov-adequate subspace does deliver. Fix the seed  $|\phi\rangle = \hat{c}_p^\dagger|0\rangle$  (the electron-addition branch; removal is identical), with weight  $\|\phi\|^2 = \langle\phi|\phi\rangle$ , and let  $A(\omega) = \sum_n |\langle n|\phi\rangle|^2 \mathcal{L}_\eta(\omega - (E_n - E_0))$  be

the exact spectral function, where  $\{|n\rangle, E_n\}$  diagonalize  $H$  in the  $(N+1)$ -particle sector and  $\mathcal{L}_\eta(x) = \frac{\eta/\pi}{x^2 + \eta^2}$ . For a configuration set  $\mathcal{S}$  let  $P = P_S$  be the orthogonal projector onto  $\text{span}(\mathcal{S})$ , let  $H_S = PHP$  with eigenpairs  $\{|\tilde{m}\rangle, \tilde{E}_m\}$ , and let  $A_S(\omega) = \sum_m |\langle \tilde{m}|\phi\rangle|^2 \mathcal{L}_\eta(\omega - (\tilde{E}_m - E_0))$  be the reconstruction. Write  $\mathcal{K}_K = \text{span}\{\phi, H\phi, \dots, H^K\phi\}$  and  $w_S = \|P\phi\|^2/\|\phi\|^2$ . Moments are taken on the point measures  $d\mu = \sum_n |\langle n|\phi\rangle|^2 \delta(\omega - (E_n - E_0))$  and  $d\mu_S = \sum_m |\langle \tilde{m}|\phi\rangle|^2 \delta(\omega - (\tilde{E}_m - E_0))$ , not on the broadened  $A$ , whose moments of order  $\geq 2$  diverge.

### 1. What the proof uses

Theorem 2 and its proof are stated in Sec. IV A. The proof uses  $P = P^\dagger = P^2$ ,  $H = H^\dagger$  and  $\mathcal{K}_K \subseteq \text{span}(\mathcal{S})$ , and nothing else. In particular it does not use that  $\mathcal{S}$  is a coordinate subspace, that it was obtained by drawing bitstrings, or that any Lanczos basis is well conditioned; it holds verbatim for the orthogonal projector onto a Krylov space, which is the form evaluated in Sec. V. The single asymmetry in Eq. (13)— $a$  runs to  $K$  while  $b$  runs to  $K+1$ —is what produces the odd reach  $2K+1$  rather than  $2K$ . Two structural remarks follow at no cost.

(a) *The node count.*  $d\mu_S$  has  $|\mathcal{S}|$  atoms, one per Ritz pair, not  $K+1$ . Equation (12) is therefore a weak quadrature statement for a rule of that size: a Gauss rule of  $n$  nodes for  $d\mu$  is exact through order  $2n-1$  [40], so the  $K=1$  row of Table II, with 280 nodes, would reach order 559 if its nodes were the Gauss nodes of  $d\mu$ . They are not, and it reaches order 3. This is the measurable content of the correction to the published proof: identifying  $A_S$  with an order- $(K+1)$  Gauss rule misdescribes both the object and its reach.

(b) *Unit captured weight is implied, not assumed.* For a coordinate subspace  $P$  is diagonal, so  $P\phi = \phi \iff \text{supp}(\phi) \subseteq \mathcal{S} \iff w_S = 1$ . Since  $\mathcal{K}_0 = \text{span}\{\phi\}$ , the hypothesis of Theorem 2 at any  $K \geq 0$  already implies  $w_S = 1$ . Every subspace admissible for moment exactness therefore sits at the extreme of the weight budget of Sec. III, where the shot term vanishes identically and the whole residual error is leakage—which is why the two statements are complementary rather than competing.

### 2. Sharpness, control, and the limits of the validation system

Table II is computed by exact diagonalization of the  $(N+1, S_z=+1)$  sector of the open  $L=6$ ,  $U/t=8$  Hubbard chain ( $D=300$ ,  $\|\phi\|^2=1/2$ ), with  $\mathcal{S} = \bigcup_{a \leq K} \text{supp}(H^a\phi)$ . Four measurements support it.

*Reach.* Through order  $2K+1$  the relative discrepancy is  $\leq 3.2 \times 10^{-15}$  for  $K=0,1$  and  $\leq 7.5 \times 10^{-15}$  for  $K=2$ . At order  $2K+2$  it is  $9.5 \times 10^{-4}$  ( $K=0$ ) and  $6.0 \times 10^{-8}$  ( $K=1$ ), eleven and seven decades above the arithmetic floor. The absolute discrepancies at that

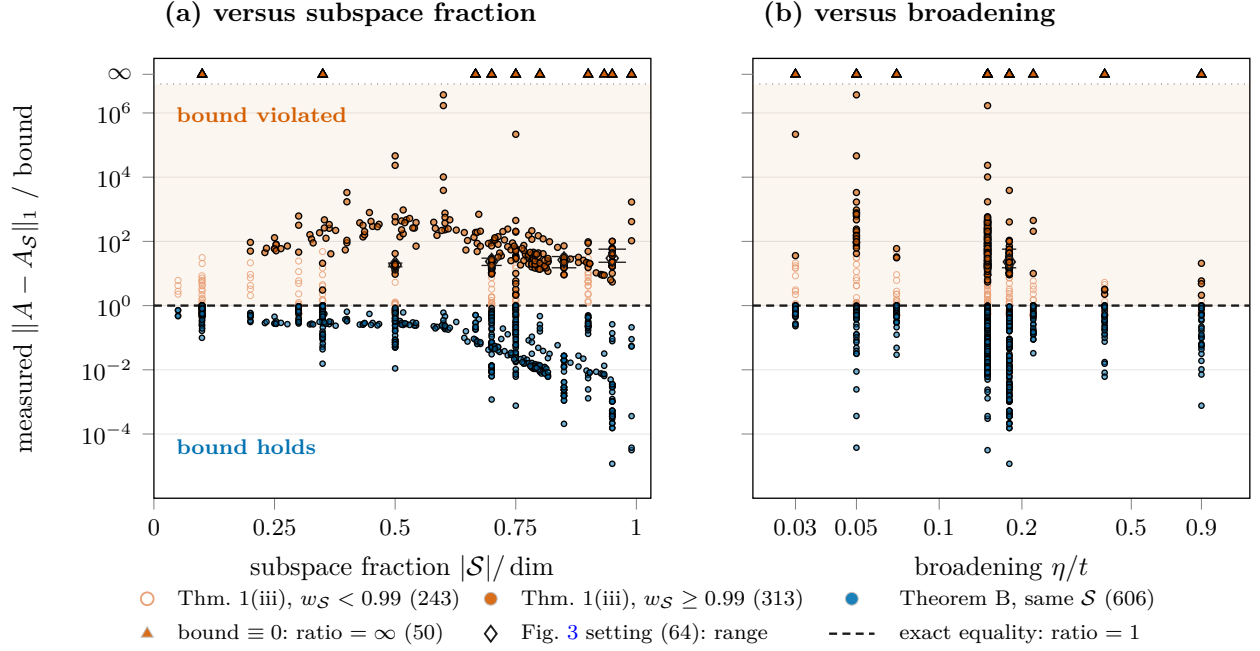


FIG. 17. **The published bound of Theorem 1(iii) fails on exactly the subspaces the paper relies on; the replacement does not.** For each subspace  $\mathcal{S}$  we plot the measured error  $\|A - A_S\|_1$  divided by the bound, so every point above the dashed line at 1 is a violation. Orange: the published weight bound  $2\|\varphi\|^2(1 - w_S)$ , split by captured weight (filled,  $w_S \geq 0.99$ ; open,  $w_S < 0.99$ ). Blue: Theorem B as stated,  $\min\{\|\varphi\|^2(1 + w_S), (\|\varphi\| + \|\varphi_S\|)(\|\varphi_Q\| + \Lambda_S(\eta)/\eta)\}$ , on the same subspaces. We pooled 621 subspaces from five machine-readable sweeps written by two independent implementations (239 + 4 from the original sweep; 162 + 112 + 104 from an independent re-implementation); 15 of them reproduce the target to within double-precision round-off (relative  $L_1 \leq 10^{-10}$ ) and define no ratio, leaving 606 live subspaces, of which 156 are finite-shot realisations at 4000 shots per snapshot and the remainder are deterministic weight-ordered or structured selections. *The published bound is violated on 468 of the 606.* On 50 of them it is identically zero: those subspaces carry the whole weight to double precision ( $w_S = 1$  exactly, and  $K_S \in \{0, 1, 2\}$ , so the remaining term is the undefined constant  $C\rho^{-K_S}$ ); their ratio is infinite and they are drawn on the row marked  $\infty$ , above the dotted rule. Among the 313 finite-ratio points with  $w_S \geq 0.99$ —the regime the paper invokes—*every one* is a violation, the mildest by a factor 2.10; with the 50 infinite ones that is 363 of 363, and the finite factors reach  $3.66 \times 10^6$ . Below  $w_S = 0.99$  the bound survives on 138 of 243 points, but only where the relative error is already of order unity, so it excludes nothing there. Diamonds mark the configuration of Fig. 3 ( $L = 8$ ,  $U/t = 8$ ,  $\eta = 0.18t$ , momentum seeds  $c_{k\uparrow}^\dagger|\Psi_0\rangle$ , subspace fractions 0.50/0.70/0.85/0.95, both branches, eight momenta): that reconstruction reproduces the deposited `sampld_akw_L8.json` to  $2.1 \times 10^{-8}$  per momentum, and its violation factor is 15.1–57.3, of which 15.1–33.7 at the published fraction 0.85. Those 64 rows sit at only four abscissae, so they are drawn as a min–median–max whisker at each one rather than as 64 overlapping markers (medians 18.5, 23.6, 26.5, 29.9 at fractions 0.50, 0.70, 0.85, 0.95); each of them is also plotted individually inside the filled orange series, which is where their scatter can be read. Theorem B is violated on none of the 606; its ratios span  $1.18 \times 10^{-5}$  to 0.9999999997. That maximum is reached on four subspaces with  $w_S = 0$  exactly, where  $A_S \equiv 0$  and  $\|A - A_S\|_1 = \|\varphi\|^2$  equals the trivial cap by algebra—the branch that carries no information—and the leakage branch  $\Lambda_S(\eta)/\eta$  is the smaller of the two on only 61 of the 606 rows: the figure shows that Theorem B *holds*, not that it is tight. In (b) the  $\eta$  axis is not a sweep within one model: the Hubbard-chain rows exist only at  $\eta/t = 0.05, 0.15, 0.18$  and the stress families (XXZ,  $t$ -V, Anderson, sparse random) only at 0.03, 0.07, 0.22, 0.40, 0.90, so no trend across the full axis may be read from it.

order,  $3.104583 \times 10^{-2}$  and  $1.340167 \times 10^{-4}$ , agree to six significant figures across all six implementations listed below and are properties of the construction, not of the arithmetic. The  $K = 2$  row carries no entry there, for a structural reason and not a numerical one: the ladder has filled the sector,  $A_S = A$  identically, and their relative  $L_1$  distance is 0 to machine arithmetic. One run of one of the six routes does report a nonzero residue in that row; the same run reports a cyclic-subspace dimension of 301 inside a sector of dimension 300 and emits floating-point overflow while forming the high powers. It is discarded,

and the corrected run is the one quoted.

*Reproducibility of the residues.* The relative discrepancies are quoted, and the absolute ones below order  $2K+2$  are not, for a reason worth recording. The exact moments  $\mu_j = \langle \phi | (H - E_0)^j | \phi \rangle$  run from  $5.0 \times 10^{-1}$  at  $j = 0$  to  $1.6 \times 10^5$  at  $j = 6$ , so an absolute residue at the top of the ladder is a roundoff quantity whose size depends on the route taken to the same identity. Across six equivalent routes—Rayleigh–Ritz on the shifted or on the unshifted matrix, crossed with exact moments obtained from the Lehmann sum, from the shifted matrix—



vector recursion, or by binomial recombination of unshifted powers—the relative residue below order  $2K+2$  stays inside  $[1.1, 1.8] \times 10^{-15}$  at  $K = 0$  and  $[1.0, 3.2] \times 10^{-15}$  at  $K = 1$ , a spread under a factor 3.3, while the absolute residue at the same orders is not stable to two significant figures. We therefore quote relative residues below  $2K+2$ , and absolute ones only at  $2K+2$ , where the discrepancy is physical and route independent.

*Negative control.* Over 300 uniformly drawn coordinate subspaces of the same size  $|\mathcal{S}| = 200$ , the zeroth moment is wrong by between  $1.86 \times 10^{-2}$  and  $4.18 \times 10^{-1}$  (median  $1.67 \times 10^{-1}$ ), i.e. by 3.7% to 84% of  $\|\phi\|^2$ , with every one of the 300 draws failing; the Krylov-support subspace of identical size is exact to  $4.4 \times 10^{-16}$ . At  $|\mathcal{S}| = 280$  the random draws fail by  $4.8 \times 10^{-5}$  to  $2.6 \times 10^{-1}$  (median  $1.12 \times 10^{-2}$ ), again without exception, against  $5.0 \times 10^{-16}$ . The discriminating variable is the identity of the determinants, not their number.

*Scale.* The Krylov ladder at  $L = 6$  occupies 0.667, 0.933 and 1.000 of the sector at  $K = 0, 1, 2$ , so it exhausts the sector at  $K = 2$  and the table has exactly three rows. This system can verify Eq. (12) and locate its exact reach; it cannot exhibit the behaviour of  $\|A - A_{\mathcal{K}_K}\|_1$  at large  $K$ , which is the regime Conjecture 1 concerns, and we make no claim there.

*Why the open chain.* The ladder terminates when the coordinate subspace contains the cyclic subspace generated by  $\phi$ —not when it fills the sector—and at that point the reconstruction is exact, because  $\text{span}(\mathcal{S})$  then carries every eigenvector of nonzero Lehmann weight. On the periodic rings of Sec. VII that happens early: at  $L = 6$  the ladder runs  $200 \rightarrow 296$  and stalls at 296 of 300 determinants, where  $A_{\mathcal{S}} = A$  to  $1.0 \times 10^{-14}$  in relative  $L_1$  and the moments are exact past order  $2K+1$  at every  $K \geq 1$ . The ring therefore offers a single row in which the reach of Eq. (12) is visible at all ( $K = 0$ : relative discrepancy  $4.4 \times 10^{-16}$  through order 1,  $4.6 \times 10^{-3}$  at order 2); the open chain offers two, and is the system on which the method validation of Sec. II is performed. Within the reach they cover, both agree: the identity is exact, sharp at the next order, and decidable from the subspace alone.

### 3. The geometric-convergence claim, and why it is left open

Conjecture 1 is easily taken for a consequence of Eq. (12). It is not one: moment exactness gives  $2K + 2$  linear identities, and  $\|A - A_{\mathcal{K}_K}\|_1$  is not a finite linear functional of them. A proof would require (i) a polynomial approximation theorem for  $x \mapsto \mathcal{L}_\eta(\omega - x)$ , uniform in  $\omega$ , on a real set containing  $\text{spec}(H)$  and  $\text{spec}(H_{\mathcal{K}_K})$ , and (ii) a tail estimate controlling the complement. For (i) the natural route is a Bernstein-ellipse bound, in which the resolution enters through the poles of the kernel at  $x = \omega \pm i\eta$ : the largest admissible ellipse has semi-minor axis below  $\eta$ , so its parameter—and with it any geometric rate—approaches 1 as  $\eta \rightarrow 0^+$ . That is a mechanism and

not a derivation: it exhibits neither  $C(\eta)$  nor  $\rho(\eta)$ , and neither appears anywhere in this work or in the deposit.

The prefactor is moreover pinned from below by the two-level family of Sec. III D. With  $H = \begin{pmatrix} 0 & g \\ g & 0 \end{pmatrix}$  and  $\phi = e_0$  one has  $\langle \phi | H | \phi \rangle = 0$ , so  $\mathcal{K}_0 = \text{span}\{\phi\}$  carries a single pole at the origin while  $A$  carries two at  $\pm g$ ; by direct integration at  $\eta = 0.15$ ,  $\|A - A_{\mathcal{K}_0}\|_1 = 1.5197, 1.8352, 1.9503, 1.9832, 1.9943, 1.9974, 1.9984$  for  $g = 1, 3, 10, 30, 100, 300, 1000$  at  $\|\phi\|^2 = 1$ , approaching the trivial ceiling  $2\|\phi\|^2 = 2$  monotonically. Any  $C(\eta)$  valid uniformly on this family therefore satisfies  $C(\eta) \geq 2\|\phi\|^2$ . A geometric bound with a trivial prefactor is not worthless—it would still decay in  $K$ —but it cannot be non-vacuous at small  $K$ , and small  $K$  is where a shot-limited subspace lives.

Conjecture 1 therefore remains open, and we name its two missing ingredients explicitly so that it can be attacked rather than left unremarked. What this work offers in its place is not a rate in  $K$  at all but a bound on  $\|A - A_{\mathcal{S}}\|_1$  itself, at the  $\eta$  of the figure and on the subspace actually in hand (Theorem 1), measured subspace by subspace, together with a measurement of where that bound is non-vacuous (Sec. V). That is a weaker statement than a rate and a stronger one than a conjecture, and it is the one the deposit can reproduce.

### 4. The sample-complexity bound

**Proposition 2** (Sampling capture). *Let  $p(x) = \frac{1}{K+1} \sum_{k=0}^K |\langle x | e^{-iHt_k} \phi \rangle|^2 / \|\phi\|^2$ , draw  $T$  bitstrings independently from  $p$ , and let  $\mathcal{S}$  be the set of distinct outcomes. For  $\varepsilon, \delta \in (0, 1)$ , if  $T \geq \varepsilon^{-1} \log(1/(\varepsilon\delta))$  then with probability at least  $1 - \delta$  the set  $\mathcal{S}$  contains every configuration  $x$  with  $p(x) \geq \varepsilon$ .*

*Proof.* Let  $\mathcal{N}_\varepsilon = \{x : p(x) \geq \varepsilon\}$ . Since  $\sum_x p(x) = 1$  and each member contributes at least  $\varepsilon$ ,  $|\mathcal{N}_\varepsilon| \leq 1/\varepsilon$ . For fixed  $x \in \mathcal{N}_\varepsilon$ ,  $\Pr[x \notin \mathcal{S}] = (1 - p(x))^T \leq (1 - \varepsilon)^T \leq e^{-\varepsilon T}$ . A union bound gives  $\Pr[\mathcal{N}_\varepsilon \not\subseteq \mathcal{S}] \leq \varepsilon^{-1} e^{-\varepsilon T}$ , which is at most  $\delta$  for the stated  $T$ .  $\square$

The cardinality bound  $|\mathcal{N}_\varepsilon| \leq 1/\varepsilon$  is deterministic, and it is what removes the circularity of the published form  $T = O(\varepsilon^{-1} \log(|\mathcal{S}|/\delta))$ , in which the random size of the sampler’s own output appears inside a bound on the number of draws that produce it. Where  $\varepsilon^{-1} \leq |\mathcal{S}|$ , the new logarithm is also the smaller one, so nothing is conceded by the repair.

The scope of  $\varepsilon$  needs the same care. It thresholds a Born probability of the time-averaged mixture, and Proposition 2 makes no statement about  $\|A - A_{\mathcal{S}}\|_1$ . Two independent obstructions stand between the two quantities. First, the weight left uncaptured is bounded only by  $|\{x : p(x) < \varepsilon\}| \cdot \varepsilon$ , which for a delocalized state is  $O(1)$ : many small probabilities can sum to most of the mixture. Second, and decisively, a subspace holding the *entire* Born support of the probe— $w_{\mathcal{S}} = 1$  exactly, nothing left to capture at any  $\varepsilon$ —still has relative  $L_1$  error 0.28 at  $L = 6$

on the open chain of Table II, and 0.43 at  $L = 6$  and 0.33 at  $L = 8$  on the periodic rings, all at  $\eta = 0.18t$  and all relative to  $\|\phi\|^2$  (Sec. III); it is the  $K = 0$  row of Table II. Proposition 2 is a statement about the sampler and about nothing else. The spectral error at fixed  $\eta$  is the subject of Theorem 1, and what the composition of the two actually costs is reported as three separate prices in Sec. VII.

### Appendix C: The leakage certificate: hypotheses, protocol, and the anatomy of the threshold

Theorem 1 holds under three hypotheses. Each is a statement about the pipeline that produced the reconstruction rather than about the theorem, so each has to be checked against the pipeline, and none of the three is checked anywhere in the deposit of versions 1–2. We check all three here, with numbers, for the sweep of Sec. V, and add the two protocol declarations and the decomposition that section quotes.

(H1) *Common origin.*  $A$  and  $A_P$  must be referred to the same  $E_0$ . A shift common to both cancels exactly—verified to 1.000000 for shifts of  $10^{-3}$ ,  $10^{-2}$  and  $10^{-1}t$ —which is how the hypothesis is satisfied here, a single  $E_0$  being used for both objects. If  $A_P$  is instead referred to  $E_0 + \delta$ , the discrepancy acquires an additive  $\|\phi_P\|^2(4/\pi)\arctan(|\delta|/2\eta)$ , verified to  $2 \times 10^{-5}$  in all 48 cases tested, *which the certificate does not control*. A ground state obtained by Rayleigh–Ritz on 60% of the  $N$ -electron sector at  $L = 8$  gives  $\delta = 6.7 \times 10^{-4}t$ , which adds 0.0024 to a bound of 1.2036—+0.20%—while multiplying the *measured* error by 16; and the dependence is not even monotone, a shift of  $10^{-2}t$  lowering the measured error at  $L = 6$ , fraction 0.85,  $\eta = 0.18t$  from  $2.7 \times 10^{-2}$  to  $1.2 \times 10^{-2}$ . The measured errors quoted beside a bound in Sec. V are therefore those of a pipeline handed the exact  $E_0$ , and they bound the error of a pipeline that estimates it from neither side. No cheap a posteriori control on  $\delta$  is available: residual (Weinstein) and Temple bounds [127, 128] computed on ground-state subspaces of 20–80% of the  $N$  sector at  $L = 6$  give arctan terms of 1.76, 1.64, 1.40 and 0.92, all vacuous against the ceiling of 2. In a shot-based pipeline  $E_0$  is itself estimated on whatever subspace the shots populated, and the certificate has never been tested that way.

(H2) *Normalization and support of the  $L_1$  norm.* The bound is on  $\int_{\mathbb{R}} |A - A_P| d\omega$  normalized by  $\|\phi\|^2 = \int_{\mathbb{R}} A d\omega$ . A relative error normalized instead by  $\int_W A$  over a finite window  $W$  and evaluated by finite quadrature—the convention of the repository, in which Sec. VII reports—is a different quantity, differing by the product of  $\int_W |A - A_P| / \int_{\mathbb{R}} |A - A_P|$  and  $\|\phi\|^2 / \int_W A$ . Both factors are measured here: the second is 1.0100, 1.0309 and 1.0373 at  $\eta = 0.05$ , 0.15 and 0.18  $t$  at  $L = 6$  and 1.0110, 1.0278 and 1.0335 at  $L = 8$ , while the first lies between 1.0040 and 1.0115 across the 204 configurations of this study. Every number in Sec. V is in the normalization of Theorem 1, and the two conventions are nowhere

interchanged.

(H3) *Exact reference.*  $A$  denotes the exact spectral function of the full  $(N \pm 1)$  sector. If the reference is itself a Lanczos reconstruction of finite depth  $n_\ell$ , the theorem does *not* bound  $\|A^{(n_\ell)} - A_P^{(n_\ell)}\|_1$ , and a measured error below the reference’s own drift measures the reference rather than the reconstruction. That drift runs from  $2.6 \times 10^{-5}$  ( $L = 12$ ,  $\eta = 0.10t$ ) to  $2.8 \times 10^{-7}$  ( $\eta = 0.18t$ ), and at  $L = 10$  the reorthogonalization gap,  $1.71 \times 10^{-5}$ , is of the same order as the depth drift. We therefore quote no measured error anywhere in Sec. V within a factor of 20 of its own reference drift: twelve rows at  $L \geq 10$  fall under that rule and are left blank rather than rounded, the weakest surviving cell— $L = 12$ ,  $\eta = 0.10t$ —standing at a factor 47. One gap is declared and not bounded. The reconstructions this paper publishes are produced by bare Lanczos at depth 150, while the certificate is evaluated on the reorthogonalized Krylov space; the difference between the two has been measured at a single point at  $L = 8$ —relative  $L_1$  of  $1.2454 \times 10^{-3}$  against  $1.2464 \times 10^{-3}$ —and never bounded, and at  $L = 12$  it has never been measured at all.

*Ranking, ties and support.* Two properties of the cut are declared with the protocol, a tie-breaking defect having cost this work numbers before. No determinant of any of the four sectors carries exactly zero accumulated Born weight, so the cut is never filled with configurations the device would never sample in the infinite-shot limit; ties at the cut involve two determinants (four at  $L = 6$ , at fractions 0.80 and 0.85) and move the bound by at most  $6.6 \times 10^{-4}$  relative. Above the support floor  $\frac{1}{2} + \frac{1}{L}$  of Sec. III D, on the other hand, the ranking selects among determinants outside  $\text{supp}(\phi)$ : at fraction 0.85 that is 30.2% of  $S$  at  $L = 6$ , 36.9% at  $L = 8$  and  $\approx 39.6\%$  at  $L = 10$ . Those contribute nothing to  $w_S$ , and they do change  $H_S$ . The depth uncertainty quoted in Sec. V is the full recomputation of  $\text{FR}^*$  with  $L = 12$  capped at  $n_\ell = 300$ : it moves by 0.0024, 0.0021, 0.0012 and 0.0000 at  $\eta = 0.10$ , 0.15, 0.18 and 0.25  $t$ .

*Anatomy of the displacement.* The percentages quoted in Sec. V E are medians over 15 of the 18 ( $\eta$ , target) cells at  $L = 8$ ; three cells are censored from above, the certificate never reaching the target inside the grid, and are excluded rather than imputed. Pooling  $L = 6$  with  $L = 8$  shifts them to 52%, 31%, 11% and 4.1%, which is why the  $L = 8$  figures are the ones quoted. One caveat travels with them, and it is why they are reported as a diagnosis and not as a quantity: of the rungs of the inequality chain only the last is computable by a user from  $(H, S, \phi, \eta)$ , so the decomposition is of an object the user never sees.

### Appendix D: The geminal witness: proof and angle sweep

*Proof of Theorem 3.* Each geminal occupies its own four spin-orbitals, so (18) is a product across the  $K$  blocks and  $\gamma$  is block diagonal with spatial natural occu-

pations  $(n_{b_i}, n_{a_i}) = (2 \cos^2 \theta, 2 \sin^2 \theta)$ ; hence  $N_u = 8K \cos^2 \theta \sin^2 \theta = 2K$  at  $\theta = \pi/4$ , and  $\mathcal{F}_1 = 4K$  by Eq. (17). The Majorana covariance is likewise block diagonal, and at  $\theta = \pi/4$  a single maximally-paired geminal is the two-determinant cat, whose covariance vanishes identically ( $\Gamma = 0$ ): every order saturates its maximum,  $\mathcal{F}_k = 2n_o = 4$  per geminal for all  $k$ , giving  $\mathcal{F}_2 = 4K$ . This equality of orders is a knife-edge of maximal pairing; away from it the orders separate ( $\theta = \pi/6$  gives  $\mathcal{F}_1 = 3$ ,  $\mathcal{F}_2 = 3.75$  per pair, both still large on the same family). In the pairing basis each geminal is a two-configuration superposition, so the product carries  $2^K$  determinants. Finally, as a product of two-orbital factors the state is a matrix-product state whose every bond crosses at most one geminal, so  $\chi = 2$  in that ordering, and contraction and Born sampling then cost  $O(K)$ . All four values are reproduced by exact diagonalization for  $K = 1, \dots, 4$ .  $\square$

Table XV continues the fiber exhibit of Table VII into a continuous sweep of the rotation angle at  $K = 3$ . It stops at  $s = 0.654$ : at the maximal-pairing angle  $s = \pi/4$  the Born distribution of  $\phi$  is exactly flat over  $2^K$  determinants, so the top- $|\mathcal{S}|$  rule does not select a subspace and no contrast measured there is well defined. Across every row the captured weight is identical at the two endpoints to  $2.2 \times 10^{-16}$ , so the entire difference sits in the leakage term, and the certificate at the rotated endpoint is the trivial bound  $1 + w_S$  in every row.

### Appendix E: One-body magic cannot bound the determinant support

**Proposition 3** (Coefficient-insensitivity obstruction). *Let  $\mu$  be any functional of a fixed finite set of reduced density matrices, continuous in the state. Then  $\mu$  cannot lower-bound the raw determinant support (the unthresholded Slater rank); like the CP tensor rank, this count is not lower-semicontinuous [94]: for every  $c > 0$  there exist states with  $\mu$  arbitrarily small and rank arbitrarily large.*

*Proof.* Rank is a coefficient-insensitive count. Take  $|\Psi_\epsilon\rangle = |D_0\rangle + \epsilon \sum_{k=1}^K |D_k\rangle$  with the  $D_k$  distinct. As  $\epsilon \rightarrow 0$  every reduced density matrix converges to that of  $|D_0\rangle$ , so any continuous  $\mu \rightarrow \mu(|D_0\rangle)$  (for  $\mathcal{F}_1$ , to 0), while the rank remains  $\Theta(K)$ .  $\square$

The obstruction is about the unthresholded rank; the operative, coefficient-sensitive support  $|\mathcal{S}|_\epsilon$  of Sec. VI collapses to 1 along the same  $\epsilon \rightarrow 0$  path, so continuity alone does not obstruct it. What obstructs  $|\mathcal{S}|_\epsilon$  itself is the orbital-rotation symmetry—no invariant functional survives it—together with the two equal-weight witnesses below, whose amplitudes are  $\Theta(1)$  and therefore threshold-robust. Two explicit witnesses rule out the natural repairs. The two-determinant cat  $(|1 \dots 10 \dots 0\rangle + |0 \dots 01 \dots 1\rangle)/\sqrt{2}$  has  $\gamma = \frac{1}{2}\mathbb{K}$ , hence  $\mathcal{F}_1 = 2n_o$  (maximal) yet Slater rank 2, so no  $\mathcal{F}_1$ -monotone bounds  $|\mathcal{S}|$  from below; a paired (BCS)

Gaussian state has Gaussian rank 1 yet fractional occupations, so no occupation functional bounds it from above. The only computable lower bounds on  $|\mathcal{S}|$  are the basis-dependent orbital-Schmidt (matrix-flattening) ranks [102]—the tensor-network bond dimension—to which any Gaussian-invariant magic measure is blind. That lower bound is exact:

**Proposition 4** (The bond dimension lower-bounds the support). *For any state and any bipartition of the spin-orbitals into (L, R), the Schmidt rank  $\chi$  across the cut (the minimal matrix-product-state bond dimension there) satisfies*

$$\chi \leq |\mathcal{S}|, \quad (\text{E1})$$

where  $|\mathcal{S}|$  is the number of fixed-basis Slater determinants of nonzero amplitude. Hence the minimal bond dimension is a rigorous, basis-dependent lower bound on the determinant support. The bound is one-sided: the geminal witness of Thm. 3 has  $\chi = 2$  yet  $|\mathcal{S}|_{\text{pair}} = 2^K$ , so  $|\mathcal{S}|$  can exceed  $\chi$  exponentially in a generic basis (and descends toward  $\chi$  in the natural-orbital basis), while the orbital-invariant one-body magic  $\mathcal{F}_1$  bounds neither (Prop. 3).

*Proof.* Write  $|\Psi\rangle = \sum_{D \in \mathcal{S}} c_D |D\rangle$  over its support. Each determinant factorizes across the cut,  $|D\rangle = |D_L\rangle \otimes |D_R\rangle$ , so the matricization  $A_{D_L, D_R} = c_D$  has at most  $|\mathcal{S}|$  nonzero entries; its rank—the Schmidt rank  $\chi$ —is therefore at most  $|\mathcal{S}|$ .  $\square$

Since a matrix-product-state simulation costs  $\text{poly}(\chi)$ , Prop. 4 says that *in the sampler's own basis a tensor network is never less compact than the sampled subspace*: this is why we make no advantage claim in one dimension, where  $\chi$  is small, and why any advantage must live in the higher-entanglement regime where  $\chi$  itself—and with it the classical tensor-network cost—grows (the chain-to-ladder jump of Fig. 8). Empirically  $\chi$  is also the operative correlate of the measured cost: across the 74 exact ground states of Fig. 7 the thresholded support  $|\mathcal{S}|_\epsilon$  tracks  $\chi$  (pooled Spearman  $\rho = 0.72$ ;  $\rho = 0.90$  within the  $n = 38$  molecular suite, where however one-body magic scores a comparable  $\rho = 0.86$ —the two resources are separated not there but by the Hubbard sweep, where  $\mathcal{F}_1$  anti-orders the support in all six strata while  $\chi$  orders it at a mean  $\rho$  of 0.948; the pooled coefficients for that sweep,  $\rho = 0.57$  and  $-0.11$ , are Simpson reversals of the six strata and are not the evidence here, Sec. VIE). Two caveats belong with that number. A rank correlation is not a statement of tightness, and per the geminal witness the bound can be exponentially loose. And the proposition is proved for the raw support: once amplitudes are thresholded the inequality need not survive—here  $|\mathcal{S}|$  is cut at 99% of the weight while  $\chi$  is cut at  $\epsilon = 0.05$ , a tighter tolerance—and  $|\mathcal{S}|_\epsilon$  falls below  $\chi$  in 7 of the 38 molecular points. What stands is that  $\chi$  is a rigorous floor on the raw support and an empirical correlate of the measured one, whereas  $\mathcal{F}_1$  orders the support with a sign that is not stable from family to family (Sec. VIE).



TABLE XV. **Angle sweep of the fiber exhibit**,  $K = 3$ ,  $\delta = 0.01$ ,  $\eta = 0.10$ ,  $|\mathcal{S}| = 4$ . Columns as in Table VII.

| $s$   | $w_S$ (both endpoints) | $\hat{\Lambda}/\eta$ al. | $\hat{\Lambda}/\eta$ rot. | cert. al.    | cert. rot.      |
|-------|------------------------|--------------------------|---------------------------|--------------|-----------------|
| 0.131 | 0.982963               | 0.328                    | 1.610                     | <b>0.913</b> | 1.983 (trivial) |
| 0.262 | 0.933013               | 0.317                    | 3.010                     | <b>1.131</b> | 1.933 (trivial) |
| 0.393 | 0.853553               | 0.299                    | 4.124                     | <b>1.312</b> | 1.854 (trivial) |
| 0.524 | 0.750000               | 0.277                    | 4.849                     | <b>1.450</b> | 1.750 (trivial) |
| 0.654 | 0.629410               | 0.252                    | 5.139                     | <b>1.543</b> | 1.629 (trivial) |

## Appendix F: The resource statistics, stratified

Section VI reports the correlation evidence within the design that produced it. This appendix gives the design, the per-stratum coefficients and the three tests, so that the decision not to use the two pooled coefficients can be checked rather than taken.

*The lattice design.* The 30 Hubbard points are six  $(L, N)$  strata swept over the same five on-site repulsions  $U = 1, 2, 4, 8, 16$ ; they are not 30 independent observations, and the effective number of independent sweeps is six. Table XVI gives each stratum. The rank correlation of  $\mathcal{F}_1$  with the static support  $|\mathcal{S}|_{\varepsilon_w}$  is  $-1$  in all six, exactly; that of  $\chi$  runs from  $+0.894$  to  $+1.000$  with mean  $+0.948$ , its departures from unity being ties in  $\chi$  and not inversions.

*Three tests, and what each assumes.* The stratified Kendall statistics are  $\tau = -1.000$  for  $\mathcal{F}_1$ , with 0 concordant and 60 discordant within-stratum pairs and no ties, and  $\tau = +1.000$  for  $\chi$ , with 50 concordant, 0 discordant and 10 tied. Because each  $\mathcal{F}_1$  stratum is a perfect reversal of five distinct points, the exact one-sided stratified permutation probability is the product of the six per-stratum chances,  $(1/120)^6 = 3.35 \times 10^{-13}$ , over a relabelling space of  $2.99 \times 10^{12}$ ; the same test applied to  $\chi$ , which has ties, gives  $1.93 \times 10^{-10}$ . Fisher’s combination of the six per-stratum probabilities is conservative here and returns  $6.56 \times 10^{-8}$  and  $1.14 \times 10^{-5}$  ( $\chi^2 = 57.4$  and  $44.7$  on 12 degrees of freedom); we quote the permutation values and record Fisher’s beside them. All are *one-sided*—a perfect reversal is one of the 120 orderings, not two—and all take a product across strata, so all assume the six strata independent. They are not six independent experiments: they are three lengths crossed with two fillings, on one code path and one  $U$  grid. What the numbers establish is that the direction is reproduced six times out of six, not that it generalizes beyond this family.

*Why neither pooled coefficient is used.* Pooled over the same 30 points the coefficients are  $\rho(\chi, |\mathcal{S}|) = +0.569$  and  $\rho(\mathcal{F}_1, |\mathcal{S}|) = -0.109$ , the second with a nominal two-sided  $p = 0.57$  and the first with  $p = 0.0010$ . Pooling six deterministic five-point sweeps with different offsets destroys the within-stratum structure that carries the signal; it is a Simpson reversal in both cases. It happens to destroy the signal in our favour on  $\chi$  and against us on  $\mathcal{F}_1$ , and that is exactly why neither may be kept: a rule that is applied to one coefficient and not to the other is

not a rule.

*The molecular set.* The  $n = 38$  molecular points are 19 molecules at two geometries, i.e. dependent pairs. The intraclass correlation of  $\log_2 |\mathcal{S}|$  within a molecule is 0.214, a design effect of 1.214 and a design-corrected effective  $n$  of 31.3. Ties are pervasive: 29 of the 38 (76%) lie in some tied group of  $|\mathcal{S}|_{\varepsilon_w}$  and 11 are pinned at the floor  $|\mathcal{S}| = 1$ , which caps the attainable Spearman coefficient at  $\rho_{\max} = 0.9863$ ; against that ceiling  $\chi$  reaches 91.4% and  $\mathcal{F}_1$  87.4%. Under a molecule-level cluster permutation ( $10^6$  resamples) both coefficients are beyond the resolution floor of the test,  $p \leq 1.0 \times 10^{-6}$ . The comparison the molecular set is asked to settle is not resolvable, and three procedures agree that it is not:  $\rho = 0.9011$  for  $\chi$  against 0.8618 for  $\mathcal{F}_1$  differ by 0.0393; a molecule-level cluster bootstrap ( $2 \times 10^4$  resamples) puts a 95% interval on the difference of  $[-0.070, +0.189]$ , which contains zero; Williams’s test for dependent correlations gives  $t = 0.94$  on 35 degrees of freedom,  $p = 0.35$ , and Steiger’s  $z = 0.94$ ,  $p = 0.35$ . Both parametric tests assume bivariate normality and untied ranks while 76% of these points sit in a tie, which is why the bootstrap is quoted first; at the cluster-level effective  $n$  of 19, Williams gives  $p = 0.53$ . Pooled over all 74 rows the pair is 0.722 against 0.599 and is likewise unresolved. Recomputed over the 71 *distinct* states—three chain rows duplicate three Hubbard rows—it is 0.746 against 0.592.

*One artefact of the deposit, recorded because it changes a digit.* The pooled  $\rho(\mathcal{F}_1, |\mathcal{S}|)$  stored in the deposit is 0.5991984 and recomputing it from the per-species values gives 0.5992028. The difference is not a bug in either: it is a tie in  $\mathcal{F}_1$  created when two rows that are the same state are stored at different precisions. The quantity is quoted to three figures throughout for that reason.

## Appendix G: Reproducibility data for the molecular suite

Table XVII lists, for every molecule of Table XIV, the equilibrium and dissociation bond lengths, the active space, the  $N$ -electron (neutral, ground-state) sector dimension  $D$ , the ground-state determinant support  $|\mathcal{S}|$  (the number of determinants carrying  $\geq 99\%$  of the FCI weight), and the minimal matrix-product-state bond dimension  $\chi$  at both geometries. These are the quantities that enter the  $\mathcal{F}_1-|\mathcal{S}|$  and  $\chi-|\mathcal{S}|$  correlations of Sec. VI, pooled over the  $n = 38$  equilibrium-and-dissociation



TABLE XVI. **The six Hubbard strata.** Each is a five-point sweep in  $U$  at fixed  $(L, N)$ . The one-sided exact permutation probability of a perfect reversal within one stratum of five distinct values is  $1/120 = 0.0083$ ; ties in  $\chi$  raise it. “distinct  $\chi$ ” counts how many of the five  $\chi$  values differ, which is what caps  $\rho(\chi, |\mathcal{S}|)$  below unity in four of the six.

| $L$ | $N$ | $\rho(\mathcal{F}_1,  \mathcal{S} )$ | $\rho(\chi,  \mathcal{S} )$ | $\chi$ over $U = 1, 2, 4, 8, 16$ | distinct $\chi$ |
|-----|-----|--------------------------------------|-----------------------------|----------------------------------|-----------------|
| 6   | 6   | -1.000                               | +1.000                      | 12, 10, 6, 5, 4                  | 5               |
| 6   | 4   | -1.000                               | +0.894                      | 11, 10, 9, 9, 9                  | 3               |
| 8   | 8   | -1.000                               | +0.949                      | 11, 11, 8, 8, 7                  | 3               |
| 8   | 6   | -1.000                               | +0.975                      | 16, 16, 15, 14, 13               | 4               |
| 10  | 10  | -1.000                               | +0.975                      | 17, 14, 10, 6, 6                 | 4               |
| 10  | 8   | -1.000                               | +0.894                      | 13, 13, 13, 11, 9                | 3               |

TABLE XVII. **Reproducibility data for the nineteen-molecule suite** (cc-pVDZ, valence bond-breaking active spaces).  $R_{\text{eq}}, R_{\text{diss}}$  in Å; for the polyatomics ( $\text{BeH}_2, \text{H}_2\text{O}, \text{NH}_3, \text{H}_4, \text{H}_6$ ) the representative varied bond length is shown and full Cartesians are in the ancillary data.  $D$ : dimension of the  $N$ -electron (neutral, ground-state) sector, in which the tabulated  $|\mathcal{S}|$  lives;  $|\mathcal{S}|$ : ground-state determinant support ( $\geq 99\%$  FCI weight);  $\chi$ : minimal MPS bond dimension across the central spin-orbital cut. The  $|\mathcal{S}|/D$  of Table XIV is a distinct ratio, taken in the  $(N \pm 1)$  spectral sector.

| Molecule             | CAS( $n_e, n_o$ ) | $R_{\text{eq}}$ | $R_{\text{diss}}$ | $D$ | $ \mathcal{S} ^{\text{eq}}$ | $ \mathcal{S} ^{\text{diss}}$ | $\chi^{\text{eq}}$ | $\chi^{\text{diss}}$ |
|----------------------|-------------------|-----------------|-------------------|-----|-----------------------------|-------------------------------|--------------------|----------------------|
| $\text{F}_2$         | (1+1,2)           | 1.412           | 3.00              | 4   | 1                           | 2                             | 2                  | 2                    |
| $\text{N}_2$         | (3+3,6)           | 1.098           | 2.40              | 400 | 7                           | 80                            | 7                  | 32                   |
| $\text{H}_2\text{O}$ | (2+2,4)           | 0.958           | 2.30              | 36  | 1                           | 13                            | 1                  | 13                   |
| $\text{HF}$          | (1+1,2)           | 0.917           | 2.30              | 4   | 1                           | 4                             | 1                  | 4                    |
| $\text{NH}_3$        | (3+3,6)           | 1.012           | 2.33              | 400 | 1                           | 68                            | 4                  | 42                   |
| $\text{CO}$          | (3+3,6)           | 1.128           | 2.40              | 400 | 8                           | 46                            | 14                 | 30                   |
| $\text{H}_6$         | (3+3,6)           | 0.900           | 2.00              | 400 | 9                           | 91                            | 8                  | 36                   |
| $\text{H}_4$         | (2+2,4)           | 0.900           | 2.00              | 36  | 2                           | 11                            | 4                  | 8                    |
| $\text{H}_2$         | (1+1,2)           | 0.741           | 2.00              | 4   | 1                           | 2                             | 1                  | 2                    |
| $\text{BeH}_2$       | (2+2,4)           | 1.334           | 2.90              | 36  | 1                           | 6                             | 1                  | 8                    |
| $\text{C}_2$         | (4+4,6)           | 1.243           | 2.60              | 225 | 13                          | 6                             | 11                 | 2                    |
| $\text{NO}$          | (4+3,6)           | 1.151           | 2.50              | 300 | 5                           | 25                            | 11                 | 7                    |
| $\text{O}_2$         | (5+3,6)           | 1.208           | 2.60              | 120 | 11                          | 4                             | 4                  | 2                    |
| $\text{CN}$          | (4+3,6)           | 1.172           | 2.50              | 300 | 27                          | 9                             | 17                 | 7                    |
| $\text{OH}$          | (3+2,4)           | 0.970           | 2.40              | 24  | 1                           | 5                             | 1                  | 4                    |
| $\text{BH}$          | (2+2,4)           | 1.232           | 2.60              | 36  | 5                           | 3                             | 2                  | 2                    |
| $\text{BeH}$         | (2+1,3)           | 1.343           | 2.70              | 9   | 1                           | 2                             | 1                  | 2                    |
| $\text{LiH}$         | (1+1,2)           | 1.596           | 3.20              | 4   | 1                           | 2                             | 1                  | 4                    |
| $\text{LiF}$         | (1+1,2)           | 1.564           | 3.20              | 4   | 1                           | 1                             | 1                  | 1                    |

points (Spearman  $\chi-|\mathcal{S}|$   $\rho = 0.90$ ,  $\mathcal{F}_1-|\mathcal{S}|$   $\rho = 0.86$ ; Pearson  $\mathcal{F}_1-\log_2 |\mathcal{S}|$   $r = 0.80$ ). The  $|\mathcal{S}|/D$  reported in Table XIV is a *distinct* quantity—the fraction of the  $(N+1)$  sector used in the spectral *reconstruction*, not the ground-state support tabulated here. Every active-space FCI energy is converged to better than  $10^{-12}$  Ha (the largest residual across all 38 points is  $1.1 \times 10^{-13}$  Ha), comfortably below the  $10^{-5}$  Ha quoted in the main text; the ground-state energies, full Cartesian geometries, and orbital active-space indices are provided as ancillary files.

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